From Grothendieck to Quillen and Thomason

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Abstract

We give a systematic account of algebraic K-theory, beginning with Grothendieck's construction of K_0 via the Grothendieck group of vector bundles and its role in the Riemann–Roch theorem, continuing through the elementary K_0 and K_1 of rings, Milnor's K_2 and the Steinberg group, and culminating in Quillen's higher algebraic K-groups defined via the plus construction and the Q-construction, together with the fundamental theorems (devissage, localisation, homotopy invariance, resolution). We then treat Waldhausen's $S_\bullet$-construction and derive Thomason's theorem that the K-theory of the category of perfect complexes on a scheme coincides with Quillen's G-theory for regular schemes, and prove Thomason–Trobaugh's localisation and descent results, including the telescope conjecture for K-theory with finite coefficients.

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1. Introduction and Historical Motivation

Algebraic K-theory is a branch of mathematics that assigns to a ring, scheme, or more general categorical structure a sequence of abelian groups K_n, $n \geq 0$, measuring subtler invariants than the classical tools of commutative algebra or algebraic geometry.

The subject originates with Grothendieck's 1957 reformulation of the Riemann–Roch theorem [BS58]: the natural receptacle for the Chern character is not ordinary cohomology but a group $\mathrm{K}(X)$ built from vector bundles on X by a universal additive process. A decade later Atiyah and Hirzebruch [AtH61] developed the topological analogue $K_{\mathrm{top}}(X)$, which became a powerful tool in homotopy theory.

The algebraic development of the higher groups K_n for $n \geq 1$ proved more difficult. Bass defined $\mathrm{K}_1(R) = \mathrm{GL}(R)^{\mathrm{ab}}$ and $\mathrm{K}_0(R)$ for general rings, and Milnor introduced $\mathrm{K}_2(R)$ and the Steinberg group [Mi71]. The decisive advance came in 1972 when Quillen gave two equivalent definitions of all higher K-groups: the *plus construction* $\mathrm{BGL}^+(R)$ and the *Q-construction* $Q\mathcal{C}\mathcal{C}$ on an exact category $\mathcal{C}$ [Qu73]. Quillen proved the fundamental theorems of algebraic K-theory – resolution, devissage, localisation, and homotopy invariance – and computed $\mathrm{K}_n(\mathbb{F}_q)$ for finite fields.

The next landmark is Waldhausen's $S_\bullet$*-construction* [Wa85], which applies to the wider class of *categories with cofibrations and weak equivalences* and is better suited to homotopy-theoretic applications such as the stable parametrised h-cobordism theorem.

Finally, Thomason and Trobaugh [TT90] gave a comprehensive treatment of the K-theory of schemes, working with perfect complexes rather than locally free sheaves, proving a general localisation theorem and a descent (Nisnevich) spectral sequence, and establishing that for a regular scheme the K-theory of perfect complexes equals Quillen's G-theory.

Plan of the paper.

- §2: K_0 of topological spaces and rings; Grothendieck–Riemann–Roch.
- §3: Bass's K_1, Whitehead's lemma, Milnor's K_2, the Steinberg group.
- §4: Quillen's plus construction; comparison with K_0 and K_1.
- §5: exact categories, the Q-construction, and the passage from exact sequences to classifying spaces.
- §6: Quillen's formal theorems, localisation, Gersten's conjecture, and the scheme-theoretic applications of the 1973 paper.
- §7: a proof-oriented reading of *Higher Algebraic K-Theory I*, with emphasis on Theorems A and B, dévissage, localization, and the projective bundle method.
- §8: commutative-diagram proofs of Quillen's formal machine.

- §9: Waldhausen's $S_\bullet$-construction and the homotopical extension of Quillen's exact-category formalism.
- §10: Thomason–Trobaugh, perfect complexes, affine replacement, localisation, and descent for schemes.
- §11: boundary phenomena, trace methods, and modern frontier results.

Throughout, R denotes a (not necessarily commutative) unital ring unless otherwise stated. All rings are associative. We work with left modules.

A guiding principle. One of the recurring themes of these notes is that algebraic K-theory is built by repeatedly passing from *objects* to *relations among decompositions*, then from those relations to *homotopies among relations*, and so on. At the level of K_0 this produces an additive decategorification: short exact sequences become equations $[B] = [A] + [C]$. At the level of K_1 one measures automorphisms and failure of determinants to capture all invertible matrices. At the level of K_2 and above, one is forced to remember coherence data — central extensions, loop spaces, simplicial constructions, and finally spectra. The higher theories of Quillen and Waldhausen should therefore be read not as disconnected technical machines, but as systematic ways of encoding the successive layers of coherence hidden inside exact sequences, extensions, and derived categories.

Three leitmotifs. The reader may keep the following dictionary in mind throughout the paper:

(a) *additivity*: exact sequences should behave like direct-sum decompositions up to correction terms encoded by homotopy;

(b) *stabilisation*: enlarging matrices, passing to infinite general linear groups, or replacing vector bundles by perfect complexes should not change the essential theory;

(c) *descent/locality*: the K-theory of a global object should be recoverable from suitably local pieces together with gluing data.

These three ideas reappear in nearly every major theorem below.

2. The Grothendieck Group K_0

2.1. The universal additive invariant

Definition 2.1. Let $\mathcal{C}$ be a small category equipped with a notion of *exact sequences* (more precisely, an exact category in the sense of §5). The **Grothendieck group** $\mathrm{K}_0(\mathcal{C})$ is the abelian group with generators $[A]$ for each object $A \in \operatorname{ob}\mathcal{C}$ and relations

$$[B] = [A] + [C]$$

for every short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{C}$.

In the most classical case $\mathcal{C} = \mathbf{Vect}(X)$, the category of algebraic (or topological) vector bundles on X.

Definition 2.2. For a compact topological space X, define

$$K^0_{\mathrm{top}}(X) \;:=\; \mathrm{K}_0\Big(\mathbf{Vect}_{\mathbb{C}}(X)\Big),$$

i.e., the Grothendieck group of complex vector bundles on X.

Construction 2.3 (Grothendieck group of a commutative monoid)**.** Let $(M, +, 0)$ be a commutative monoid. Define the **group completion** M^{gp} as the set of formal differences $[a] - [b]$ with $a, b \in M$, subject to

$$[a] - [b] = [a'] - [b'] \iff \exists\, c \in M,\ a + b' + c = a' + b + c.$$

This is the universal map from M to an abelian group.

Concretely, the set $\mathbf{Vect}(X)/\cong$ of isomorphism classes of vector bundles forms a commutative monoid under $\oplus$, and $K^0_{\mathrm{top}}(X) = (\mathbf{Vect}(X)/\cong)^{\mathrm{gp}}$.

Remark 2.4 (Why group completion is the right universal operation)*.* The passage from a commutative monoid to its group completion is the minimal way of forcing subtraction to exist. In K-theory this is exactly what one needs in order to compare two objects only after adding a common summand. Thus virtual bundles $[E] - [F]$ should be thought of as formal records of *stable* comparison: E and F may fail to be isomorphic, but they may become isomorphic after adding an auxiliary bundle. This stable viewpoint is what later makes Bott periodicity, stabilisation of general linear groups, and the use of perfect complexes so natural.

Proposition 2.5. *For any pointed compact space (X, x_0), let $\widetilde{\mathrm{K}}^0_{\mathrm{top}}(X)$ denote the kernel of the rank homomorphism $\mathrm{K}^0_{\mathrm{top}}(X) \to \mathbb{Z}$. Then*

$$\widetilde{\mathrm{K}}^0_{\mathrm{top}}(X) = [X, \mathbb{Z} \times \mathrm{BU}],$$

where BU *is the classifying space of the infinite unitary group, and $[-,-]$ denotes based homotopy classes of maps.*

Proof. By the classification of vector bundles, a rank-n complex bundle on X is classified by a map $X \to \mathrm{BU}(n)$. Taking the colimit over n gives the classifying space $\mathrm{BU} = \mathrm{colim}_n\, \mathrm{BU}(n)$. A virtual bundle $[E] - [F]$ of virtual rank zero gives a well-defined element of $[X, \mathbb{Z} \times \mathrm{BU}]$. The bijection follows from the homotopy equivalence $\mathbb{Z} \times \mathrm{BU} \simeq \Omega^2(\mathbb{Z} \times \mathrm{BU})$ given by Bott periodicity [Bo59] and the universal property of the Grothendieck group. □

2.2. K_0 of a ring

Definition 2.6. For a ring R, let $\mathbf{Proj}(R)$ denote the exact category of finitely generated projective (left) R-modules. Define

$$\mathrm{K}_0(R) \;:=\; \mathrm{K}_0(\mathbf{Proj}(R)).$$

In concrete terms, $\mathrm{K}_0(R)$ measures projective modules modulo the two basic ambiguities that dominate stable module theory: decomposition by direct sum and replacement by short exact extensions. It is the universal additive invariant of finitely generated projective modules.

Example 2.7 (Fields and local rings)**.** If R is a field, every finite-dimensional R-vector space is determined up to isomorphism by its dimension, so $\mathrm{K}_0(R) \cong \mathbb{Z}$. More generally, if R is a local ring, every finitely generated projective R-module is free, hence again

$$\mathrm{K}_0(R) \cong \mathbb{Z},$$

generated by the class of R itself. In this sense K_0 begins to detect genuinely new arithmetic or geometric information only when projective modules cease to be classified by rank alone.

Example 2.8. If R is a principal ideal domain (e.g. $R = \mathbb{Z}$), every finitely generated projective module is free, so

$$\mathrm{K}_0(R) \cong \mathbb{Z}, \quad [P] \mapsto \mathrm{rank}(P).$$

Example 2.9. Let $\mathcal{O}_F$ be the ring of integers of a number field F. Then

$$\mathrm{K}_0(\mathcal{O}_F) \cong \mathbb{Z} \oplus \mathrm{Cl}(\mathcal{O}_F),$$

where $\mathrm{Cl}(\mathcal{O}_F)$ is the ideal class group. Indeed, every finitely generated projective $\mathcal{O}_F$-module of rank r is isomorphic to $\mathcal{O}_F^{r-1} \oplus I$ for a unique ideal class $[I]$, by the structure theorem for projective modules over Dedekind domains.

Remark 2.10 (Interpretation of the class-group summand)*.* The decomposition in theorem 2.9 is extremely instructive. The $\mathbb{Z}$-summand records the coarse numerical invariant (rank), while the class-group summand records precisely the obstruction to being free after fixing the rank. In modern language, K_0 separates the *discrete size* of a projective module from the *twisting* that remains even after the size has been fixed. This is one of the first hints that K-theory packages arithmetic information invisible to naive linear algebra.

2.3. Grothendieck–Riemann–Roch

Let $f : X \to Y$ be a proper morphism of smooth quasi-projective varieties over a field k.

Definition 2.11. The **Grothendieck group of coherent sheaves** $\mathrm{G}_0(X) := \mathrm{K}_0(\mathrm{Coh}(X))$ is the Grothendieck group of the exact category of coherent $\mathcal{O}_X$-modules. For smooth X, the natural map $\mathrm{K}_0(X) := \mathrm{K}_0(\mathbf{Vect}(X)) \to \mathrm{G}_0(X)$ is an isomorphism (resolution by locally free sheaves).

Theorem 2.12 (Grothendieck–Riemann–Roch, [BS58])**.** *Let $f : X \to Y$ be a proper morphism of smooth varieties. There is a direct image map $f_! : \mathrm{K}_0(X) \to \mathrm{K}_0(Y)$ defined by*

$$f_!([E]) = \sum_{i \geq 0} (-1)^i [R^i f_*(E)],$$

and the following diagram commutes:

$$\begin{tikzcd} \mathrm{K}_0(X) \arrow[r, "f_!"] \arrow[d, "\mathrm{ch}(\,\cdot\,)\cdot\mathrm{td}(T_X)"'] & \mathrm{K}_0(Y) \arrow[d, "\mathrm{ch}(\,\cdot\,)\cdot\mathrm{td}(T_Y)"] \\ H^*(X,\mathbb{Q}) \arrow[r, "f_*"] & H^*(Y,\mathbb{Q}), \end{tikzcd}$$

where ch *is the Chern character and* td *is the Todd class.*

Proof. The key steps are:

(i) *Reduction to the case* $Y = \mathrm{pt}$ (Riemann–Roch without denominators for f an embedding).

(ii) *Projective bundle formula*: if E is a rank-r bundle on Y and $f : \mathbb{P}(E) \to Y$ is the projective bundle, then $f_!$ is computable via the relative Euler sequence.

(iii) *Closed embedding*: $f : X \hookrightarrow Y$ a closed regular immersion. One constructs the Gysin map $f^! : \mathrm{K}_0(Y) \to \mathrm{K}_0(X)$ using the deformation to the normal cone and the Koszul complex for the normal bundle $N_{X/Y}$. The formula $\mathrm{ch}(f_!(\xi)) = f_*(\mathrm{ch}(\xi) \cdot \mathrm{td}^{-1}(N_{X/Y}))$ is verified by the splitting principle.

(iv) *General proper* f: factor as closed immersion followed by a projection $X \hookrightarrow \mathbb{P}^n \times Y \to Y$ and combine.

□

Remark 2.13 (Conceptual meaning of Grothendieck–Riemann–Roch). The theorem says that K_0 is not merely an algebraic receptacle for vector bundles: it is the correct source for functorial pushforward formulas. Ordinary cohomology sees characteristic classes, but only after applying the Chern character and introducing the Todd correction term. K-theory, by contrast, sees the exact additive behaviour of sheaf cohomology directly. From this perspective, the Chern character is not the definition of theory but a comparison map from the integral and categorical world of vector bundles to the rational and graded-commutative world of cohomology.

3. Elementary K_1 and K_2

3.1. The general linear group and K_1

For $n \geq 1$, let $\mathrm{GL}_n(R)$ be the group of invertible $n \times n$ matrices over R. The stabilisation maps $\mathrm{GL}_n(R) \to \mathrm{GL}_{n+1}(R)$, $A \mapsto \left(\begin{smallmatrix} A & 0 \\ 0 & 1 \end{smallmatrix}\right)$ give the *infinite general linear group*

$$\mathrm{GL}(R) := \operatorname{colim}_n \mathrm{GL}_n(R).$$

Definition 3.1. An **elementary matrix** $e_{ij}(\lambda)$ ($i \neq j$, $\lambda \in R$) is the identity matrix with λ in position (i, j). The **elementary subgroup** $\mathrm{E}(R) \leq \mathrm{GL}(R)$ is the (normal) subgroup generated by all $e_{ij}(\lambda)$.

Theorem 3.2 (Whitehead's Lemma)**.** $\mathrm{E}(R) = [\mathrm{GL}(R), \mathrm{GL}(R)]$*, i.e. the elementary subgroup equals the commutator subgroup of* $\mathrm{GL}(R)$*.*

Proof. We first show $\mathrm{E}(R) \supseteq [\mathrm{GL}(R), \mathrm{GL}(R)]$. Since $\mathrm{E}(R)$ is normal in $\mathrm{GL}(R)$ (a straightforward matrix computation), the quotient $\mathrm{GL}(R)/\mathrm{E}(R)$ is abelian, so $[\mathrm{GL}(R), \mathrm{GL}(R)] \leq \mathrm{E}(R)$.

For the reverse inclusion $\mathrm{E}(R) \leq [\mathrm{GL}(R), \mathrm{GL}(R)]$, we use the identity

$$e_{ij}(\lambda) = [e_{ik}(\lambda), e_{kj}(1)] \quad (i, j, k \text{ distinct}),$$

which holds for $n \geq 3$ and shows every elementary matrix is a commutator in $\mathrm{GL}_n(R)$ for sufficiently large n, hence in $\mathrm{GL}(R)$. □

Definition 3.3.

$$\mathrm{K}_1(R) := \mathrm{GL}(R)/\mathrm{E}(R) = \mathrm{GL}(R)^{\mathrm{ab}}.$$

Example 3.4. For a commutative ring R, the determinant gives a split surjection $\det : \mathrm{K}_1(R) \twoheadrightarrow R^\times$, with kernel the *special Whitehead group* $\mathrm{SK}_1(R)$. If R is a field or more generally a Euclidean domain, $\mathrm{SK}_1(R) = 0$ and $\mathrm{K}_1(R) \cong R^\times$.

Example 3.5. For $R = \mathbb{Z}$, $\mathrm{K}_1(\mathbb{Z}) \cong \mathbb{Z}^\times = \{\pm 1\}$. This reflects the fact that the only units in $\mathbb{Z}$ are ± 1 and that $\mathrm{SL}_n(\mathbb{Z})$ is generated by elementary matrices (Gaussian elimination over $\mathbb{Z}$).

Remark 3.6 (Why K_1 is already subtler than determinants suggest)*.* For a commutative ring, the determinant captures a large part of K_1, but in general K_1 measures the stable failure of invertible matrices to be built from elementary row operations. In this sense it is a receptacle for *stable automorphism theory.* The appearance of the quotient $\mathrm{GL}(R)/\mathrm{E}(R)$ is therefore not accidental: one mods out precisely by the matrix operations that should count as homologically trivial.

3.2. The Steinberg group and K_2

Definition 3.7. The **Steinberg group** $\mathrm{St}(R)$ is the group with generators $x_{ij}(\lambda)$ for $i \neq j$, $\lambda \in R$, and relations

$$x_{ij}(\lambda)\, x_{ij}(\mu) = x_{ij}(\lambda + \mu), \tag{1}$$

$$[x_{ij}(\lambda), x_{kl}(\mu)] = 1 \quad \text{if } j \neq k, i \neq l, \tag{2}$$

$$[x_{ij}(\lambda), x_{jk}(\mu)] = x_{ik}(\lambda\mu) \quad \text{if } i \neq k. \tag{3}$$

There is a natural surjective homomorphism $\phi : \mathrm{St}(R) \to \mathrm{E}(R)$, $x_{ij}(\lambda) \mapsto e_{ij}(\lambda)$.

Theorem 3.8 (Milnor [Mi71])**.** *The kernel* $\mathrm{K}_2(R) := \ker(\phi : \mathrm{St}(R) \to \mathrm{E}(R))$ *is central in* $\mathrm{St}(R)$*, and*

$$0 \to \mathrm{K}_2(R) \to \mathrm{St}(R) \xrightarrow{\phi} \mathrm{E}(R) \to 1$$

is a central extension. Moreover, this is the universal *central extension of* $\mathrm{E}(R)$*.*

Proof. Step 1: $\mathrm{K}_2(R)$ *is central.* Let $w_{ij}(\lambda) := x_{ij}(\lambda)x_{ji}(-\lambda^{-1})x_{ij}(\lambda)$ for $\lambda \in R^\times$. One computes (using the Steinberg relations) that for any $x_{kl}(\mu)$ and any $\alpha \in \ker\phi$,

$$[x_{kl}(\mu), \alpha] \;=\; \phi(x_{kl}(\mu))^{-1}\,\alpha\,\phi(x_{kl}(\mu))\,\alpha^{-1}.$$

Since $\alpha \in \ker\phi$, $\phi(\alpha) = 1$, and this equals 1 after careful bookkeeping, proving centrality.

Step 2: Universality. Suppose $1 \to A \to \widetilde{E} \to \mathrm{E}(R) \to 1$ is any central extension. We must show there is a unique section $\sigma : \mathrm{St}(R) \to \widetilde{E}$ over $\mathrm{E}(R)$. Choose lifts $\tilde{x}_{ij}(\lambda)$ of $e_{ij}(\lambda)$; the Steinberg relations can be verified to hold in $\widetilde{E}$ since A is central and the relations hold in $\mathrm{E}(R)$. This gives σ and its uniqueness follows from the fact that $\mathrm{St}(R)$ is generated by the $x_{ij}(\lambda)$. □

Example 3.9 (Moore's theorem)**.** For a number field F with ring of integers $\mathcal{O}_F$, $\mathrm{K}_2(\mathcal{O}_F)$ is finite (Borel's theorem on the ranks of higher K-groups gives $\mathrm{K}_n(\mathcal{O}_F) \otimes \mathbb{Q} \neq 0$ only for $n = 4k+1$, $k \geq 1$, and $n = 0$).

Remark 3.10 (Why K_2 enters through a universal central extension)*.* The step from K_1 to K_2 is the first moment where higher coherence becomes unavoidable. The elementary subgroup $\mathrm{E}(R)$ is generated by elementary matrices, but the relations among those generators are not exhausted by naive commutator calculus inside $\mathrm{GL}(R)$. The Steinberg group remembers those relations before they collapse in $\mathrm{E}(R)$, and the kernel $\mathrm{K}_2(R)$ measures the residual defect. In modern language, K_2 is the first genuinely homotopical correction term beyond decategorified linear algebra.

3.3. Milnor K-theory

Definition 3.11. Let F be a field. The **Milnor K-ring** $\mathrm{K}_*^M(F) = T_{\mathbb{Z}}(F^\times)/I$ is the tensor algebra on $F^\times$ modulo the two-sided ideal I generated by $\{a \otimes (1-a) \mid a \in F^\times, a \neq 1\}$. Write elements of degree n as $\{a_1, \ldots, a_n\} := a_1 \otimes \cdots \otimes a_n$ (Milnor symbols).

Remark 3.12. The map $\{a\} \mapsto [-a] \in \mathrm{K}_1(F) = F^\times$ and the Steinberg symbol $\{a, b\} \mapsto$ the class of the Dennis–Stein symbol in $\mathrm{K}_2(F)$ give isomorphisms $\mathrm{K}_1^M(F) \cong \mathrm{K}_1(F)$ and $\mathrm{K}_2^M(F) \cong \mathrm{K}_2(F)$ (Matsumoto's theorem).

Theorem 3.13 (Matsumoto [Mi71])**.** *For any field F,*

$$\mathrm{K}_2(F) \;\cong\; \Big(F^\times \otimes_{\mathbb{Z}} F^\times\Big)\Big/\langle a \otimes (1-a) \mid a \neq 0, 1\rangle.$$

Proof. Define the *Steinberg symbol* $c : F^\times \times F^\times \to \mathrm{K}_2(F)$ by

$$c(a, b) = x_{12}(a)\,x_{21}(-b^{-1})\,x_{12}(a^{-1})\,x_{21}(1)\,x_{12}(1)\,x_{21}(-b) \quad \in \mathrm{K}_2(F).$$

One verifies: (i) c is bimultiplicative; (ii) $c(a, 1-a) = 1$ (using a direct computation in $\mathrm{St}(F)$); (iii) c is surjective (the w_{ij} and central elements generate $\mathrm{K}_2(F)$). The universal property of the presentation gives an isomorphism. □

4. Quillen's Plus Construction

4.1. What Quillen's 1973 paper actually accomplishes

Quillen's *Higher Algebraic K-Theory I* is not just the paper that defines the higher K-groups. It is already a remarkably complete research program. The paper is divided into *two parts*. The first develops the general categorical and homotopy-theoretic machine: classifying spaces of small categories, the homotopy-fibre criterion now called *Quillen's Theorem B*, exact categories, the Q-construction, and the formal theorems of devissage, resolution, and localisation. The second part shows that these abstract tools immediately have strong arithmetic and geometric consequences: the homotopy property for regular rings, computations for finite fields and certain skew fields, the codimension filtration on coherent sheaves, Gersten-type statements, Bloch's formula, and the projective bundle theorem for schemes. In other words, the paper is simultaneously a foundation, a toolbox, and a source of major applications.

A very important point for the modern reader is that Quillen's proof strategy is already recognisably *higher-categorical* in spirit, even though the language predates model categories and ∞-categories. One repeatedly replaces algebraic questions by questions about classifying spaces of categories, identifies fibres by means of comma categories, and then extracts long exact sequences on homotopy groups. Theorems A and B are the hidden engine of the paper: Theorem A converts contractibility of certain over-categories into homotopy equivalences, while Theorem B identifies the homotopy fibre of a functor under a precise base-change hypothesis. Devissage and resolution are Theorem A arguments; localisation is a Theorem B argument.

4.2. Acyclic maps and perfect normal subgroups

Definition 4.1. A map $f : X \to Y$ of CW complexes is **acyclic** if $\widetilde{H}_*(f; \mathbb{Z}[Y]) = 0$, i.e. $H_*(X; \mathcal{L}) \xrightarrow{\sim} H_*(Y; \mathcal{L})$ for every local coefficient system $\mathcal{L}$ on Y.

Definition 4.2. A group P is **perfect** if $P = [P, P]$. A normal subgroup $N \trianglelefteq G$ is perfect if $N = [N, N]$.

Theorem 4.3 (Quillen's Plus Construction, [Qu73])**.** *Let X be a connected CW complex and $N \trianglelefteq \pi_1(X)$ a perfect normal subgroup. There exists a CW complex X^+ and an acyclic cofibration $i : X \hookrightarrow X^+$ such that $\pi_1(i) : \pi_1(X) \to \pi_1(X^+) = \pi_1(X)/N$ is the quotient map. The pair (X^+, i) is unique up to homotopy equivalence under X.*

Proof. Existence. Since N is normally generated by some set of elements $\{g_\alpha\}$, attach a 2-cell e^2_α to X for each generator, with attaching map representing g_α. Let X_1 be the resulting complex; $\pi_1(X_1) = \pi_1(X)/N$. The inclusion $X \hookrightarrow X_1$ kills N in π_1 but may create new H_2. We kill these by attaching 3-cells. More

precisely: the long exact sequence of the pair (X_1, X) in homology gives

$$H_2(X) \to H_2(X_1) \to H_2(X_1, X) \to H_1(X) \to \cdots .$$

Since $N = [N, N]$, $H_1(N; \mathbb{Z}) = N^{\mathrm{ab}} = 0$, and a spectral sequence argument (Lyndon–Hochschild–Serre) shows $H_2(X_1, X; \mathbb{Z}) \cong H_2(BN; \mathbb{Z})$. Attach 3-cells to kill each generator of this group. Iterating produces X^+.

Acyclicity. By construction the attaching maps are chosen so that $\widetilde{H}_*(i; \mathbb{Z}) = 0$. For twisted coefficients $\mathcal{L}$, a Serre spectral sequence argument using the fact that the fibres of $X^+ \to B(\pi_1(X)/N)$ are acyclic completes the proof.

Uniqueness. Any two such complexes X^+, $X^{+\prime}$ receive acyclic maps from X inducing the same map on π_1. The acyclicity and the universal property of the Whitehead tower force a homotopy equivalence $X^+ \simeq X^{+\prime}$ under X. □

Remark 4.4 (The geometric intuition behind the plus construction). The striking feature of Quillen's plus construction is that it changes the fundamental group while preserving integral homology. One should think of it as a controlled way of quotienting by the "unstable noise" carried by the perfect subgroup $\mathrm{E}(R)$ without losing the higher homological information that K-theory is meant to remember. This is why $\mathrm{BGL}(R)^+$ can support new higher homotopy groups even though its ordinary homology agrees with that of $\mathrm{BGL}(R)$.

4.3. Higher K-groups via the plus construction

Lemma 4.5 (Whitehead's lemma for the stable linear group)**.** *Let* $\mathrm{E}(R) = \operatorname{colim}_n \mathrm{E}_n(R) \subseteq \mathrm{GL}(R)$ *be the stable elementary subgroup. Then*

$$\mathrm{E}(R) = [\mathrm{GL}(R), \mathrm{GL}(R)] \qquad \textit{and} \qquad \mathrm{E}(R) = [\mathrm{E}(R), \mathrm{E}(R)].$$

Proof. In the stable group one may choose three distinct indices i, j, k, and the Steinberg commutator relation

$$[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$$

shows that every elementary matrix belongs to $[\mathrm{E}(R), \mathrm{E}(R)]$; hence $\mathrm{E}(R)$ is perfect. Conversely, the quotient $\mathrm{GL}(R)/\mathrm{E}(R)$ is the stable Whitehead group, so every commutator in $\mathrm{GL}(R)$ dies there and therefore lies in $\mathrm{E}(R)$. Since the reverse inclusion is obvious, we obtain $[\mathrm{GL}(R), \mathrm{GL}(R)] = \mathrm{E}(R)$. □

Definition 4.6. For a unital ring R, define

$$\mathrm{BGL}^+(R) := \Big(\mathrm{BGL}(R)\Big)^+$$

where the plus is taken with respect to the perfect normal subgroup $\mathrm{E}(R) = [\mathrm{GL}(R), \mathrm{GL}(R)] \trianglelefteq \mathrm{GL}(R) = \pi_1(\mathrm{BGL}(R))$. Quillen's K-theory space of R is

$$\mathcal{K}(R) := \mathrm{K}_0(R) \times \mathrm{BGL}^+(R),$$

where $\mathrm{K}_0(R)$ is viewed as a discrete space. For $n \geq 0$ define

$$\mathrm{K}_n(R) := \pi_n\Big(\mathcal{K}(R)\Big).$$

Proposition 4.7 (Comparison with the classical groups)**.** *With this definition:*

(i) $\pi_0\mathcal{K}(R) = \mathrm{K}_0(R)$.

(ii) $\mathrm{K}_1(R) = \pi_1\mathcal{K}(R) = \mathrm{GL}(R)/\mathrm{E}(R)$, *recovering Bass's* K_1.

(iii) There is a canonical isomorphism

$$\mathrm{K}_2(R) = \pi_2\mathcal{K}(R) = \pi_2\Big(\mathrm{BGL}^+(R)\Big) \cong \ker\Big(\mathrm{St}(R) \to \mathrm{E}(R)\Big).$$

Proof. (i) is immediate from the definition. For (ii), the plus construction changes the fundamental group by quotienting out the chosen perfect normal subgroup, so

$$\pi_1\Big(\mathrm{BGL}^+(R)\Big) = \pi_1\Big(\mathrm{BGL}(R)\Big)/\mathrm{E}(R) = \mathrm{GL}(R)/\mathrm{E}(R).$$

For (iii), the discrete factor $\mathrm{K}_0(R)$ contributes nothing in degree ≥ 1. Because $\mathrm{BGL}(R) \to \mathrm{BGL}^+(R)$ is acyclic, the universal cover of $\mathrm{BGL}^+(R)$ has the same integral homology as $\mathrm{BE}(R)$. Since $\mathrm{E}(R)$ is perfect by theorem 4.5, its universal central extension is the Steinberg extension

$$1 \to \ker\Big(\mathrm{St}(R) \to \mathrm{E}(R)\Big) \to \mathrm{St}(R) \to \mathrm{E}(R) \to 1.$$

The kernel identifies with $H_2(\mathrm{E}(R);\mathbb{Z})$, and by the Hurewicz theorem this is exactly $\pi_2\Big(\mathrm{BGL}^+(R)\Big)$. □

Remark 4.8 (Why $\mathrm{K}_0(R)$ sits outside $\mathrm{BGL}(R)^+$)*.* The plus construction is designed to produce the higher groups from the stable automorphism space. It captures K_1 and above directly, but $\mathrm{K}_0(R)$ has a different origin: it is already the group completion of isomorphism classes of projective modules under direct sum. Quillen's space $\mathcal{K}(R) = \mathrm{K}_0(R) \times \mathrm{BGL}(R)^+$ therefore packages the classical Grothendieck group together with the new higher homotopy groups in a single infinite-loop space.

Remark 4.9 (A first comparison of the two machines)*.* The plus construction starts from the stable linear group and modifies its classifying space. The Q-construction starts instead from the exact category of projective modules itself and encodes exact sequences categorically. The fact that these two very different recipes produce the same theory is one of Quillen's deepest insights: algebraic K-theory is simultaneously a theory of stable automorphisms and a theory of exact structures. The first viewpoint is particularly well adapted to explicit low-degree calculations and comparison with classical topology; the second is the source of the formal theorems such as devissage, resolution, and localisation.

5. Exact Categories and the Q-Construction

5.1. Exact categories

Definition 5.1 (Quillen [Qu73])**.** An **exact category** is an additive category $\mathcal{C}$ equipped with a class $\mathcal{E}$ of sequences

$$A \xrightarrow{i} B \xrightarrow{p} C$$

called **conflations** (or short exact sequences), with i an *admissible monomorphism* (inflation) and p an *admissible epimorphism* (deflation), satisfying:

(E1) The class $\mathcal{E}$ is closed under isomorphism.

(E2) The composition of inflations is an inflation; dually for deflations.

(E3) Admissible epimorphisms are stable under base change, and admissible monomorphisms are stable under cobase change.

(E4) For every conflation $A \hookrightarrow B \twoheadrightarrow C$, $A \hookrightarrow B$ is the kernel of $B \twoheadrightarrow C$ and $B \twoheadrightarrow C$ is the cokernel of $A \hookrightarrow B$.

(E5) The identity id_A and id_0 are inflations; their duals are deflations.

Example 5.2.

(i) Any abelian category with all short exact sequences is exact.

(ii) $\mathbf{Proj}(R)$ with the split exact sequences is exact (a split-exact category).

(iii) $\mathbf{Proj}(R)$ with the short exact sequences of R-modules (automatically split, since projective modules have the lifting property) coincides with (ii).

(iv) Any extension-closed full subcategory of an abelian category inherits an exact structure.

Remark 5.3 (Why exact categories are Quillen's natural habitat)*.* An exact category is deliberately weaker than an abelian category: one does not ask for all kernels and cokernels, only for a distinguished class of short exact sequences that behaves as if it came from an ambient abelian world. This is exactly the level of generality needed for projective modules, vector bundles, coherent sheaves with chosen resolutions, and many other naturally occurring categories that are not abelian themselves. Quillen's insight was that the formal proofs of the fundamental theorems depend only on this exact structure, not on the full abelian apparatus.

5.2. Characteristic exact sequences and Quillen additivity

Let $\mathcal{E}(\mathcal{C})$ denote the exact category whose objects are conflations

$$A \hookrightarrow B \twoheadrightarrow C$$

in $\mathcal{C}$ and whose morphisms are commutative diagrams of conflations. There are exact functors

$$s, t, q : \mathcal{E}(\mathcal{C}) \longrightarrow \mathcal{C},$$

sending a conflation to its source, middle term, and quotient. Quillen proves that the middle term is, from the viewpoint of K-theory, the *sum* of the two ends.

Theorem 5.4 (Quillen additivity, [Qu73])**.** *The functor*

$$(s, q) : Q\mathcal{E}(\mathcal{C}) \longrightarrow Q\mathcal{C} \times Q\mathcal{C}$$

is a homotopy equivalence. Consequently the map induced by the middle-term functor satisfies

$$t_* = s_* + q_* : \mathrm{K}_n(\mathcal{E}(\mathcal{C})) \longrightarrow \mathrm{K}_n(\mathcal{C})$$

for every $n \geq 0$.

Proof idea. This is one of the first places where Quillen's Theorem A enters decisively. Fix a pair of objects (A, C) of $\mathcal{C}$. The fibre category of (s, q) over (A, C) consists of extensions of C by A together with the extra data needed to live inside the Q-construction. Quillen isolates full subcategories by requiring certain structure maps to be surjective or injective, and shows that the successive inclusions admit adjoints. These reductions shrink the fibre to a category with an explicit initial object, hence a contractible classifying space. Theorem A then gives the homotopy equivalence. □

Remark 5.5 (Why this theorem matters). The statement looks innocuous, but it is really the categorical form of the identity $[B] = [A] + [C]$ attached to a short exact sequence $0 \to A \to B \to C \to 0$. In Quillen's paper, it is the bridge from exact sequences to operations on K-theory spaces. Later, Waldhausen's additivity theorem will vastly generalise this principle, but the original seed is already here.

5.3. The Q-construction

Definition 5.6 (Quillen [Qu73])**.** Let $\mathcal{C}$ be a small exact category. The **Q-construction** $Q\mathcal{C}\mathcal{C}$ is the category whose:

- **objects** are the objects of $\mathcal{C}$;
- **morphisms** $A \to B$ are isomorphism classes of diagrams

  $$A \xleftarrow{j} C \xrightarrow{i} B$$

  where j is an admissible epimorphism and i is an admissible monomorphism in $\mathcal{C}$;

- **composition**: given $A \xleftarrow{j} C \xrightarrow{i} B$ and $B \xleftarrow{j'} D \xrightarrow{i'} E$, form the pullback

$$\begin{tikzcd}
 & C \times_B D \arrow[dl, "j''"'] \arrow[dr, "i''"] & \\
C \arrow[d, "j"'] \arrow[dr, "i"] & & D \arrow[dl, "j'"'] \arrow[d, "i'"] \\
A & B & E
\end{tikzcd}$$

and take the composite roof $A \xleftarrow{j \circ j''} C \times_B D \xrightarrow{i' \circ i''} E$.

Definition 5.7. For $n \geq 1$, define

$$\mathrm{K}_n(\mathcal{C}) := \pi_n\Big(|\mathrm{N}(Q\mathcal{C}\mathcal{C})|, [0]\Big),$$

where $[0]$ is the zero object viewed as a basepoint in $Q\mathcal{C}\mathcal{C}$, N denotes the nerve of a category, and $|-|$ is geometric realisation. Set $\mathrm{K}_0(\mathcal{C}) := \pi_1(|\mathrm{N}(Q\mathcal{C}\mathcal{C})|, [0])$.

Remark 5.8 (What a morphism in $Q\mathcal{C}$ really means). A roof

$$A \xleftarrow{\text{adm. epi}} C \xrightarrow{\text{adm. mono}} B$$

should be read as a *controlled passage from A to a subquotient of B*. The intermediate object C carries both a quotient map onto the source and an inclusion into the target. This is why $Q\mathcal{C}$ encodes extension theory rather than mere isomorphism theory: composition is defined by pullback because one must match successive subquotients compatibly.

Proposition 5.9 (The fundamental group of $BQ\mathcal{C}$)**.** *For any small exact category $\mathcal{C}$,*

$$\pi_1\Big(|\mathrm{N}(Q\mathcal{C}\mathcal{C})|, [0]\Big) \cong \mathrm{K}_0(\mathcal{C}).$$

Idea of proof. A loop at the zero object is represented by a zig-zag of roofs beginning and ending at 0. Such zig-zags generate the group completion of isomorphism classes of objects under direct sum. The additional identifications coming from two-simplexes in $NQ\mathcal{C}$ are exactly the relations induced by conflations $A \hookrightarrow B \twoheadrightarrow C$, namely $[B] = [A] + [C]$. Quillen proves that no further relations occur, so the resulting fundamental group is the Grothendieck group. □

Theorem 5.10 (Comparison of the Q- and $+$-models)**.** *For $\mathcal{C} = \mathbf{Proj}(R)$, the loop space of the Q-construction agrees with the matrix-space model of algebraic K-theory:*

$$|\mathrm{N}(Q\mathcal{C}\mathbf{Proj}(R))| \;\simeq\; \Omega\Big(\mathcal{K}(R)\Big) = \Omega\Big(\mathrm{K}_0(R) \times \mathrm{BGL}^+(R)\Big).$$

Consequently

$$\mathrm{K}_n(\mathbf{Proj}(R)) \cong \mathrm{K}_n(R) \quad \textit{for all } n \geq 0.$$

Modern argument with historical warning. The equivalence between the Q-construction and the stable linear-group model is unquestionably part of Quillen's program, but the introduction of [Qu73] explicitly states that the proof is *not* contained in *Higher Algebraic K-Theory I*. What the 1973 paper develops in full detail is the Q-construction and the exact-category machinery; the comparison with the $\mathrm{BGL}(R)^+$ model had been announced earlier and was used in the background for some applications.

From the modern point of view, the comparison proceeds in two stages. First, because $\mathbf{Proj}(R)$ is split exact, one compares $Q\mathbf{Proj}(R)$ with the group completion of the symmetric monoidal category of finitely generated projective modules under direct sum. The resulting loop space is $|\mathrm{N}(Q\mathcal{C}\mathbf{Proj}(R))|$. Second, after passing to a cofinal family of free modules R^n, the automorphism groups become $\mathrm{GL}_n(R)$ and their colimit is $\mathrm{GL}(R)$. The map from the group-completion model to $\mathrm{BGL}(R)$ is a homology equivalence whose kernel on π_1 is the perfect subgroup $\mathrm{E}(R)$. The universal property of the plus construction therefore factors this map through $\mathrm{BGL}(R)^+$ and upgrades it to a homotopy equivalence. Reintroducing the discrete $\mathrm{K}_0(R)$-factor yields the stated equivalence.

The key point for the present notes is conceptual: the Q-model is the exact-category model that drives the proofs in [Qu73], while the $+$-model is the matrix-space model that connects higher algebraic K-theory to the classical groups. □

Remark 5.11 (Why the Q-construction is indispensable). Once K-theory is phrased in terms of exact categories, the classical structural theorems become accessible because exact functors and exact subcategories can be manipulated directly. This is the decisive advantage of the categorical approach over the bare plus construction: one can state and prove localisation, devissage, and resolution in the natural language of exact sequences rather than trying to recover them from matrix groups alone.

5.4. A closer reading of *Higher Algebraic K-Theory I*

Quillen's 1973 paper has an unusually rigid architecture. Section 1 develops the homotopy technology of classifying spaces and proves Theorems A and B. Section 2 introduces exact categories and the Q-construction. Sections 3–5 prove exactness, resolution, devissage, and localisation. Only then does Quillen turn, in Sections 6–8, to filtered rings, regular rings, schemes, and projective bundles. The important lesson is that the paper is not a list of isolated computations: each application is routed through one of the two homotopy-theoretic mechanisms established in Section 1, together with the categorical formalism of Section 2.

Proposition 5.12 (How Theorem A really functions in the paper)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Quillen's proof of Theorem A is based on a simplicial replacement $S(F)$ equipped with two projection functors*

$$P_1 : S(F) \longrightarrow \mathcal{C}, \qquad P_2 : S(F) \longrightarrow \mathcal{D}.$$

The projection P_1 is always a homotopy equivalence, while the fibre of BP_2 over an object $d \in \mathcal{D}$ is modeled by the comma category F/d. Hence contractibility of all the categories F/d forces $BF : B\mathcal{C} \to B\mathcal{D}$ to be a homotopy equivalence.

Proof close to Quillen's argument. The simplicial replacement $S(F)$ is built so that a simplex records a composable chain in $\mathcal{C}$ together with a compatible chain in $\mathcal{D}$ lying over it. Projection to the $\mathcal{C}$-chain has a homotopy inverse up to contractible choice, which is why BP_1 is a homotopy equivalence. Projection to the terminal object in the $\mathcal{D}$-chain yields P_2. After realising in one simplicial direction and using the bisimplicial-realisation lemma, Quillen identifies the fibre of BP_2 over d with $B(F/d)$. If every F/d is contractible, then each fibre is contractible and BP_2 is a homotopy equivalence. Since BF is identified with the composite of BP_1^{-1} and BP_2, the result follows. □

Proposition 5.13 (How Theorem B produces long exact sequences)**.** *Suppose $F : \mathcal{C} \to \mathcal{D}$ has the property that for every morphism $d \to d'$ in $\mathcal{D}$, the induced functor $F/d \to F/d'$ is a homotopy equivalence. Then Quillen's proof upgrades the previous picture from a mere fibre computation to a homotopy-fibration statement:*

$$B(F/d) \longrightarrow B\mathcal{C} \longrightarrow B\mathcal{D}.$$

This is the exact homotopy sequence mechanism used later in localisation theorems.

Proof close to Quillen's argument. The same simplicial replacement $S(F)$ is used, but now the projection to $\mathcal{D}$ is no longer expected to be a homotopy equivalence. Instead Quillen proves, by a lemma on the realisation of simplicial spaces, that the map BP_2 is a quasi-fibration whenever the transition functors $F/d \to F/d'$ are homotopy equivalences. The fibre over d is again $B(F/d)$. Passing to the homotopy exact sequence of a quasi-fibration yields the long exact sequence on homotopy groups. This is why localisation in K-theory is fundamentally a Theorem B statement rather than a Theorem A statement. □

Proposition 5.14 (The universal property of the Q-construction)**.** *Let $\mathcal{M}$ be a small exact category. To specify a functor $H : Q\mathcal{M} \to \mathcal{C}$ is equivalent to specifying, for every object $M \in \mathcal{M}$, an object $H(M) \in \mathcal{C}$, and for every admissible monomorphism $i : M' \hookrightarrow M$ and admissible epimorphism $j : M \twoheadrightarrow M''$, morphisms*

$$i^! : H(M') \to H(M), \qquad j^! : H(M'') \to H(M),$$

compatible with composition and with bicartesian squares.

Explanation. A morphism in $Q\mathcal{M}$ is a roof $M \xleftarrow{j} N \xrightarrow{i} M'$ with left arrow an admissible epimorphism and right arrow an admissible monomorphism. The proposed data tell us how H should act on the two pieces of the roof; compatibility with composition and with pullback squares guarantees that the resulting value of H is independent of the chosen representative and behaves functorially. This is exactly the universal property Quillen isolates before computing the fundamental group of $BQ\mathcal{M}$. □

Remark 5.15 (Why $\pi_1(BQ\mathcal{M})$ is computed by covering spaces). Quillen does not identify $\pi_1(BQ\mathcal{M})$ by manipulating explicit singular loops. Instead he classifies covering spaces of $BQ\mathcal{M}$ by morphism-inverting functors $Q\mathcal{M} \to \mathbf{Set}$ and shows that these are the same thing as actions of the Grothendieck group $\mathrm{K}_0(\mathcal{M})$. The passage from a conflation $M' \hookrightarrow M \twoheadrightarrow M''$ to the relation $[M] = [M'] + [M'']$ is therefore encoded at once in the monodromy of covering spaces. This is one of the cleverest points in the paper: the definition of higher K-groups is homotopical, but the computation of π_1 is reduced to a purely categorical form of Galois theory.

Remark 5.16 (Historical warning about the comparison with $\mathrm{BGL}(R)^+$). The introduction of [Qu73] explicitly says that the theorem identifying the groups constructed from $Q\mathbf{Proj}(R)$ with those obtained from the $\mathrm{BGL}(R)$ H-space model had been announced earlier but was not proved in *Higher Algebraic K-Theory I*. So a reader who wants to stay faithful to that article should keep two levels separate. Internally, the paper proves its structural results from exact categories, classifying spaces, and Theorems A and B. Externally, the matrix-space comparison ties those results back to the plus-construction model. Making that distinction keeps the historical logic of Quillen's work completely clear.

Proposition 5.17 (Why localisation is the model Theorem B application)**.** *In Quillen's localisation theorem, the exact functor Qs from the category of objects supported on a closed subcategory to the ambient category factors through a comma category $0\backslash Qs$. The proof is then reduced to two statements: first, base-change along any morphism induces a homotopy equivalence on the relevant comma categories; second, the special fibre category $0\backslash Qs$ is homotopy equivalent to the exact category one wants on the closed part. Once these are established, Theorem B identifies the homotopy fibre and hence produces the localisation long exact sequence.*

Relation with Quillen's original proof. This is exactly the pattern Quillen follows in Section 5. He factors the relevant functor through $0\backslash Qs$, checks the base-change hypothesis of Theorem B by reducing to injective or surjective maps and then to a manageable full subcategory, and finally proves by a Theorem A argument that the inclusion of this smaller subcategory is a homotopy equivalence. The resulting fibration sequence is therefore not an abstract afterthought: it is the direct homotopical shadow of the exact-category manipulations carried out inside the proof. □

6. Fundamental Theorems of Algebraic K-Theory

The theorems in this section explain why K-theory is computable at all. They do not merely produce abstract equivalences; they provide the basic reduction procedures of the subject. Devissage replaces a complicated exact category by a smaller one with the same composition factors. Resolution allows one to replace arbitrary objects by better behaved ones. Homotopy invariance explains

why polynomial extension often does not change K-theory. Localisation splits global information into a closed part and an open complement. Together, these results form the working toolkit behind virtually every concrete computation in algebraic and geometric K-theory.

6.1. Theorems A and B behind the scenes

The proofs of Quillen's structural theorems are not ad hoc. They all follow one of two homotopy-theoretic templates established in §1 of [Qu73]. In a *Theorem A* argument one studies an exact functor F by showing that each comma category F/d is contractible; this is the mechanism behind devissage and resolution. In a *Theorem B* argument one verifies a base-change condition ensuring that a comma category models the homotopy fibre of BF; this is the mechanism behind localisation. Seen this way, the fundamental results are all consequences of a single idea: translate exact algebra into the geometry of classifying spaces, then read off the resulting fibre sequences.

6.2. Devissage

Theorem 6.1 (Devissage, [Qu73])**.** *Let $\mathcal{A}$ be an abelian category and $\mathcal{B} \subseteq \mathcal{A}$ a full exact subcategory closed under subobjects and quotients such that every object of $\mathcal{A}$ has a finite filtration with quotients in $\mathcal{B}$. Then the inclusion $\mathcal{B} \hookrightarrow \mathcal{A}$ induces an isomorphism $\mathrm{K}_*(\mathcal{B}) \xrightarrow{\sim} \mathrm{K}_*(\mathcal{A})$.*

Proof. We use the *Quillen Theorem A*: a functor $F : \mathcal{C} \to \mathcal{D}$ is a homotopy equivalence on nerves if and only if for every object $d \in \mathcal{D}$, the over-category F/d is contractible.

Define the inclusion $j : Q\mathcal{C}\mathcal{B} \hookrightarrow Q\mathcal{C}\mathcal{A}$. For an object $A \in \mathcal{A}$, the over-category j/A has objects: $(B \in \mathcal{B}, A \leftarrow C \to B)$ in $Q\mathcal{C}\mathcal{A}$. By the assumption on filtrations, every such A admits a finite filtration $0 = A_0 \subset A_1 \subset \cdots \subset A_n = A$ with $A_i/A_{i-1} \in \mathcal{B}$. One constructs an explicit contracting homotopy of j/A using this filtration, showing it is contractible. Theorem A then gives $|\mathrm{N}\, j|$ is a homotopy equivalence. □

Corollary 6.2. *For a Dedekind domain R with fraction field F,*

$$\mathrm{G}_0(R) \cong \mathrm{K}_0(R) \oplus \mathrm{K}_0(\mathbf{k}(R)),$$

where $\mathrm{G}_0(R) = \mathrm{K}_0(\mathrm{Coh}(R))$ and the second summand records torsion via the residue fields.

6.3. Resolution theorem

Theorem 6.3 (Resolution, [Qu73])**.** *Let $\mathcal{P} \subseteq \mathcal{A}$ be a full exact subcategory of an exact category such that:*

(i) Every object of $\mathcal{A}$ has a finite resolution by objects of $\mathcal{P}$.

(ii) $\mathcal{P}$ is closed under extensions in $\mathcal{A}$.

(iii) $\mathcal{P}$ is closed under kernels of admissible epimorphisms in $\mathcal{A}$.

Then $\mathrm{K}_*(\mathcal{P}) \xrightarrow{\sim} \mathrm{K}_*(\mathcal{A})$.

Proof. One applies Quillen's Theorem A to the inclusion $\mathcal{QCP} \hookrightarrow \mathcal{QCA}$, showing that the under-category $A/\mathcal{QCP}$ is contractible for every $A \in \mathcal{A}$. The key point is that the finite resolution hypothesis provides a canonical terminating sequence in $\mathcal{P}$ that can be used to build the contraction. More precisely, given a finite resolution

$$0 \to P_n \to \cdots \to P_0 \to A \to 0$$

with $P_i \in \mathcal{P}$, the object $P_0 \twoheadrightarrow A$ is a terminal object in the relevant subcategory of $A/\mathcal{QCP}$ after stabilisation, and an induction on resolution length completes the argument. □

Corollary 6.4 (Regular rings)**.** *If R is a regular ring (every finitely generated R-module has a finite projective resolution), then the inclusion* $\mathbf{Proj}(R) \hookrightarrow \mathbf{Mod}^{\mathrm{fg}}(R)$ *induces an isomorphism*

$$\mathrm{K}_*(R) \xrightarrow{\sim} \mathrm{G}_*(R).$$

6.4. Homotopy invariance

Theorem 6.5 (Homotopy invariance, [Qu73])**.** *For any regular ring R, the projection $R[t] \to R$ (evaluation at $t = 0$) induces an isomorphism*

$$\mathrm{K}_*(R[t]) \xrightarrow{\sim} \mathrm{K}_*(R).$$

More generally, for a regular scheme X, the projection $X \times \mathbb{A}^1 \to X$ induces $\mathrm{K}_*(X \times \mathbb{A}^1) \xrightarrow{\sim} \mathrm{K}_*(X)$.

Proof faithful to Quillen's method. The actual argument in [Qu73] is subtler than a direct comparison of the two evaluation maps $t \mapsto 0$ and $t \mapsto 1$. Quillen first proves the statement for G-theory of regular Noetherian rings by introducing exact categories of filtered modules and comparing them with modules over the associated graded ring. For the t-adic filtration on $R[t]$, the associated graded ring is again polynomial and the filtration quotients are controlled by copies of R. Devissage then identifies the relevant graded pieces, while the localization theorem shows that the passage from filtered to graded objects does not change K-theory.

Once this is done, the resolution theorem upgrades the statement from G-theory to K-theory for regular rings, because finitely generated modules admit finite projective resolutions. The scheme-theoretic statement follows by applying the same strategy Zariski locally and using Quillen's localization formalism to glue the affine computations. □

Remark 6.6 (Why this theorem is deeper than it looks)*.* The conclusion $\mathrm{K}(R[t]) \simeq \mathrm{K}(R)$ suggests that adding an affine coordinate should be harmless. But for Quillen this was not a formal slogan: it emerged from a delicate analysis of filtered categories and their associated graded pieces. This is one of the first

places in the subject where one clearly sees the modern principle that geometric deformation should induce homotopy invariance only after strong finiteness or regularity hypotheses are imposed.

Theorem 6.7 (Fundamental theorem for regular rings, [Qu73]). *If A is a regular ring, then for every $i \geq 0$ one has canonical isomorphisms*

$$\mathrm{K}_i(A[t]) \cong \mathrm{K}_i(A), \qquad \mathrm{K}_i(A[t, t^{-1}]) \cong \mathrm{K}_i(A) \oplus \mathrm{K}_{i-1}(A).$$

Proof. The first is the homotopy-invariance theorem just proved. For the Laurent polynomial ring one applies localisation to the multiplicative system generated by t in $A[t]$. The closed part is supported at the divisor (t) and is identified, by devissage, with the K-theory of A. The open part is $A[t, t^{-1}]$. The unit t furnishes a natural splitting of the boundary map, so the long exact localisation sequence breaks into short exact sequences, yielding the direct-sum decomposition above. □

Remark 6.8 (Why this is the prototype of Bass–Quillen phenomena). The theorem explains why the Laurent variable contributes one copy of the old K-theory shifted by one degree. In modern language, G_m behaves like a one-dimensional suspension coordinate for regular algebraic K-theory. This decomposition is the algebraic antecedent of many later formulas involving nil-terms, Bass delooping, and nonconnective K-theory.

6.5. Localisation sequence

Theorem 6.9 (Quillen Localisation, [Qu73]). *Let R be a Noetherian ring and $S \subseteq R$ a multiplicative set. Let $\mathcal{N}(R, S)$ denote the exact category of finitely generated R-modules M with $S^{-1}M = 0$ (i.e. S-torsion modules). There is a long exact sequence*

$$\cdots \to \mathrm{K}_n(\mathcal{N}) \to \mathrm{K}_n(R) \to \mathrm{K}_n(S^{-1}R) \to \mathrm{K}_{n-1}(\mathcal{N}) \to \cdots$$

and for regular R, a localisation sequence in G*-theory:*

$$\cdots \to \mathrm{G}_n(\mathcal{N}) \to \mathrm{G}_n(R) \to \mathrm{G}_n(S^{-1}R) \to \mathrm{G}_{n-1}(\mathcal{N}) \to \cdots .$$

Proof. Consider the functors

$$F : \mathcal{QCN} \to \mathcal{QC}(\mathbf{Mod}^{\mathrm{fg}}(R)), \qquad G : \mathcal{QC}(\mathbf{Mod}^{\mathrm{fg}}(R)) \to \mathcal{QC}(\mathbf{Mod}^{\mathrm{fg}}(S^{-1}R)),$$

where G is induced by localisation $M \mapsto S^{-1}M$. Then $G \circ F = 0$.

By Quillen's Theorem B applied to G, the homotopy fibre of $|\mathrm{N}\, G|$ over $[0]$ is $|\mathrm{N}(G/[0])|$. One identifies $G/[0]$ with a category equivalent to $\mathcal{QCN}$ using devissage and the definition of the morphisms in $\mathcal{QC}$. This gives the fibration sequence

$$|\mathrm{N}\, \mathcal{QCN}| \to |\mathrm{N}\, \mathcal{QC}\mathrm{Mod}^{\mathrm{fg}}(R)| \to |\mathrm{N}\, \mathcal{QC}\mathrm{Mod}^{\mathrm{fg}}(S^{-1}R)|,$$

and the associated long exact sequence on homotopy groups is the desired localisation sequence. □

6.6. Computation: K-theory of finite fields

Theorem 6.10 (Quillen [Qu72]). *For the finite field $\mathbb{F}_q$ with $q = p^r$,*

$$\mathrm{K}_n(\mathbb{F}_q) \cong \begin{cases} \mathbb{Z} & n = 0, \\ \mathbb{Z}/(q^k - 1)\mathbb{Z} & n = 2k-1,\ k \geq 1, \\ 0 & n = 2k,\ k \geq 1. \end{cases}$$

Proof. Quillen computes $\mathrm{BGL}^+(\mathbb{F}_q)$ using the Adams operations in topological K-theory. Specifically, the Adams operation ψ^q acts on BU via the q-th Adams operation. Quillen shows that the fixed-point set $\mathrm{BU}^{h\psi^q}$ of the homotopy action of ψ^q on BU is homotopy equivalent to $\mathrm{BGL}^+(\mathbb{F}_q)$ (this is the *Brauer lift*). The homotopy groups of $\mathrm{BU}^{h\psi^q}$ are computed from the fibration

$$\mathrm{BU}^{h\psi^q} \to \mathrm{BU} \xrightarrow{\psi^q - 1} \mathrm{BU},$$

whose long exact sequence on homotopy gives the stated answer, using $\pi_{2k-1}(\mathrm{BU}) = \mathbb{Z}$ and $\pi_{2k}(\mathrm{BU}) = 0$ together with the eigenvalue of ψ^q on $\pi_{2k-1}(\mathrm{BU}) = \mathbb{Z}$ being q^k. □

Remark 6.11 (The Brauer lift and Bott periodicity). The computation for finite fields is striking because it imports genuinely topological input into algebraic K-theory. The Brauer lift identifies the Frobenius action on linear algebra over $\mathbb{F}_q$ with the Adams operation ψ^q on complex topological K-theory. Bott periodicity then turns what would otherwise be an intractable algebraic problem into the simple calculation of the cokernel and kernel of $\psi^q - 1$ on the even homotopy groups of BU. This is one of the earliest and clearest demonstrations that higher algebraic K-theory is inherently homotopy-theoretic.

6.7. Another landmark: projective modules over polynomial rings

Theorem 6.12 (Quillen [Qu76]). *Let k be a field. Then every finitely generated projective $k[t_1, \ldots, t_n]$-module is free.*

Historical outline of proof. Quillen proves the theorem by combining homotopy-invariance ideas with a local-global analysis on affine space. Very roughly, one shows that the locus over which a projective module is extended from the base field is stable under the localizations needed to cover affine space, and one patches these local descriptions to deduce global freeness. Suslin independently obtained a proof by different methods, so the result is now universally known as the Quillen–Suslin theorem. □

Remark 6.13 (K-theoretic meaning). From the viewpoint of K_0, the theorem says that polynomial extension over a field does not create new projective classes at all: every class comes from a free module. In this sense it is the most concrete module-theoretic avatar of Quillen's broader regularity philosophy, namely that affine homotopies should not change algebraic K-theory in the regular world.

6.8. Schemes, filtration by support, and Bloch's formula

One of the deepest parts of [Qu73] is the passage from rings to schemes via coherent sheaves. For a noetherian scheme X, Quillen defines $\mathrm{G}_i(X)$ using the abelian category of coherent sheaves and filters that category by codimension of support. The resulting spectral sequence is the first systematic bridge between higher K-theory and cycle theory.

Theorem 6.14 (Quillen's codimension spectral sequence, [Qu73])**.** *Let X be a noetherian scheme of finite Krull dimension. Then there is a convergent spectral sequence*

$$E_1^{p,q}(X) = \bigoplus_{x \in X^{(p)}} \mathrm{K}_{-p-q}\Big(k(x)\Big) \Longrightarrow \mathrm{G}_{-p-q}(X),$$

where $X^{(p)}$ denotes the set of points of codimension p.

Construction sketch. Let $\mathcal{M}^p(X)$ be the Serre subcategory of coherent sheaves whose support has codimension at least p. The filtration

$$\mathcal{M}^0(X) \supseteq \mathcal{M}^1(X) \supseteq \mathcal{M}^2(X) \supseteq \cdots$$

gives successive quotients controlled by sheaves supported at points of exact codimension p. Applying localisation to the inclusions $\mathcal{M}^{p+1}(X) \subseteq \mathcal{M}^p(X)$ yields an exact couple, hence a spectral sequence. Devissage identifies the quotient term with the direct sum of the K-theories of the residue fields $k(x)$ for $x \in X^{(p)}$. □

Remark 6.15 (Gersten and Bloch in Quillen's paper)*.* For regular schemes of finite type over a field, Quillen proves important cases of Gersten's conjecture and uses the resulting exactness to identify codimension-p cycles modulo rational (or linear, in Quillen's original formulation) equivalence with sheaf cohomology of the Zariski sheaf $\mathcal{K}_p$. This is the route to Bloch's formula. What matters conceptually is that the codimension filtration turns higher K-theory into a contravariant receptacle for cycle-theoretic information, anticipating the later development of motivic cohomology.

6.9. Gersten's conjecture in Quillen's equicharacteristic framework

The geometric heart of Section 7 of [Qu73] is not merely the existence of the codimension spectral sequence, but the possibility of forcing its first differential to be as explicit as the divisor map. Quillen isolates this in terms of the exact categories

$$\mathcal{M}^p(X) \subseteq \mathcal{M}(X)$$

of coherent sheaves whose support has codimension at least p. For a regular local ring A this filtration produces a tower of exact categories

$$\mathcal{M}^0(A) \supseteq \mathcal{M}^1(A) \supseteq \mathcal{M}^2(A) \supseteq \cdots,$$

and Gersten's conjecture asks that each inclusion

$$\mathcal{M}^{p+1}(A) \hookrightarrow \mathcal{M}^p(A)$$

induce the zero map on K-groups. When this vanishing holds, the E_1-page ceases to be a formal bookkeeping device and becomes an actual Gersten complex.

Proposition 6.16 (Quillen's exactness criterion for the coniveau filtration)**.** *Let A be a noetherian regular local ring and let $n \geq 0$. Suppose that for every $p \geq 0$ the inclusion*

$$\mathcal{M}^{p+1}(A) \hookrightarrow \mathcal{M}^p(A)$$

induces the zero map on K_n. Then the codimension spectral sequence degenerates along its first nontrivial row into the exact Gersten complex

$$0 \longrightarrow K_n(A) \longrightarrow K_n(\operatorname{Frac} A) \xrightarrow{\partial} \bigoplus_{x \in (\operatorname{Spec} A)^{(1)}} K_{n-1}(k(x)) \xrightarrow{\partial} \bigoplus_{x \in (\operatorname{Spec} A)^{(2)}} K_{n-2}(k(x)) \longrightarrow \cdots,$$

and, after Zariski sheafification on a regular scheme, yields a flasque resolution of the K-theory sheaf $\mathcal{K}_n$.

Expanded proof. The localisation sequence for the pair $\mathcal{M}^{p+1}(A) \subseteq \mathcal{M}^p(A)$ gives exact rows

$$\cdots \to K_{n+1}(\mathcal{M}^p/\mathcal{M}^{p+1}) \to K_n(\mathcal{M}^{p+1}) \to K_n(\mathcal{M}^p) \to K_n(\mathcal{M}^p/\mathcal{M}^{p+1}) \to \cdots.$$

By dévissage, the quotient category $\mathcal{M}^p/\mathcal{M}^{p+1}$ has the same K-theory as the direct sum over points of codimension exactly p, namely

$$K_n(\mathcal{M}^p/\mathcal{M}^{p+1}) \cong \bigoplus_{x \in (\operatorname{Spec} A)^{(p)}} K_n\big(k(x)\big).$$

If the map $K_n(\mathcal{M}^{p+1}) \to K_n(\mathcal{M}^p)$ is zero, then exactness shortens the long exact sequence into short exact pieces, and the exact-couple differential identifies with the residue map from codimension p to codimension $p+1$. Iterating this for all p produces the displayed complex. For a regular scheme one performs the same construction Zariski-locally and then sheafifies. The terms are products of skyscraper sheaves supported at points, hence flasque, so one obtains a flasque resolution of $\mathcal{K}_n$. □

Theorem 6.17 (Quillen's equicharacteristic Gersten theorem)**.** *The vanishing condition of Proposition 6.16 holds in the following cases.*

(a) If R is a finite type algebra over a field k, S is a finite set of primes of R at which R is regular, and $A = S^{-1}R$ is the resulting regular semi-local ring.

(b) If $A = k[[x_1, \ldots, x_n]]$ is a formal power series ring over a field, or more generally a ring of convergent power series over a complete valued field.

Consequently, in these cases the Gersten complex for Quillen K-theory is exact.

Quillen's variant of Noether normalization. The decisive geometric input in Theorem 6.17 is the following refined normalization lemma. It is the place where Quillen forces the codimension-one problem into relative dimension one over an affine base.

Lemma 6.18 (Normalization lemma in Quillen's Gersten proof)**.** *Let R be a smooth finite type k-algebra of dimension r, let $t \in R$ be a regular element, and let $S \subseteq \operatorname{Spec} R$ be a finite set. Then there exist elements*

$$x_1, \dots, x_{r-1} \in R$$

algebraically independent over k such that, for $B := k[x_1, \dots, x_{r-1}] \subseteq R$,

(i) R/tR is finite over B;

(ii) R is smooth over B at every point of S.

Proof. Step 1: reduce to maximal ideals and encode tangent directions. For each prime in S, choose a maximal ideal containing it; enlarging S does not weaken the statement, so we may assume that S consists of maximal ideals. Let $\Omega := \Omega_{R/k}$. Because R is smooth of dimension r over k, Ω is a projective R-module of rank r. For R to be smooth over a subalgebra $B = k[x_1, \dots, x_{r-1}]$ at a maximal ideal $\mathfrak{m} \in S$ it is enough that the differentials $dx_1, \dots, dx_{r-1}$ be linearly independent in

$$\Omega \otimes_R k(\mathfrak{m}),$$

for then $\Omega_{R/B}$ has rank one at $\mathfrak{m}$, which is the local criterion for smoothness of relative dimension one.

Let

$$J := \bigcap_{\mathfrak{m} \in S} \mathfrak{m}.$$

By the Chinese remainder theorem, after choosing classes in the cotangent spaces $\mathfrak{m}/\mathfrak{m}^2$ and lifting them compatibly through R/J^N for $N \gg 0$, one can find a finite-dimensional k-subspace $V \subseteq R$ with the following properties:

(a) V generates R as a k-algebra;

(b) for each $\mathfrak{m} \in S$ there exist elements

$$v_{\mathfrak{m},1}, \dots, v_{\mathfrak{m},r} \in V$$

whose differentials form a basis of $\Omega \otimes_R k(\mathfrak{m})$ and which vanish at every other point of S.

The second condition is the reservoir of degree-one perturbations that will later force smoothness over the chosen polynomial subalgebra.

Step 2: filter R/tR by the degree in V. Define an increasing filtration by

$$F_n(R/tR) := \text{the } k\text{-span of monomials of degree at most } n \text{ in elements of } V.$$

Because V generates R, this is an exhaustive finite-type filtration. Consider the associated graded ring

$$\mathrm{gr}_F(R/tR) = \bigoplus_{n\geq 0} F_n(R/tR)/F_{n-1}(R/tR).$$

Quillen's geometric observation is that

$$\mathbf{Proj}\Big(\bigoplus_{n\geq 0} F_n(R/tR)\Big)$$

is the projective closure of $\mathrm{Spec}(R/tR)$ inside the projective space attached to V. Since t is a regular element and R has dimension r, the quotient R/tR has dimension $r-1$. Hence the projective closure also has dimension $r-1$, and its divisor at infinity is

$$\mathbf{Proj}(\mathrm{gr}_F(R/tR)),$$

which therefore has dimension $r-2$. Consequently

$$\dim \mathrm{gr}_F(R/tR) = r-1.$$

Step 3: choose homogeneous parameters for the graded ring. Pick a homogeneous system of parameters

$$z_1, \ldots, z_{r-1} \in \mathrm{gr}_F(R/tR)$$

of positive degrees such that $\mathrm{gr}_F(R/tR)$ is finite over the polynomial ring $k[z_1, \ldots, z_{r-1}]$. Replacing each z_i by a sufficiently large positive power does not destroy finiteness, so we may suppose that every z_i has degree at least 2. Choose lifts $y_i \in R$ of the classes z_i. The standard filtered–graded argument now shows that R/tR is finite over the subalgebra $k[y_1, \ldots, y_{r-1}]$: a finite homogeneous generating set of the graded ring lifts to a filtered generating set of the original ring, and one proves finiteness by induction on the filtration degree.

Step 4: perturb the lifts without changing their leading terms. The lifts y_i need not have differentials independent at the points of S. Here the choice of V is used. One may choose degree-one corrections $w_i \in V$ so that the modified elements

$$x_i := y_i + w_i$$

have differentials $dx_1, \ldots, dx_{r-1}$ linearly independent in $\Omega \otimes_R k(\mathfrak{m})$ for every $\mathfrak{m} \in S$. Because each w_i has filtration degree 1 whereas each z_i has degree at least 2, the initial form of x_i in the associated graded ring is still z_i. Therefore the same filtered–graded argument shows that R/tR is finite over

$$B := k[x_1, \ldots, x_{r-1}].$$

Step 5: check algebraic independence and smoothness. Since R/tR is finite over the image of $k[T_1, \ldots, T_{r-1}] \to R/tR$, $T_i \mapsto x_i \bmod t$, that image has Krull dimension $r-1$. A quotient of a polynomial ring in $r-1$ variables

can have dimension $r-1$ only if its defining ideal is zero; hence the elements $x_1, \ldots, x_{r-1}$ are algebraically independent over k. Finally, by construction the differentials $dx_1, \ldots, dx_{r-1}$ are independent at every point of S. Since R is smooth over k, this is exactly the Jacobian criterion asserting that R is smooth over $B = k[x_1, \ldots, x_{r-1}]$ at the points of S. This proves both (i) and (ii). □

Remark 6.19. The normalization lemma is the exact place where the Gersten problem is converted into a problem about a smooth morphism of relative dimension one. Everything after it is one-dimensional geometry plus additivity: without the lemma there is no reason for the principal-kernel argument to appear.

Expanded proof following Quillen. We explain the finite-type case, from which the formal-power-series case is obtained by the same pattern with Weierstrass preparation replacing Noether normalization.

Step 1: spread out and reduce to the smooth case. Quillen first replaces the ground field by a finitely generated subfield of the prime field and then by the prime field itself. After shrinking around the chosen set S, one may assume that R is smooth over a perfect field k at every point of S. This reduction is important because smoothness is what turns the coniveau problem into a geometric problem about relative curves. In Quillen's paper this is the first major simplification of the conjecture.

Step 2: reduce the conjectural vanishing to a codimension-one transfer. Fix $p \geq 0$. Quillen rewrites the map

$$K_*\big(\mathcal{M}^{p+1}(A)\big) \longrightarrow K_*\big(\mathcal{M}^p(A)\big)$$

as a filtered colimit over open neighborhoods and then over regular elements $t \in R$. One is reduced to showing that for every regular element t there exists $f \in R$ not vanishing on S such that the exact functor

$$\mathcal{M}^p(R/tR) \longrightarrow \mathcal{M}^p(R_f)$$

induces the zero map on K-groups. This is the decisive reduction: all higher-categorical complexity is condensed into proving that a single codimension-one pushforward vanishes.

Step 3: apply the normalization lemma and obtain a relative curve. Lemma 6.18 produces algebraically independent elements $x_1, \ldots, x_{r-1} \in R$ and a polynomial subalgebra $B = k[x_1, \ldots, x_{r-1}]$ such that R/tR is finite over B and R is smooth over B at the points of S. Geometrically this means that, after shrinking around S, the smooth r-dimensional scheme $\operatorname{Spec} R$ is viewed as a scheme smooth of relative dimension one over the affine $(r-1)$-space $\operatorname{Spec} B$. This is the decisive reduction: the Gersten problem is pushed into the geometry of a family of curves, where the closed fibre can be cut out by a principal ideal.

Step 4: base change to the closed fibre and principalize the kernel. Set $B' := R/tR$ and $R' := R \otimes_B B'$. There is then a cartesian square of rings

$$\begin{tikzcd} R' \arrow[r] \arrow[d, "u'"'] & R \arrow[d, "u"] \\ B' \arrow[r] & B, \end{tikzcd}$$

in which the horizontal arrows are finite. Because u is smooth of relative dimension one along S, the base change u' is again smooth of relative dimension one along the finite set S' lying over S. A standard fact from the theory of smooth curves then shows that the kernel

$$I := \ker(R' \to B')$$

is principal at the points of S', hence principal on a neighborhood of S'. Since R'/R is finite, that neighborhood contains the inverse image of a neighborhood of S in $\operatorname{Spec} R$; after replacing R by R_f for a suitable f not vanishing on S, Quillen arranges simultaneously that

$$I_f \cong R'_f \qquad \text{as } R'_f\text{-modules},$$

and that R'_f is smooth, hence flat, over B'. This is the technical pivot of the whole proof: the ideal of the closed fibre is not merely principal but, as a module, indistinguishable from the ambient base-changed algebra.

Step 5: use additivity to force vanishing. For any object $M \in \mathcal{M}^p(B')$ there is an exact sequence of R_f-modules

$$0 \longrightarrow I_f \otimes_{B'} M \longrightarrow R'_f \otimes_{B'} M \longrightarrow M_f \longrightarrow 0.$$

Equivalently, there is a short exact sequence of exact functors

$$0 \longrightarrow I_f \otimes_{B'} (-) \longrightarrow R'_f \otimes_{B'} (-) \longrightarrow (-)_f \longrightarrow 0$$

from $\mathcal{M}^p(B')$ to $\mathcal{M}^p(R_f)$. Because R'_f is flat over B', both tensor-product functors preserve exact sequences and the codimension-p support condition. Since $I_f \cong R'_f$, the first two exact functors induce the same homomorphism on K-groups. Quillen's additivity theorem therefore gives

$$(R'_f \otimes_{B'} -)_* = (I_f \otimes_{B'} -)_* + ((-)_f)_*$$

on K-theory, and the equality of the first two summands forces $((-)_f)_* = 0$. This is exactly the vanishing required in Step 2. Hence the inclusions $\mathcal{M}^{p+1} \hookrightarrow \mathcal{M}^p$ induce zero on K-groups, and Gersten exactness follows.

For the formal-power-series case, the same scheme works after replacing the normalization lemma by Weierstrass preparation: after a change of variables, A/tA becomes finite over a smaller power series ring, and the same principal-kernel argument applies. $\square$

Remark 6.20 (Why the conjecture matters for Bloch's formula). The theorem does more than settle a local exactness statement. Once the Gersten complex is exact, the first nontrivial differential in the coniveau spectral sequence becomes the divisor map from codimension-$(p-1)$ points to codimension-p points. Proposition 5.14 of Quillen's paper then identifies its image with the subgroup of codimension-p cycles linearly equivalent to zero, so that the E_2-term in bidegree $(p, -p)$ becomes the Chow-type group of codimension-p cycles modulo linear equivalence. This is the exact bridge from the codimension filtration to Bloch's formula and, historically, to the later emergence of higher Chow groups and motivic cohomology.

6.10. Jouanolou's device and affine replacement

A technical point that improves the overall narrative is that Quillen's scheme-theoretic K'-theory is already accompanied by an *affine replacement principle*. In §4.2 of [Qu73] Quillen recalls Jouanolou's observation that for a quasi-projective scheme X over a field there exists a torsor

$$p : P \longrightarrow X$$

under a vector bundle such that P itself is affine. Since such torsors are Zariski-locally trivial with fibre an affine space, homotopy invariance and Zariski descent imply that passing from X to P does not change the K'-groups that Quillen studies. Thus many geometric arguments may be carried out on an affine model even when the original scheme is not affine.

This point is conceptually important for what comes later. Jouanolou's device is an early sign that the correct form of algebraic K-theory of schemes is not tied to a single presentation of the scheme. One may replace X by a vector-bundle torsor with better geometric properties, perform the computation there, and descend the answer back to X. In the later literature this philosophy becomes systematic; in the present notes it serves as a bridge from Quillen's concrete calculations to Thomason's more flexible use of perfect complexes and affine replacement techniques.

6.11. Quillen's projective bundle theorem

Theorem 6.21 (Projective bundle theorem, [Qu73]). *Let E be a vector bundle of rank r over a quasi-compact scheme S, and put $X = \mathbb{P}(E) = \operatorname{Proj}(\operatorname{Sym} E)$. Let $f : X \to S$ be the structural morphism and let $z = [\mathcal{O}_X(-1)] \in \mathrm{K}_0(X)$. Then for each $q \geq 0$ the map*

$$\bigoplus_{i=0}^{r-1} \mathrm{K}_q(S) \longrightarrow \mathrm{K}_q(X), \qquad (a_0, \ldots, a_{r-1}) \longmapsto \sum_{i=0}^{r-1} z^i\, f^*(a_i)$$

is an isomorphism.

Proof idea faithful to Quillen's argument. Quillen does not merely imitate the classical Grothendieck-group proof. He introduces exact subcategories of vector

bundles on X controlled by the vanishing of higher direct images and by regularity conditions along the fibres. A canonical resolution of a regular sheaf on projective space allows him to compare these subcategories inductively. The resulting exact functors produce mutually inverse maps between $\mathrm{K}_q(X)$ and $\bigoplus_{i=0}^{r-1} \mathrm{K}_q(S)$. The line bundle class $z = [\mathcal{O}_X(-1)]$ is the algebraic shadow of the Bott element that keeps track of the projective direction. □

Remark 6.22 (Why this theorem is a decisive geometric test). This result shows that Quillen's higher groups behave with respect to projective bundles exactly as Grothendieck's K_0 suggested they should. It is therefore a strong consistency check on higher K-theory as a genuine cohomological invariant of schemes. In the same section Quillen pushes the method further to the projective line over noncommutative rings and to Severi–Brauer schemes, already indicating the flexibility of the theory far beyond the commutative setting.

7. A proof-oriented dossier on Quillen's *Higher Algebraic K-Theory I*

The previous sections explained what Quillen's constructions do and why they are the organising centre of higher algebraic K-theory. We now go back and isolate the *proof architecture* of the 1973 paper itself. The aim of the present section is not merely to state the final theorems once again, but to exhibit the mechanism by which they are proved. In Quillen's article the decisive point is that one first builds a piece of homotopy technology for classifying spaces of categories and only afterwards feeds exact categories into that machine. Once this is understood, the celebrated formal theorems cease to look miraculous: they are repeated applications of the same two templates.

Template A. To prove that a functor induces an equivalence on classifying spaces, one proves that each suitable comma category is contractible and applies Theorem A.

Template B. To obtain a long exact sequence, one factors a functor through a comma category and verifies the base-change hypothesis of Theorem B, thereby identifying a homotopy fibre.

Everything in Sections 2–5 of [Qu73] is built from these two patterns together with the exact-category language of admissible monomorphisms and epimorphisms.

7.1. The simplicial replacement behind Theorems A and B

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between small categories. Quillen introduces a category, which one may denote by $S(F)$, whose objects are triples (X, Y, v) with $X \in \mathcal{C}$, $Y \in \mathcal{D}$, and $v : Y \to F(X)$ a morphism in $\mathcal{D}$. A morphism

$$(X, Y, v) \longrightarrow (X', Y', v')$$

consists of a pair of arrows $u : X \to X'$ in $\mathcal{C}$ and $w : Y \to Y'$ in $\mathcal{D}$ such that the square

$$\begin{tikzcd} Y \arrow[r, "w"] \arrow[d, "v"'] & Y' \arrow[d, "v'"] \\ F(X) \arrow[r, "F(u)"'] & F(X') \end{tikzcd}$$

commutes. There are projection functors

$$P_1 : S(F) \to \mathcal{C}, \qquad P_2 : S(F) \to \mathcal{D},$$

given by $P_1(X, Y, v) = X$ and $P_2(X, Y, v) = Y$.

The point of $S(F)$ is that P_1 is easy and P_2 carries the fibre information. More precisely, fixing $X \in \mathcal{C}$, the category $X\backslash P_1$ has an initial object, namely $(X, F(X), \mathrm{id}_{F(X)})$. Hence $X\backslash P_1$ is contractible. By Theorem A, $BP_1 : BS(F) \to B\mathcal{C}$ is therefore a homotopy equivalence. On the other hand, fixing $Y \in \mathcal{D}$, one checks that the fibre category $Y\backslash P_2$ is equivalent to the comma category F/Y. Thus P_2 is the place where the categories F/Y naturally occur.

The entire force of Theorems A and B comes from combining these two observations with a bisimplicial analysis of $S(F)$. In words: $S(F)$ is large enough that projection to $\mathcal{C}$ is harmless, but rigid enough that projection to $\mathcal{D}$ remembers the over-categories of F.

Theorem 7.1 (Quillen's Theorem A in expanded form)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Assume that for every object $d \in \mathcal{D}$ the comma category F/d is contractible. Then the induced map of classifying spaces*

$$BF : B\mathcal{C} \longrightarrow B\mathcal{D}$$

is a homotopy equivalence.

Proof in steps. Step 1: build the replacement category. Introduce $S(F)$ as above. The projection $P_1 : S(F) \to \mathcal{C}$ has the property that for each $X \in \mathcal{C}$ the comma category $X\backslash P_1$ has the initial object $(X, F(X), \mathrm{id}_{F(X)})$. Hence every $X\backslash P_1$ is contractible. By Theorem A applied to P_1 itself, we deduce that BP_1 is a homotopy equivalence.

Step 2: identify the fibres of P_2. Fix $d \in \mathcal{D}$. An object of $d\backslash P_2$ is an object (X, Y, v) of $S(F)$ together with a morphism $d \to Y$ in $\mathcal{D}$. Composing with $v : Y \to F(X)$ produces a morphism $d \to F(X)$, equivalently an object of the comma category $d\backslash F$. In Quillen's conventions one writes this as F/d or $d\backslash F$ depending on variance; the essential point is that the fibre is a comma category built from F. The obvious comparison functors are inverse equivalences, so $d\backslash P_2 \simeq F/d$.

Step 3: translate the hypothesis. By assumption, each F/d is contractible. Therefore each comma category $d\backslash P_2$ is contractible. Applying Theorem A to P_2 gives that $BP_2 : BS(F) \to B\mathcal{D}$ is a homotopy equivalence.

Step 4: compare BF with the two projections. The functor F factors through $S(F)$ up to canonical natural transformation: one sends $X \in \mathcal{C}$ to

$(X, F(X), \mathrm{id}_{F(X)})$. On classifying spaces this identifies BF with $BP_2 \circ (BP_1)^{-1}$ up to homotopy. Since both BP_1 and BP_2 are homotopy equivalences, so is BF. □

The preceding proof explains why Theorem A appears everywhere in Quillen's paper: one almost never proves contractibility of a classifying space directly. Instead one proves contractibility of a family of comma categories that encode all possible ways an object of the target can be resolved by objects coming from the source.

Theorem 7.2 (Quillen's Theorem B in expanded form)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be a functor such that for every morphism $d \to d'$ in $\mathcal{D}$ the induced functor $F/d \to F/d'$ is a homotopy equivalence. Then for every object $d \in \mathcal{D}$ there is a homotopy fibration sequence*

$$B(F/d) \longrightarrow B\mathcal{C} \xrightarrow{BF} B\mathcal{D},$$

and therefore a long exact sequence on homotopy groups.

Proof in steps. Step 1: keep the same replacement category. One again uses $S(F)$ with projections P_1 and P_2. As above, BP_1 is always a homotopy equivalence.

Step 2: why Theorem A is no longer enough. Now the categories F/d need not be contractible. So one cannot expect BP_2 to be a homotopy equivalence. The right question is instead whether BP_2 behaves like a fibration whose fibre over d is $B(F/d)$.

Step 3: simplicial-space criterion. Quillen realises the nerve of $S(F)$ in one simplicial direction and obtains a simplicial space whose n-simplices are disjoint unions of classifying spaces of categories of the form F/d_0. The face maps are induced by the transition functors associated to arrows in $\mathcal{D}$. By the hypothesis, every such transition functor is a homotopy equivalence. Quillen's lemma on realisation of simplicial spaces then implies that the realised map is a quasi-fibration.

Step 4: identify the fibre. The fibre over $d \in B\mathcal{D}$ is equivalent to the classifying space of the category F/d. The quasi-fibration statement therefore produces the homotopy exact sequence

$$\cdots \to \pi_{i+1}(B\mathcal{D}, d) \to \pi_i(B(F/d)) \to \pi_i(B\mathcal{C}) \to \pi_i(B\mathcal{D}) \to \cdots .$$

Replacing $B\mathcal{C}$ by $BS(F)$ via BP_1 yields the stated homotopy fibration sequence. □

Remark 7.3. Theorem B is the precise reason localisation theorems in K-theory produce long exact sequences rather than mere isomorphisms. It is not an abstract general principle added afterwards; it is built into Section 1 of Quillen's paper from the start. The later proofs are elaborate only because they must verify the hypotheses of Theorem B in concrete exact-category situations.

7.2. A closer look at the simplicial-space input to Theorem B

The proof of Theorem B in [Qu73] is not a formal abstract nonsense argument. What enters is a specific lemma about simplicial spaces whose face operators are fibrewise homotopy equivalences. Since this is the point at which many modern expositions become too compressed, it is worth isolating the mechanism.

Lemma 7.4 (Quillen's realisation criterion, schematic form)**.** *Let $X_\bullet \to Y_\bullet$ be a map of simplicial spaces such that for every simplicial degree n and every simplex $\sigma \in Y_n$, the fibre $X_n(\sigma)$ depends functorially on σ, and every simplicial operator induces a homotopy equivalence between the corresponding fibres. Then after geometric realisation the induced map*

$$|X_\bullet| \longrightarrow |Y_\bullet|$$

behaves as a quasi-fibration, with fibre over a vertex identified up to homotopy with the fibre in simplicial degree 0*.*

Explanation of why this applies. Quillen filters the realisation by skeleta. On each skeletal layer one is attaching cells indexed by simplices of $Y_\bullet$, and along each such cell the fibre is controlled by the simplicial fibre attached to the initial vertex. The hypothesis that all transition maps between fibres are homotopy equivalences allows one to glue these local identifications compatibly. The Dold–Thom criterion for quasi-fibrations then implies that the realised map has the expected homotopy fibre. The theorem is not asserting a strict Serre-fibration statement; it is precisely tailored to the classifying-space situation where one has coherent fibrewise equivalences rather than literal local triviality. □

In the case of $P_2 : S(F) \to \mathcal{D}$, the simplicial space in question is the simplicial replacement of the nerve of $\mathcal{D}$ whose n-simplices are disjoint unions of nerves of the comma categories F/d_0. The ith face map either forgets an intermediate arrow or precomposes by a morphism in $\mathcal{D}$. Theorem B assumes exactly that these induced functors on comma categories are homotopy equivalences. Thus the abstract-looking fibre condition in the theorem is nothing more than the statement that every simplicial operator acts harmlessly on the relevant fibres.

Remark 7.5. For readers used to modern homotopy-theoretic language, one can summarise the situation as follows: $S(F)$ is a categorical model for a homotopy colimit of the diagram $d \mapsto F/d$ over $\mathcal{D}$. Theorem B says that if this diagram is homotopically locally constant along morphisms of $\mathcal{D}$, then its homotopy colimit projects to $B\mathcal{D}$ with fibre $B(F/d)$. Quillen formulates and proves this without model categories or derivators, but the conceptual pattern is already completely visible.

7.3. Theorems A and B almost line by line: Quillen's bisimplicial replacement

The replacement category $S(F)$ is only half of the real argument. In the proof of Theorem A, Quillen also introduces a bisimplicial set, usually denoted $T(F)$,

whose (p,q)-simplices are pairs of strings

$$\Big(Y_p \to \cdots \to Y_0 \to F(X_0),\ X_0 \to X_1 \to \cdots \to X_q\Big).$$

The p-direction moves in $\mathcal{D}$ and the q-direction moves in $\mathcal{C}$. The diagonal of $T(F)$ is the nerve of $S(F)$, so the geometric realisation of $T(F)$ is canonically $BS(F)$. What makes $T(F)$ useful is that one can realise in either simplicial direction first, and each choice reveals a different part of the proof.

Proposition 7.6 (The double realisation picture behind Theorem A)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. For fixed q, the p-realisation of the simplicial space $T(F)_{\bullet,q}$ is a disjoint union of classifying spaces of categories of the form $\mathcal{D}/F(X_0)$, indexed by q-simplices $X_0 \to \cdots \to X_q$ of $N\mathcal{C}$. For fixed p, the q-realisation of $T(F)_{p,\bullet}$ is a disjoint union of classifying spaces of categories of the form $Y_0\backslash F$, indexed by p-simplices $Y_p \to \cdots \to Y_0$ of $N\mathcal{D}$. In particular:*

(a) *the projection to $N\mathcal{C}$ is a homotopy equivalence because every category $\mathcal{D}/F(X_0)$ has a final object $\mathrm{id}_{F(X_0)}$;*

(b) *if each comma category $Y\backslash F$ is contractible, the projection to $N\mathcal{D}$ is also a homotopy equivalence.*

Hence BF is a homotopy equivalence.

Proof in steps closer to Quillen's actual construction. Step 1: realise in the p-direction. Fix a composable string $X_0 \to \cdots \to X_q$ in $\mathcal{C}$. The datum of the left-hand string in $T(F)_{p,q}$ is exactly a p-simplex in the nerve of the over-category $\mathcal{D}/F(X_0)$. Therefore

$$|T(F)_{\bullet,q}| \cong \coprod_{(X_0\to\cdots\to X_q)} B(\mathcal{D}/F(X_0)).$$

Since $\mathcal{D}/F(X_0)$ has a final object, each summand is contractible. The induced map to $(N\mathcal{C})_q$ is therefore a homotopy equivalence for every q.

Step 2: realise in the q-direction. Fix a composable string $Y_p \to \cdots \to Y_0$ in $\mathcal{D}$. The data in the q-direction amount to a q-simplex in the nerve of the comma category $Y_0\backslash F$. Thus

$$|T(F)_{p,\bullet}| \cong \coprod_{(Y_p\to\cdots\to Y_0)} B(Y_0\backslash F).$$

If each $Y\backslash F$ is contractible, then the induced map to $(N\mathcal{D})_p$ is a homotopy equivalence for every p.

Step 3: compare the two realisations. Because the diagonal of $T(F)$ is the nerve of $S(F)$, both iterated realisations compute $BS(F)$. Combining the equivalence from Step 1 with the equivalence from Step 2 shows that the two projections $BS(F) \to B\mathcal{C}$ and $BS(F) \to B\mathcal{D}$ are homotopy equivalences. This is precisely Theorem A. □

Remark 7.7. This is the point at which Quillen's proof is more than a clever manipulation of comma categories. The bisimplicial object $T(F)$ gives a two-sided interpolation between source and target. Realising first in one direction or the other is what allows one to read off either contractibility statements (Theorem A) or fibre statements (Theorem B) from the same underlying construction.

Proposition 7.8 (Why the quasi-fibration lemma is exactly what Theorem B needs)**.** *Let I be a small category and let $X : I \to \mathbf{Top}$ be a functor such that for every arrow $i \to i'$ the induced map $X_i \to X_{i'}$ is a homotopy equivalence. Realising the simplicial space*

$$[p] \longmapsto \coprod_{i_0 \to \cdots \to i_p} X_{i_0}$$

produces a map $g : E \to BI$ which is a quasi-fibration. Moreover the fibre over the vertex i is homotopy equivalent to X_i.

Proof in steps following Quillen's skeletal induction. Filter BI by skeleta F_p. Over the interior of each nondegenerate p-simplex the map g is a product projection, so the only difficulty is gluing across the boundary. Remove the barycentres of the p-cells to obtain an open set $U \subset F_p$ that deformation-retracts onto F_{p-1}. The radial deformation from a punctured simplex to its boundary induces, on fibres, the transition maps attached to face operators. By hypothesis those transition maps are homotopy equivalences. The Dold–Thom gluing criterion for quasi-fibrations therefore applies inductively from F_{p-1} to F_p, proving that g is a quasi-fibration on all of BI. The fibre over a vertex i is visibly X_i. □

Corollary 7.9 (Theorem B in its operative form)**.** *With the notation of Theorem 7.2, the map $BP_2 : BS(F) \to B\mathcal{D}$ is the realisation of a simplicial-space map of the form above with fibre over $d \in \mathcal{D}$ given by $B(F/d)$. Hence the base-change hypothesis in Theorem B is exactly the condition needed to turn BP_2 into a quasi-fibration.*

Proof. Apply Proposition 7.8 to the functor $d \mapsto B(F/d)$. The face maps of the simplicial replacement are induced by the transition functors $F/d \to F/d'$. If these are homotopy equivalences for every arrow $d \to d'$, the realised map is a quasi-fibration with fibre $B(F/d)$ over d. Since BP_1 is already a homotopy equivalence, this is precisely the homotopy-fibre description asserted by Theorem B. □

7.4. The Q-construction in explicit categorical form

Let $\mathcal{M}$ be a small exact category. The objects of $Q\mathcal{M}$ are the same as those of $\mathcal{M}$. A morphism from M to M' is represented by a diagram

$$M \xleftarrow{j} N \xrightarrow{i} M',$$

where j is an admissible epimorphism and i is an admissible monomorphism. Two such roofs are identified if they are isomorphic in the obvious way. Composition is defined by taking the pullback-pushout factorisation of the middle term; the exactness axioms are precisely what make this associative.

The elementary calculus of morphisms in $Q\mathcal{M}$ is the first genuinely new piece of categorical algebra in the paper.

Lemma 7.10 (Factorisation in $Q\mathcal{M}$)**.** *Every morphism $u : M \to M'$ in $Q\mathcal{M}$ admits two canonical factorizations, unique up to unique isomorphism:*

$$u = i^! \, j_!, \qquad u = j'_! \, i'^!,$$

where i, i' are admissible monomorphisms and j, j' are admissible epimorphisms. Moreover a morphism in $Q\mathcal{M}$ is both injective and surjective if and only if it comes from an isomorphism in $\mathcal{M}$.

Proof. Take a representative roof $M \xleftarrow{j} N \xrightarrow{i} M'$. The factorisation $u = i^! j_!$ is built into the definition. To obtain the opposite order, form the bicartesian square determined by i and j inside the ambient exact category. Exactness guarantees that pullback and pushout agree for such diagrams, so the resulting object yields a second roof in which the injective and surjective pieces have changed order. Uniqueness up to unique isomorphism is a direct diagram chase with universal properties of the bicartesian square. The final statement follows because a morphism that is simultaneously represented by an admissible monomorphism and an admissible epimorphism must have trivial kernel and cokernel, hence is an isomorphism. □

Proposition 7.11 (Universal property of $Q\mathcal{M}$)**.** *To give a functor $H : Q\mathcal{M} \to \mathcal{C}$ is equivalent to giving:*

(a) *an object $H(M) \in \mathcal{C}$ for each $M \in \mathcal{M}$;*

(b) *for each admissible monomorphism $i : M' \hookrightarrow M$, a morphism $i^! : H(M') \to H(M)$;*

(c) *for each admissible epimorphism $j : M \twoheadrightarrow M''$, a morphism $j_! : H(M'') \to H(M)$;*

such that composition of admissible monos and epis is respected and such that in every bicartesian square the two induced morphisms agree.

Proof in steps. Step 1: necessity. If H is already a functor on $Q\mathcal{M}$, apply it to the morphisms in $Q\mathcal{M}$ induced by admissible monomorphisms and admissible epimorphisms. Functoriality supplies all the required compatibilities.

Step 2: sufficiency on representatives. Suppose the data are given. For a roof $M \xleftarrow{j} N \xrightarrow{i} M'$ define

$$H(u) := i^! \circ j_! : H(M) \longrightarrow H(M').$$

Because every morphism in $Q\mathcal{M}$ has injective-surjective factorisations, this is the only possible formula.

Step 3: independence of the chosen roof. If two roofs are isomorphic, the factorisation lemma and the compatibility with bicartesian squares show that the corresponding composites agree. This is exactly where Quillen uses the square relation in the statement of the universal property.

Step 4: compatibility with composition. Compose two roofs by forming the pullback-pushout object. The bicartesian-square relation shows that the map attached to the composite equals the composite of the maps attached to the two factors. Hence one obtains a well-defined functor. □

This universal property explains why $Q\mathcal{M}$, and not merely the set of objects of $\mathcal{M}$ modulo exact-sequence relations, is the correct enhancement of K_0: it is the free categorical gadget in which admissible monos and epis act functorially and compatibly.

7.5. Why $\pi_1(BQ\mathcal{M}) \cong K_0(\mathcal{M})$

The identification of the fundamental group with the Grothendieck group is one of the most beautiful parts of Quillen's paper. Rather than compute loops directly, Quillen computes the category of covering spaces of $BQ\mathcal{M}$.

Theorem 7.12 (Quillen's computation of the fundamental group)**.** *Let $\mathcal{M}$ be a small exact category and let 0 be a zero object. Then there is a canonical isomorphism*

$$\pi_1(BQ\mathcal{M}, 0) \cong K_0(\mathcal{M}).$$

Consequently

$$K_i(\mathcal{M}) := \pi_{i+1}(BQ\mathcal{M}, 0)$$

extends the usual Grothendieck group.

Proof in steps following Quillen's strategy. Step 1: coverings of a classifying space. For a connected small category $\mathcal{E}$, covering spaces of $B\mathcal{E}$ are classified by functors $\mathcal{E} \to \mathbf{Set}$ that invert all morphisms. Such a functor is the categorical shadow of the monodromy action of $\pi_1(B\mathcal{E})$ on the fibre over the base point.

Step 2: specialise to $Q\mathcal{M}$. Let Γ be the full subcategory of morphism-inverting functors $F : Q\mathcal{M} \to \mathbf{Set}$ with the normalisation $F(0) = S$ and $F(i^!) = \mathrm{id}_S$ for every map $0 \to M$. Every covering functor is isomorphic to one of this form, so it suffices to analyse Γ.

Step 3: build functors from $K_0(\mathcal{M})$-sets. Given a $K_0(\mathcal{M})$-set S, define a functor F_S by

$$F_S(M) = S, \qquad F_S(i^!) = \mathrm{id}_S, \qquad F_S(j_!) = \text{action of } [\ker(j)] \text{ on } S.$$

The exact-sequence relation $[M] = [M'] + [M'']$ in K_0 is exactly what is needed to check functoriality on composite roofs.

Step 4: recover the action from a functor. Conversely, if $F \in \Gamma$, then for an admissible epimorphism $j : M \twoheadrightarrow M''$ the automorphism $F(j_!)$ depends

only on the class of $\ker(j)$ in $K_0(\mathcal{M})$. Indeed, for every exact sequence $0 \to M' \to M \to M'' \to 0$, the universal property of $Q\mathcal{M}$ implies that $F(j_!)$ is multiplicative in short exact sequences. Hence the assignment $[M'] \mapsto F(j_!)$ extends uniquely to a homomorphism

$$K_0(\mathcal{M}) \longrightarrow \mathrm{Aut}(F(0)).$$

Thus $F(0)$ becomes a $K_0(\mathcal{M})$-set.

Step 5: equivalence of categories. The two constructions are inverse equivalences between $K_0(\mathcal{M})$-sets and the category of covering functors of $BQ\mathcal{M}$. Therefore the automorphism group of the universal cover, namely $\pi_1(BQ\mathcal{M}, 0)$, is canonically $K_0(\mathcal{M})$. □

Remark 7.13. This proof matters philosophically. It shows that K_0 is not merely the first truncation of higher K-theory: it is literally the fundamental group governing coverings of the K-theory space. In Quillen's formulation, the Grothendieck group is already encoded in the topology of $BQ\mathcal{M}$ before one ever defines the higher groups.

7.6. Characteristic exact sequences and additivity in steps

Let $\mathcal{E}(\mathcal{M})$ denote the exact category of short exact sequences in $\mathcal{M}$. An object is a conflation

$$0 \longrightarrow sE \longrightarrow tE \longrightarrow qE \longrightarrow 0.$$

The source, total object, and quotient define exact functors

$$s, t, q : \mathcal{E}(\mathcal{M}) \longrightarrow \mathcal{M}.$$

Quillen's Theorem 2 states that $(s, q) : Q\mathcal{E}(\mathcal{M}) \to Q\mathcal{M} \times Q\mathcal{M}$ is a homotopy equivalence. The standard additivity identity $t_* = s_* + q_*$ on K-groups is an immediate corollary, but the proof of the homotopy equivalence is itself a model example of Theorem A.

Theorem 7.14 (Characteristic exact sequences)**.** *The functor*

$$(s, q) : Q\mathcal{E}(\mathcal{M}) \longrightarrow Q\mathcal{M} \times Q\mathcal{M}$$

is a homotopy equivalence.

Proof in steps. Fix $(M, N) \in Q\mathcal{M} \times Q\mathcal{M}$. By Theorem A it suffices to prove that the comma category $(s, q)/(M, N)$ is contractible.

Step 1: describe the fibre category. An object of $(s, q)/(M, N)$ is a short exact sequence E together with morphisms $u : sE \to M$ and $v : qE \to N$ in $Q\mathcal{M}$. Quillen introduces two full subcategories: first $\mathcal{C}'$, where u is represented by a surjection, and then $\mathcal{C}''$, where in addition v is represented by an injection.

Step 2: simplify by adjoints. The inclusions

$$\mathcal{C}'' \hookrightarrow \mathcal{C}' \hookrightarrow (s, q)/(M, N)$$

have left adjoints. Concretely, given a general object, one alters it first by a pushout so that u becomes surjective, then dually by a pullback so that v becomes injective. The universal properties of the pushout and pullback produce the required adjunctions.

Step 3: identify the smallest subcategory. The category $\mathcal{C}''$ has an initial object, namely the split short exact sequence

$$0 \longrightarrow M \longrightarrow M \oplus N \longrightarrow N \longrightarrow 0$$

with the obvious structure maps. Therefore $\mathcal{C}''$ is contractible.

Step 4: conclude. Since $\mathcal{C}'' \to \mathcal{C}' \to (s,q)/(M,N)$ are homotopy equivalences and $\mathcal{C}''$ is contractible, the whole comma category is contractible. Theorem A then gives the result. □

Corollary 7.15 (Additivity)**.** *On K-groups one has*

$$t_* = s_* + q_* : K_i(\mathcal{E}(\mathcal{M})) \longrightarrow K_i(\mathcal{M}).$$

Proof. The exact functor sending (M', M'') to the split sequence $0 \to M' \to M' \oplus M'' \to M'' \to 0$ defines a homotopy inverse to (s,q) on K-theory spaces. Composing with t gives the direct-sum functor. Since K-theory sends direct sums to addition, one obtains $t_* = s_* + q_*$. □

7.7. What exactly the additivity theorem is measuring

The additivity theorem is sometimes paraphrased by saying that "the K-theory class of a short exact sequence is the sum of the classes of its ends". This is correct, but it conceals the more subtle point proved by Quillen: the entire category of extensions carries no more homotopy information than the ordered pair consisting of source and quotient. In particular, the middle term contributes no independent homotopy-theoretic variable.

Proposition 7.16 (Why the split sequence controls the fibre)**.** *Let (M,N) be a pair of objects of $\mathcal{M}$. In the comma category $(s,q)/(M,N)$, every object is connected by a zig-zag of morphisms to the split sequence*

$$0 \longrightarrow M \longrightarrow M \oplus N \longrightarrow N \longrightarrow 0.$$

Consequently the classifying space of $(s,q)/(M,N)$ is contractible.

Proof in steps. Start with an extension

$$0 \to A \to B \to C \to 0$$

endowed with comparison morphisms $A \to M$ and $C \to N$ in $Q\mathcal{M}$. By the first adjunction step in Quillen's proof one may replace the given object by one in which the map to M is represented by an admissible epimorphism; geometrically, one enlarges the extension by a pushout so that the source is forced to map onto M. Dually one then passes, via a pullback, to a homotopy-equivalent object for which the map from the quotient to N is represented by

an admissible monomorphism. Once both normal forms are achieved, there is a canonical morphism from the split extension on (M, N) to the chosen object. This exhibits the split extension as an initial object of the terminally simplified subcategory. □

This is the real content behind the additivity identity. The theorem is not merely a formal consequence of the relation $[B] = [A] + [C]$ in K_0; rather, it states that the whole *moduli category of extensions* retracts homotopically onto the category of split extensions. That statement is much stronger and is exactly what later made additivity one of the central axioms in Waldhausen K-theory.

7.8. Resolution in steps

The resolution theorem is Quillen's method for replacing a large exact category by a smaller one provided every object admits sufficiently many resolutions by objects of the smaller category.

Theorem 7.17 (Resolution theorem in expanded form)**.** *Let $\mathcal{P} \subseteq \mathcal{M}$ be a full subcategory of an exact category. Assume:*

(i) *$\mathcal{P}$ is closed under extensions;*

(ii) *if $0 \to M' \to M \to M'' \to 0$ is exact in $\mathcal{M}$ and $M'' \in \mathcal{P}$, then there exists an admissible epimorphism $P \twoheadrightarrow M''$ with $P \in \mathcal{P}$ whose pullback along $M \to M''$ again lies in $\mathcal{P}$;*

(iii) *every object of $\mathcal{M}$ admits a finite $\mathcal{P}$-resolution.*

Then the inclusion $Q\mathcal{P} \to Q\mathcal{M}$ induces a homotopy equivalence and hence isomorphisms $K_i(\mathcal{P}) \cong K_i(\mathcal{M})$ for all $i \geq 0$.

Proof in steps. The proof proceeds by induction on the resolution length.

Step 1: the one-step case. Suppose first every object $M \in \mathcal{M}$ fits into an exact sequence

$$0 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

with $P_0, P_1 \in \mathcal{P}$. Consider the inclusion $f : Q\mathcal{P} \to Q\mathcal{M}$. By Theorem A it suffices to show that for each $M \in \mathcal{M}$ the category f/M is contractible.

Step 2: describe f/M. An object of f/M is a morphism $P \to M$ in $Q\mathcal{M}$ with $P \in \mathcal{P}$. Such a morphism is represented by a subquotient of M that is isomorphic to an object of $\mathcal{P}$. Quillen replaces this category by a simpler category of short exact sequences ending in M with middle term and kernel in $\mathcal{P}$.

Step 3: contractibility. Using the hypothesis that every object has a one-step $\mathcal{P}$-resolution, Quillen constructs an initial object in a homotopy-equivalent subcategory of f/M. The idea is that any other resolution of M by objects of $\mathcal{P}$ admits a morphism from a chosen universal one, obtained by a pullback-pushout argument. Hence f/M is contractible.

Step 4: induction on the length. Let $\mathcal{P}_n$ be the full subcategory of objects admitting a $\mathcal{P}$-resolution of length at most n. Quillen proves that each $\mathcal{P}_n$ is closed under extensions and that the inclusion $\mathcal{P}_n \subseteq \mathcal{P}_{n+1}$ satisfies the one-step hypotheses. Therefore $K_i(\mathcal{P}_n) \to K_i(\mathcal{P}_{n+1})$ is an isomorphism for every n. Passing to the colimit over n yields $K_i(\mathcal{P}) \cong K_i(\mathcal{M})$. □

Corollary 7.18. *If A is a regular noetherian ring, then the inclusion of finitely generated projective A-modules into finitely generated A-modules induces an isomorphism*

$$K_i(A) \cong K'_i(A)$$

for all $i \geq 0$.

Proof. Every finitely generated module over a regular noetherian ring has finite projective dimension, so the resolution theorem applies with $\mathcal{P} = \mathbf{Proj}(A)$ and $\mathcal{M} = \mathrm{Mod}^{\mathrm{fg}}(A)$. □

7.9. Devissage in steps

Devissage is the opposite of resolution. Instead of approximating a large category by one with better objects, one proves that a smaller category already captures the K-theory because every object of the ambient category can be filtered with successive quotients in the smaller one.

Theorem 7.19 (Devissage in expanded form)**.** *Let $\mathcal{B}$ be a full abelian subcategory of an abelian category $\mathcal{A}$, closed under subobjects, quotients, and finite direct sums. Assume every object $M \in \mathcal{A}$ admits a finite filtration*

$$0 = M_0 \subset M_1 \subset \cdots \subset M_n = M$$

with subquotients $M_i/M_{i-1} \in \mathcal{B}$. Then the inclusion $Q\mathcal{B} \to Q\mathcal{A}$ is a homotopy equivalence.

Proof in steps, close to Quillen's argument. Step 1: apply Theorem A. Let $f : Q\mathcal{B} \to Q\mathcal{A}$ denote the inclusion. It is enough to show that f/M is contractible for every object $M \in \mathcal{A}$.

Step 2: identify f/M with an ordered set of layers. An object of f/M is a morphism $N \to M$ in $Q\mathcal{A}$ with $N \in \mathcal{B}$. Such a morphism is encoded by a layer (M_0, M_1) in M for which $M_1/M_0 \in \mathcal{B}$. The category f/M is thus equivalent to the ordered set $J(M)$ of such layers. The order relation is inclusion of layers.

Step 3: reduce the filtration length. Suppose $M' \subseteq M$ and $M/M' \in \mathcal{B}$. Quillen defines functors

$$r : J(M) \to J(M'), \qquad s : J(M) \to J(M)$$

by intersecting a layer with M' and by enlarging it minimally so that the quotient by the lower term lies in $\mathcal{B}$. These functors satisfy $r \circ i = \mathrm{id}$ for the natural inclusion $i : J(M') \hookrightarrow J(M)$, and there are natural transformations

$$i \circ r \Longrightarrow s \Longleftarrow \mathrm{id}_{J(M)}.$$

Therefore i is a homotopy equivalence.

Step 4: iterate along the chosen filtration. Repeatedly remove the top quotient of a filtration of M by objects of $\mathcal{B}$. Since each step induces a homotopy equivalence on the corresponding ordered sets of layers, one reduces to the case $M \in \mathcal{B}$, where $J(M)$ has a terminal object $(0, M)$ and is therefore contractible. Thus f/M is contractible for every M. □

Remark 7.20. The proof is especially elegant because the comma category collapses to a partially ordered set. Once this reduction is made, homotopy theory becomes a question about manipulating filtrations. This is the cleanest illustration of Quillen's principle that higher K-theory should reduce hard homotopy questions to controlled categorical combinatorics.

7.10. Localization in steps

Localization is the most delicate of the formal theorems because it is genuinely a Theorem B statement. Let $\mathcal{B}$ be a Serre subcategory of an abelian category $\mathcal{A}$, and let $\mathcal{A}/\mathcal{B}$ be the Gabriel quotient. Denote by

$$e : \mathcal{B} \to \mathcal{A}, \qquad s : \mathcal{A} \to \mathcal{A}/\mathcal{B}$$

the canonical exact functors.

Theorem 7.21 (Localization in expanded form)**.** *There is a long exact sequence*

$$\cdots \longrightarrow K_{i+1}(\mathcal{A}/\mathcal{B}) \xrightarrow{\partial} K_i(\mathcal{B}) \xrightarrow{e_*} K_i(\mathcal{A}) \xrightarrow{s_*} K_i(\mathcal{A}/\mathcal{B}) \longrightarrow K_{i-1}(\mathcal{B}) \longrightarrow \cdots .$$

Proof in steps following Quillen's Section 5. Step 1: factor through a comma category. Because every object of $\mathcal{B}$ maps to zero in $\mathcal{A}/\mathcal{B}$, the composite $Qs \circ Qe$ is naturally isomorphic to the constant functor with value 0. Hence Qe factors through the comma category $0\backslash Qs$:

$$Q\mathcal{B} \longrightarrow 0\backslash Qs \longrightarrow Q\mathcal{A}.$$

Here an object of $0\backslash Qs$ is an object $M \in \mathcal{A}$ together with a morphism $0 \to s(M)$ in $Q(\mathcal{A}/\mathcal{B})$.

Step 2: identify the potential fibre. The functor $Q\mathcal{B} \to 0\backslash Qs$ sends $B \in \mathcal{B}$ to the object $e(B)$ with its canonical zero map in the quotient. Quillen proves this functor is a homotopy equivalence by replacing $0\backslash Qs$ with a full subcategory where the arrow $0 \to s(M)$ is represented in a very rigid way and then constructing a left adjoint. The argument is a Theorem A style contraction.

Step 3: verify the base-change hypothesis of Theorem B. Let $u : V \to V'$ be a morphism in $Q(\mathcal{A}/\mathcal{B})$. One must show that the induced functor

$$u^* : V'\backslash Qs \longrightarrow V\backslash Qs$$

is a homotopy equivalence. Quillen factors u into the injective and surjective cases and handles each separately. In each case he identifies a full subcategory

on which the pullback functor admits an adjoint or inverse up to homotopy, again reducing the problem to an ordered set or comma category with an evident contraction.

Step 4: apply Theorem B. Once Step 3 is established, Theorem B applies to the functor $Qs : Q\mathcal{A} \to Q(\mathcal{A}/\mathcal{B})$ and shows that the homotopy fibre over 0 is $B(0\backslash Qs)$, hence also $BQ\mathcal{B}$ by Step 2. The homotopy exact sequence of this fibration gives the localization long exact sequence. □

Remark 7.22 (About the boundary map). Quillen explicitly remarks that the proof, though completely sufficient to produce the exact sequence, does not yet explain the boundary homomorphism ∂ very geometrically. This is historically important: the theorem is already powerful, but it also points to later work on more conceptual descriptions of localisation, notably in Waldhausen and Thomason–Trobaugh.

7.11. The localization theorem in Quillen's own reduction pattern

The localization proof in Section 5 of [Qu73] is one of the most compressed passages in the paper, even though it is structurally one of the most important. The argument becomes much more transparent once one isolates the three auxiliary categories that Quillen introduces and the exact role each one plays.

Definition 7.23. Let $\mathcal{B} \subseteq \mathcal{A}$ be a Serre subcategory and $s : \mathcal{A} \to \mathcal{A}/\mathcal{B}$ the quotient functor. For an object $V \in \mathcal{A}/\mathcal{B}$, write $V\backslash Qs$ for the comma category of objects (M, u) with $M \in \mathcal{A}$ and $u : V \to sM$ a morphism in $Q(\mathcal{A}/\mathcal{B})$. Define $\mathcal{F}_V \subseteq V\backslash Qs$ to be the full subcategory on those (M, u) for which u is an isomorphism in $Q(\mathcal{A}/\mathcal{B})$.

When $V = 0$, the condition that $0 \to sM$ be an isomorphism is equivalent to $sM = 0$, hence to $M \in \mathcal{B}$. Therefore $\mathcal{F}_0 \cong Q\mathcal{B}$. The whole proof is designed to show that every comma category $V\backslash Qs$ is homotopy equivalent to this manageable core after base change.

Lemma 7.24 (First reduction in the localization proof)**.** *For every $V \in \mathcal{A}/\mathcal{B}$, the inclusion*

$$\mathcal{F}_V \hookrightarrow V\backslash Qs$$

induces a homotopy equivalence on classifying spaces.

Proof in steps faithful to Quillen's reductions. Apply Theorem A to the inclusion. It is enough to show that for each object (M, u) of $V\backslash Qs$, the comma category of objects of $\mathcal{F}_V$ over (M, u) is contractible. Quillen identifies this comma category with an ordered set of *layers* (M_0, M_1) inside M such that the morphism u is represented by the quotient M_1/M_0. This ordered set is nonempty because every morphism in the Gabriel quotient can be represented by a subquotient, and it is filtering because two such layers admit a common refinement by intersection and sum. A filtering category has contractible classifying space. Hence Theorem A gives the claim. □

The second auxiliary category is the one that connects objects of the quotient category back to the Serre subcategory $\mathcal{B}$.

Definition 7.25. Fix an object $N \in \mathcal{A}/\mathcal{B}$. Let $\mathcal{E}_N$ be the category whose objects are pairs (M, h) where $M \in \mathcal{A}$ and $h : sM \to N$ is an isomorphism in the quotient. Equivalently, before passing to the quotient, one may think of h as a morphism in $\mathcal{A}$ whose kernel and cokernel belong to $\mathcal{B}$. A morphism $(M, h) \to (M', h')$ is represented by a morphism $u : M \to M'$ in $Q\mathcal{A}$ such that $h' = h \circ su$.

Each object (M, h) has a kernel in $\mathcal{B}$, well-defined up to canonical isomorphism, and Quillen uses this to define a functor

$$k_N : \mathcal{E}_N \longrightarrow Q\mathcal{B}, \qquad (M, h) \longmapsto \ker(h).$$

The difficulty is to show that this functor is a homotopy equivalence.

Lemma 7.26 (Kernel functor in the localization proof)**.** *For every $N \in \mathcal{A}/\mathcal{B}$, the functor $k_N : \mathcal{E}_N \to Q\mathcal{B}$ is a homotopy equivalence.*

Proof strategy in steps. Quillen proves this by interposing one more category, usually denoted $\mathcal{H}_N$, whose objects keep track simultaneously of a representative (M, h) and of a chosen layer inside M representing the kernel contribution. There are functors

$$\mathcal{H}_N \longrightarrow \mathcal{E}_N, \qquad \mathcal{H}_N \longrightarrow Q\mathcal{B},$$

and both are shown to be homotopy equivalences by Theorem A.

Step 1: from $\mathcal{H}_N$ to $\mathcal{E}_N$. For a fixed (M, h), the fibre category consists again of suitable layers inside M compatible with h. Exactly as in Lemma 7.24, this fibre is filtering and therefore contractible.

Step 2: from $\mathcal{H}_N$ to $Q\mathcal{B}$. Fix $B \in \mathcal{B}$. The fibre over B is described by mod-$\mathcal{B}$ isomorphisms whose chosen kernel layer is isomorphic to B. Quillen reduces this fibre to a category with an initial object by a sequence of diagram chases in $\mathcal{A}$, again invoking Theorem A.

Step 3: conclude. Since both comparison functors are homotopy equivalences, so is k_N. □

The point of these lemmas is that the hypotheses of Theorem B can now be checked on the comma categories of Qs.

Proposition 7.27 (Base-change verification for localization)**.** *For every morphism $u : V' \to V$ in $Q(\mathcal{A}/\mathcal{B})$, the induced functor*

$$u^* : V \backslash Qs \longrightarrow V' \backslash Qs$$

is a homotopy equivalence.

Proof in steps mirroring Quillen's reductions. Factor u into an admissible epimorphism followed by an admissible monomorphism in the quotient. By passage to the opposite category, it suffices to treat the injective case. Quillen then

reduces further to the special injective map $i_V : V' \hookrightarrow V$ attached to a chosen layer. On the subcategories $\mathcal{F}_V$ this functor is controlled by the kernel category $\mathcal{E}_N$ of the quotient $N = V/V'$. Lemma 7.26 shows that the effect of u^* on these core subcategories is a homotopy equivalence, and Lemma 7.24 transfers the result to all of $V\backslash Qs$. □

Corollary 7.28 (How Theorem B yields localization)**.** *The factorisation of $Qe : Q\mathcal{B} \to Q\mathcal{A}$ through $0\backslash Qs$ together with Proposition 7.27 and the identification $0\backslash Qs \simeq Q\mathcal{B}$ satisfy the hypotheses of Theorem B. Consequently there is a homotopy fibration*

$$BQ\mathcal{B} \longrightarrow BQ\mathcal{A} \longrightarrow BQ(\mathcal{A}/\mathcal{B}),$$

and hence the localization long exact sequence in K-theory.

Remark 7.29. This makes clear why the localization theorem is the paradigmatic application of Theorem B in the whole paper. Theorem A is used internally to simplify auxiliary comma categories; Theorem B is the step that turns those simplifications into the actual long exact sequence.

7.12. Regular rings, homotopy invariance, and the fundamental theorem

The formal theorems of Sections 3–5 are then fed into the ring-theoretic applications of Section 6. The philosophy is that filtered or graded structures allow one to compare a ring with a simpler associated graded object, and regularity ensures the comparison takes place in ordinary K-theory rather than only in G-theory.

Theorem 7.30 (Homotopy invariance for regular rings)**.** *If A is a regular noetherian ring, then the map induced by evaluation at $t = 0$*

$$K_i(A) \longrightarrow K_i(A[t])$$

is an isomorphism for every $i \geq 0$.

Proof strategy in steps. Quillen proves first a graded statement: for a positively graded noetherian ring, the category of graded modules has the same K-theory as modules over the degree-zero piece, after adjoining the translation operator. One then passes from filtered rings to graded rings by the Rees-ring construction. Applying localisation to the central element defining the Rees deformation shows that the K-theory of the filtered ring agrees with that of its associated graded ring. Taking the filtration on $A[t]$ by degree in t yields the homotopy invariance statement. Regularity is used to identify K-theory and G-theory via the resolution theorem. □

Theorem 7.31 (Fundamental theorem for regular rings)**.** *If A is a regular noetherian ring, then*

$$K_i(A[t, t^{-1}]) \cong K_i(A) \oplus K_{i-1}(A)$$

for all $i \geq 1$.

Proof strategy in steps. The localization sequence for the pair $(A[t], (t))$ compares the K-theory of $A[t]$, the closed subscheme defined by $t = 0$, and the open complement where t becomes invertible. By devissage, the closed part contributes the K-theory of A. By homotopy invariance, the ambient term is again $K_i(A)$. Writing out the long exact sequence and splitting it by the map induced from the unit section yields the direct-sum decomposition above. □

Remark 7.32. This theorem is the prototype of what later became Bass' negative K-theory and the systematic study of Bass fundamental exact sequences. In Quillen's paper it already appears in a form that shows the close relation between localisation, homotopy invariance, and the passage from affine space to the punctured affine line.

7.13. From filtered rings to homotopy invariance: the graded calculation unpacked

The homotopy invariance theorem in Section 6 of [Qu73] is easy to misread if one only looks at the final statement. Quillen does not attack $A[t]$ directly. He proves a graded result, packages filtered objects by means of the Rees construction, and only then specialises to the polynomial ring.

Proposition 7.33 (The graded comparison underlying homotopy invariance)**.** *Let $A = \bigoplus_{n\geq 0} A_n$ be a positively graded noetherian ring. Then the exact category of finitely generated graded A-modules is generated, up to devissage, by the degree-zero part. More precisely, after adjoining the translation functor $M \mapsto M(1)$, its K-theory is identified with $K(A_0)$ together with the bookkeeping variable recording the shift.*

Proof in steps. For a graded module M, let $F^pM = \bigoplus_{n\geq p} M_n$. The successive quotients $F^pM/F^{p+1}M$ are concentrated in one degree and therefore come from A_0-modules, shifted by p. Devissage applied to the filtration by the F^pM reduces the K-theory of graded modules to the K-theory of the full subcategory of degree-concentrated modules. That subcategory is equivalent to finitely generated A_0-modules, while the grading shift contributes the Laurent-polynomial variable that later becomes the Bott-like parameter in the graded calculation. □

Now let A be a filtered noetherian ring with associated graded ring $\mathrm{gr}(A)$ and Rees ring

$$R(A) = \bigoplus_{n\geq 0} F^nA\, t^n \subseteq A[t].$$

Specialising t to 1 recovers A, while specialising t to 0 recovers $\mathrm{gr}(A)$. The key localisation sequence for the central element t compares the K-theory of $R(A)$, of the closed fibre $R(A)/(t) \cong \mathrm{gr}(A)$, and of the localisation $R(A)[t^{-1}]$, which identifies with a Laurent extension of A. Because the graded K-theory of $R(A)$ is controlled by the degree-zero piece, one can pass from $\mathrm{gr}(A)$ back to A.

Corollary 7.34 (Homotopy invariance revisited)**.** *If A is regular noetherian, then $K_i(A) \to K_i(A[t])$ is an isomorphism for all $i \geq 0$.*

Expanded proof. Equip $A[t]$ with the filtration by degree in t. Its associated graded ring is canonically isomorphic to $A[t]$ again, now viewed as positively graded with degree-zero part A. The graded comparison above and the Rees-ring localisation argument identify the G-theory of $A[t]$ with the G-theory of A. Since A is regular, the resolution theorem identifies K-theory with G-theory on both rings, giving the desired isomorphism. □

7.14. Schemes: codimension filtration and Bloch's formula

For a noetherian scheme X, Quillen defines G-theory from the abelian category of coherent sheaves. Filtering coherent sheaves by codimension of support yields a spectral sequence whose first page is a sum over points of fixed codimension. In the regular finite-type case this leads to Gersten-type exactness and to the identification of codimension-p cycles modulo rational (linear) equivalence with a sheaf cohomology group built from the K-theory sheaf.

Theorem 7.35 (Codimension spectral sequence, schematic form)**.** *Let X be noetherian. Filtering coherent sheaves by codimension of support yields a spectral sequence*

$$E_1^{p,q}(X) = \bigoplus_{x \in X^{(p)}} K_{-p-q}(k(x)) \Longrightarrow G_{-p-q}(X).$$

If X is regular and of finite type over a field, this leads to

$$\mathrm{CH}^p(X) \cong H^p(X, \mathcal{K}_p),$$

Quillen's form of Bloch's formula.

Idea of proof in steps. The filtration by support defines a filtration on the exact category of coherent sheaves. The successive quotients are controlled by coherent modules supported exactly at points of fixed codimension. Devissage identifies their K-theory with the K-theory of the residue fields. The spectral sequence is the exact-couple machine associated to this filtration. In the regular case, Gersten-type exactness identifies the first differential with the divisor maps, and the resulting degree-p cohomology computes codimension-p cycles modulo rational equivalence. □

Remark 7.36. From the viewpoint of the present notes, this is where Quillen's work begins to touch the later world of Bloch, motivic cohomology, and higher Chow groups. The spectral sequence already makes visible the principle that cycle groups should arise as shadows of a higher K-theoretic theory.

7.15. Projective bundles as the geometric culmination of the paper

The last section of [Qu73] is sometimes omitted in quick summaries, but it is one of the most revealing parts of the paper. After the formal theorems and the codimension spectral sequence, Quillen shows that higher K-theory already has enough geometric strength to reproduce the projective bundle theorem and its noncommutative variants.

Theorem 7.37 (Projective bundle theorem, Quillen)**.** *Let X be a noetherian scheme and let E be a vector bundle of rank r on X. Then there is a natural isomorphism*

$$K_i(\mathbb{P}(E)) \cong \bigoplus_{j=0}^{r-1} K_i(X)$$

for every $i \geq 0$, where the summands are generated by tensoring with powers of the tautological line bundle.

Proof strategy in steps. Step 1: reduce to the universal case. By Zariski descent and the compatibility of K-theory with exact functors, it is enough to treat the projective space bundle $\mathbb{P}^{r-1}_X$ locally on X.

Step 2: use canonical resolutions. Quillen constructs, for a coherent sheaf on projective space that is sufficiently regular in the sense of Mumford, a canonical finite resolution by direct sums of twists $\mathcal{O}(-j)$. This produces a filtration of the exact category of such sheaves whose successive pieces are controlled by coherent sheaves on the base.

Step 3: apply additivity and devissage. Each layer in the canonical resolution contributes one copy of the K-theory of X. Because the resolution is functorial enough for Quillen's additivity theorem to apply, the K-theory of $\mathbb{P}(E)$ splits as the direct sum of these layers.

Step 4: identify the generators. The summand indexed by j is represented by tensoring with $\mathcal{O}_{\mathbb{P}(E)}(-j)$ and then pulling back from X. The basis of summands is therefore the same one familiar from Grothendieck's projective bundle formula for K_0, now promoted to all higher K-groups. □

Remark 7.38. This theorem is a useful corrective to a common caricature of Quillen's paper. The paper is not only about abstract homotopy groups of exact categories; it already recovers nontrivial geometric structure theorems for schemes. In that sense Section 8 is the point where the general theory proves that it deserves to be called a geometric cohomology theory.

7.16. Canonical resolutions on projective bundles in Quillen's own style

Section 8 of [Qu73] is deeper than a bare statement of the projective bundle theorem. Its real engine is a canonical resolution for regular sheaves on a projective bundle. Once this is written down, the projective bundle theorem becomes a triangular-algebra argument on K-groups.

Let $f : X = \mathbb{P}(E) \to S$ with E of rank r, and write $\mathcal{O}_X(1)$ for the tautological line bundle. Following Quillen, an X-module F is called *regular* if

$$R^q f_*(F(-q)) = 0 \qquad (q > 0).$$

For example, the standard Cech calculation shows that $\mathcal{O}_X(n) \otimes f^*N$ is regular for $n \geq 0$ and every quasi-coherent N on S.

Proposition 7.39 (Canonical resolution of a regular sheaf)**.** *If F is a regular sheaf on $X = \mathbb{P}(E)$, there exist exact functors T_i from regular sheaves on X to sheaves on S and a canonical exact sequence*

$$\begin{aligned} 0 &\longrightarrow \mathcal{O}_X(-r+1) \otimes T_{r-1}(F) \longrightarrow \cdots \longrightarrow \mathcal{O}_X(-1) \otimes T_1(F) \\ &\longrightarrow \mathcal{O}_X \otimes T_0(F) \longrightarrow F \longrightarrow 0. \end{aligned}$$

Moreover the objects $T_i(F)$ are determined by F up to unique isomorphism.

Proof in steps following Quillen's recursion. Step 1: surjection from the zeroth direct image. For regular F, Quillen proves that the canonical map

$$f^* f_*(F) \longrightarrow F$$

is surjective. This is obtained from the regularity of positive twists together with the exactness of the graded module of direct images.

Step 2: define the recursive kernels. Set $Z_{-1}(F) = F$. Having defined $Z_{n-1}(F)$, let

$$T_n(F) := f_*(Z_{n-1}(F)(n)),$$

and define $Z_n(F)$ to be the kernel of the canonical evaluation map

$$\mathcal{O}_X(-n) \otimes T_n(F) \longrightarrow Z_{n-1}(F).$$

By construction these assignments are additive functors of F.

Step 3: preserve regularity. Using the long exact sequence of higher direct images and the stability properties of regularity under short exact sequences, Quillen shows inductively that each $Z_n(F)(n+1)$ is again regular and that

$$f_*(Z_n(F)(n)) = 0 \qquad (n \geq 0).$$

The key point is that regularity kills the higher obstruction terms, so the recursive kernels remain under control.

Step 4: termination after $r-1$ steps. A second induction using the cohomology of projective space shows that $Z_{r-1}(F) = 0$. Splicing the short exact sequences

$$0 \to Z_n(F) \to \mathcal{O}_X(-n) \otimes T_n(F) \to Z_{n-1}(F) \to 0$$

together yields the stated canonical resolution. □

Remark 7.40. This is one of the hidden gems of Quillen's paper. The projective bundle theorem is not proved by a single abstract descent argument; it is built on a functorial finite resolution whose terms are the standard twists $\mathcal{O}_X(-i)$. That is what makes the eventual K-theory matrix triangular.

Proposition 7.41 (Why the projective bundle matrix is triangular)**.** *For $0 \leq n \leq r-1$, define exact functors*

$$U_n : \mathbf{Proj}(S) \longrightarrow \mathbf{Proj}(X), \qquad U_n(N) = \mathcal{O}_X(-n) \otimes f^* N,$$

and, on a suitable exact subcategory of vector bundles on X, define

$$V_n(F) = f_*(F(n)).$$

Then the composite V_nU_m vanishes for $m > n$ and is naturally isomorphic to the identity for $m = n$. Consequently the matrix $(V_nU_m)_{0\le n,m\le r-1}$ is upper triangular with identities on its diagonal.

Proof in steps. The direct-image calculation for twists on projective bundles shows that $R^qf_*(\mathcal{O}_X(n-m)\otimes f^*N)$ vanishes in the range relevant to the chosen subcategory. When $m > n$, the twist is too negative and the zeroth direct image vanishes; when $m = n$, one gets $f_*f^*N \cong N$. The higher-direct-image vanishings needed to make V_n exact are provided by the regularity lemmas preceding the projective bundle theorem. Thus the matrix of composites is indeed triangular with 1's on the diagonal. □

Theorem 7.42 (Projective bundle theorem with Quillen's injectivity–surjectivity split)**.** *Let $u = \bigoplus_{n=0}^{r-1}(U_n)_*$. Then*

$$u : \bigoplus_{n=0}^{r-1} K_q(S) \longrightarrow K_q(X)$$

is an isomorphism for every $q \ge 0$.

Proof in steps close to Section 8 of Quillen's paper. Step 1: injectivity. Form the map $v = \bigoplus_{n=0}^{r-1}(V_n)_*$ on the stabilized exact subcategory of vector bundles for which all the V_n are defined. By Proposition 7.41, the composite vu is represented by an upper triangular matrix with identity diagonal entries. Therefore vu is an isomorphism, and u is injective.

Step 2: surjectivity from the canonical resolution. The exact functors T_n in Proposition 7.39 give a second map

$$t : K_q(X) \longrightarrow \bigoplus_{n=0}^{r-1} K_q(S).$$

Additivity applied to the canonical resolution shows that the composite ut agrees with the map on K-groups induced by inclusion of the exact subcategory of regular bundles into the ambient exact category of vector bundles on X.

Step 3: stabilization of exact subcategories. Quillen proves, by twisting and using his earlier additivity theorem, that these inclusions are isomorphisms on K-groups. Hence ut is an isomorphism. Since u is already injective, it follows that u is surjective, hence bijective. □

Remark 7.43. The structure of the proof is striking. Injectivity comes from the triangular direct-image matrix; surjectivity comes from the canonical resolution and additivity. This is considerably more refined than simply asserting that the classes $1, z, \dots, z^{r-1}$ span $K_q(X)$.

7.17. What this achieves conceptually

We may now state explicitly what the proof dossier reveals. Quillen's paper is not merely a source of definitions; it is a remarkably coherent demonstration that higher algebraic K-theory is governed by a small number of homotopy-theoretic moves applied to categories with exact sequences. Theorems A and B are the two engines. The Q-construction is the categorical transmission that allows those engines to interact with exact algebra. Additivity, resolution, devissage, and localisation are the four principal outputs. Homotopy invariance, the fundamental theorem, the codimension spectral sequence, and Bloch's formula are the first major applications.

For a modern reader, the decisive insight is therefore the following. Quillen did not discover a collection of isolated identities satisfied by some mysterious groups K_i. He discovered a *method*: encode exact algebra in a classifying space, analyse the induced functors by comma categories, and extract either equivalences or fibre sequences according to whether Theorem A or Theorem B applies. Everything after that is elaboration, extension, and refinement of this method.

8. Diagrammatic supplements on Quillen's formal theorems

The preceding sections already isolate the correct proof architecture of [Qu73]. What we add here is a more diagrammatic version of the core arguments, so that the decisive universal constructions appear on the page and not only in prose. The goal is not to replace Quillen's original text, which is compressed but complete, but to unpack the exact categorical moves by which Theorems A and B produce additivity, resolution, dévissage, and localisation.

8.1. Theorem A and the bisimplicial replacement written out

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Quillen's replacement category $S(F)$ has as objects triples (X, Y, v) with $X \in \mathcal{C}$, $Y \in \mathcal{D}$, and $v : Y \to F(X)$ in $\mathcal{D}$; a morphism $(X, Y, v) \to (X', Y', v')$ is a pair (a, b) with $a : X \to X'$ in $\mathcal{C}$, $b : Y \to Y'$ in $\mathcal{D}$, and $F(a) \circ v = v' \circ b$. The fundamental commutative square is

$$\begin{tikzcd} S(F) \arrow[r, "P_1"] \arrow[d, "P_2"'] & \mathcal{C} \arrow[d, "F"] \\ \mathcal{D} \arrow[r, equal] & \mathcal{D}, \end{tikzcd}$$

with $P_1(X, Y, v) = X$ and $P_2(X, Y, v) = Y$.

To understand why this square controls both Theorem A and Theorem B,

one introduces the bisimplicial set $T(F)$ whose (p,q)-simplices are diagrams

$$Y_p \longrightarrow \cdots \longrightarrow Y_1 \longrightarrow Y_0 \xrightarrow{v} F(X_0)$$

$$X_0 \longrightarrow X_1 \longrightarrow \cdots \longrightarrow X_q$$

with the top row lying in $\mathcal{D}$ and the bottom row in $\mathcal{C}$. Realising first in the horizontal direction gives a simplicial space whose q-simplices are indexed by q-simplices of $N\mathcal{C}$ and whose fibres are the classifying spaces $B(\mathcal{D}/F(X_0))$. Realising first in the vertical direction gives a simplicial space indexed by simplices of $N\mathcal{D}$ with fibres $B(Y_0\backslash F)$.

Proposition 8.1 (Diagrammatic form of Theorem A)**.** *If each comma category $Y\backslash F$ is contractible, then $BF : B\mathcal{C} \to B\mathcal{D}$ is a homotopy equivalence.*

Proof. For fixed q, the p-simplices of $T(F)_{\bullet,q}$ form the nerve of the over-category $\mathcal{D}/F(X_0)$, hence

$$|T(F)_{\bullet,q}| \cong \coprod_{(X_0\to\cdots\to X_q)} B(\mathcal{D}/F(X_0)).$$

Since $\mathcal{D}/F(X_0)$ has the final object $\mathrm{id}_{F(X_0)}$, each summand is contractible. Therefore the map $|T(F)_{\bullet,q}| \to (N\mathcal{C})_q$ is a homotopy equivalence for every q, so after realising in the q-direction we obtain a homotopy equivalence

$$|\operatorname{diag} T(F)| \xrightarrow{\sim} B\mathcal{C}.$$

On the other hand, for fixed p one has

$$|T(F)_{p,\bullet}| \cong \coprod_{(Y_p\to\cdots\to Y_0)} B(Y_0\backslash F).$$

By hypothesis each $Y_0\backslash F$ is contractible, so the map $|T(F)_{p,\bullet}| \to (N\mathcal{D})_p$ is a homotopy equivalence for every p, and therefore

$$|\operatorname{diag} T(F)| \xrightarrow{\sim} B\mathcal{D}.$$

These two equivalences identify $|\operatorname{diag} T(F)|$ with both $B\mathcal{C}$ and $B\mathcal{D}$, and the induced map between them is precisely BF. Hence BF is a homotopy equivalence. □

8.2. Theorem B as a homotopy-cartesian square

The real content of Theorem B is not merely that a certain fibre has the correct homotopy type, but that the entire square of categories below becomes homotopy-cartesian after passage to classifying spaces:

$$\begin{tikzcd} Y\backslash F \arrow[r, "j"] \arrow[d, "f'"'] & \mathcal{C} \arrow[d, "F"] \\ Y\backslash\mathcal{D} \arrow[r, "j'"'] & \mathcal{D}. \end{tikzcd}$$

Here $j(X,v) = X$, $f'(X,v) = (F(X),v)$, and $j'(Y',v) = Y'$. Quillen's condition is that for every arrow $u : Y \to Y'$ in $\mathcal{D}$ the induced functor $u^* : Y'\backslash F \to Y\backslash F$ is a homotopy equivalence.

Proposition 8.2 (Diagrammatic form of Theorem B)**.** *Under the base-change hypothesis above, the square*

$$\begin{tikzcd} B(Y\backslash F) \arrow[r, "Bj"] \arrow[d, "Bf'"'] & B\mathcal{C} \arrow[d, "BF"] \\ B(Y\backslash \mathcal{D}) \arrow[r, "Bj'"'] & B\mathcal{D} \end{tikzcd}$$

is homotopy-cartesian. Since $Y\backslash\mathcal{D}$ has initial object (Y, id_Y), the lower-left corner is contractible, and thus the homotopy fibre of BF over Y is homotopy equivalent to $B(Y\backslash F)$.

Proof. We again use the bisimplicial replacement $T(F)$. For each simplex $Y_p \to \cdots \to Y_0$ of $N\mathcal{D}$ consider the topological space $X_{Y_\bullet} := B(Y_0\backslash F)$. The simplicial operators act on these spaces through the base-change functors induced by arrows in $\mathcal{D}$. By hypothesis each of those maps is a homotopy equivalence. Proposition 7.8 therefore applies to the simplicial space

$$[p] \longmapsto \coprod_{Y_0\to\cdots\to Y_p} B(Y_0\backslash F),$$

showing that its realisation maps to $B\mathcal{D}$ as a quasi-fibration whose fibre over Y is $B(Y\backslash F)$. But this realised simplicial space is exactly $BS(F)$, and the map to $B\mathcal{D}$ is BP_2. Since $BP_1 : BS(F) \to B\mathcal{C}$ is always a homotopy equivalence, the quasi-fibration statement transports along BP_1 and yields a homotopy-cartesian square

$$\begin{tikzcd} B(Y\backslash F) \arrow[r] \arrow[d] & BS(F) \arrow[d, "BP_2"] \\ * \arrow[r] & B\mathcal{D}. \end{tikzcd}$$

Composing horizontally with the homotopy equivalence BP_1 gives the stated square with $B\mathcal{C}$ in the upper-right corner. Since $Y\backslash\mathcal{D}$ has an initial object, $B(Y\backslash\mathcal{D})$ is contractible, and the fibre identification follows. □

8.3. Additivity with the universal pushout–pullback diagrams

Let $\mathcal{E}(\mathcal{M})$ be the exact category of short exact sequences in an exact category $\mathcal{M}$, with functors s, t, q to source, middle term, and quotient. Theorem 7.14 says that $(s,q) : Q\mathcal{E}(\mathcal{M}) \to Q\mathcal{M} \times Q\mathcal{M}$ is a homotopy equivalence. The key point is that, over a fixed pair (M,N), every extension can be reduced by universal constructions to the split extension.

Proposition 8.3 (Full fibrewise proof of additivity)**.** *For each pair $(M, N) \in \mathcal{M} \times \mathcal{M}$, the comma category $(s, q)/(M, N)$ is contractible. Hence (s, q) is a homotopy equivalence by Theorem A.*

Proof. An object of $(s, q)/(M, N)$ consists of an exact sequence

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

together with roofs in $Q\mathcal{M}$

$$A \xleftarrow{\alpha} U \xrightarrow{u} M, \qquad C \xleftarrow{\beta} V \xrightarrow{v} N,$$

where α and β are admissible epimorphisms and u, v are admissible monomorphisms.

We first force the left comparison to be represented by an admissible epimorphism $A \twoheadrightarrow M$. Pull back the extension along $\alpha : U \twoheadrightarrow A$ and then push out along $u : U \hookrightarrow M$:

$$\begin{tikzcd}
0 \arrow[r] & U \arrow[r] \arrow[d, "u"'] & B \times_A U \arrow[r] \arrow[d] & C \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & M \arrow[r] & B_1 \arrow[r] & C \arrow[r] & 0.
\end{tikzcd}$$

The bottom row defines an object of $(s, q)/(M, N)$ equipped with a morphism to the original object. This construction is functorial and left adjoint to the inclusion of the full subcategory $\mathcal{C}_1 \subseteq (s, q)/(M, N)$ consisting of those objects whose source map is literally represented by an admissible epimorphism $A \twoheadrightarrow M$. Hence $B\mathcal{C}_1 \to B((s, q)/(M, N))$ is a homotopy equivalence.

Next we force the right comparison to be represented by an admissible monomorphism $C \hookrightarrow N$. Starting from an object of $\mathcal{C}_1$, pull back along $\beta : V \twoheadrightarrow C$ and then push out along $v : V \hookrightarrow N$:

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[d, equal] & B \times_C V \arrow[r] \arrow[d] & V \arrow[r] \arrow[d, "v"] & 0 \\
0 \arrow[r] & A \arrow[r] & B_2 \arrow[r] & N \arrow[r] & 0.
\end{tikzcd}$$

This defines a functor left adjoint to the inclusion of the full subcategory $\mathcal{C}_2 \subseteq \mathcal{C}_1$ consisting of those objects for which the quotient map is literally represented by an admissible monomorphism $C \hookrightarrow N$. Again the inclusion $\mathcal{C}_2 \hookrightarrow \mathcal{C}_1$ is a homotopy equivalence.

It therefore suffices to prove that $\mathcal{C}_2$ is contractible. In $\mathcal{C}_2$ an object is an exact sequence together with an admissible epimorphism $A \twoheadrightarrow M$ and an admissible monomorphism $C \hookrightarrow N$. The split sequence

$$0 \longrightarrow M \longrightarrow M \oplus N \longrightarrow N \longrightarrow 0$$

belongs to $\mathcal{C}_2$. Given any other object of $\mathcal{C}_2$, the maps $A \twoheadrightarrow M$ and $C \hookrightarrow N$ determine a unique morphism of exact sequences from the split sequence into it:

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r] \arrow[d] & M \oplus N \arrow[r] \arrow[d, dashed] & N \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & A \arrow[r] & B \arrow[r] & C \arrow[r] & 0.
\end{tikzcd}$$

The left vertical arrow is the chosen admissible epimorphism $A \twoheadrightarrow M$ read in the source direction of $Q\mathcal{M}$, the right vertical arrow is the chosen admissible monomorphism $C \hookrightarrow N$, and the middle arrow is forced by exactness and the universal property of biproducts. Thus the split sequence is an initial object of $\mathcal{C}_2$. Therefore $\mathcal{C}_2$ is contractible, and so is $(s, q)/(M, N)$. $\square$

8.4. Resolution in the one-step case, with the comparison diagram

The conceptual heart of the resolution theorem is already visible in the case where every object M of $\mathcal{M}$ admits a one-step resolution by objects of $\mathcal{P}$.

Proposition 8.4 (One-step resolution)**.** *Assume every $M \in \mathcal{M}$ fits into an exact sequence*

$$0 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

with $P_0, P_1 \in \mathcal{P}$, and assume $\mathcal{P}$ is extension-closed. Then the inclusion $Q\mathcal{P} \hookrightarrow Q\mathcal{M}$ is a homotopy equivalence.

Proof. Let $f : Q\mathcal{P} \to Q\mathcal{M}$ be the inclusion. By Theorem A it is enough to show that f/M is contractible for each $M \in \mathcal{M}$.

Fix once and for all a resolution

$$\varepsilon_M : 0 \longrightarrow P_1(M) \longrightarrow P_0(M) \xrightarrow{\pi_M} M \longrightarrow 0.$$

Let $\mathcal{R}(M)$ be the full subcategory of f/M whose objects are roofs represented by exact sequences

$$0 \longrightarrow P' \longrightarrow P \xrightarrow{\rho} M \longrightarrow 0 \qquad (P, P' \in \mathcal{P}).$$

Every object $P \to M$ of f/M admits such a replacement: represent it by a roof $P \xleftarrow{\alpha} U \xrightarrow{i} M$ with α an admissible epimorphism. Pull back the fixed resolution ε_M along $i : U \hookrightarrow M$, obtaining

$$\begin{tikzcd}
0 \arrow[r] & P_1(M) \arrow[r] \arrow[d, equal] & P_0(M) \times_M U \arrow[r] \arrow[d] & U \arrow[r] \arrow[d, "\alpha"] & 0 \\
0 \arrow[r] & P_1(M) \arrow[r] & P_0(M) \arrow[r, "\pi_M"] & M \arrow[r] & 0.
\end{tikzcd}$$

Since $\mathcal{P}$ is closed under extensions and kernels of admissible epimorphisms occurring in the resolution hypothesis, the middle term belongs to $\mathcal{P}$. This gives

a functor from f/M to $\mathcal{R}(M)$ left adjoint to the inclusion; hence $B\mathcal{R}(M) \simeq B(f/M)$.

It remains to show that $\mathcal{R}(M)$ has an initial object, namely the chosen resolution ε_M. Let

$$0 \longrightarrow P' \longrightarrow P \xrightarrow{\rho} M \longrightarrow 0$$

be any object of $\mathcal{R}(M)$. Because ρ is an admissible epimorphism, the map $\pi_M : P_0(M) \to M$ lifts to a morphism $\lambda : P_0(M) \to P$. Since $P_1(M) = \ker(\pi_M)$ and $P' = \ker(\rho)$, exactness forces λ to carry $P_1(M)$ into P', giving a morphism of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & P_1(M) \arrow[r] \arrow[d] & P_0(M) \arrow[r, "\pi_M"] \arrow[d, "\lambda"] & M \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & P' \arrow[r] & P \arrow[r, "\rho"] & M \arrow[r] & 0.
\end{tikzcd}$$

This is precisely a morphism from ε_M to the given object of $\mathcal{R}(M)$. Uniqueness follows because any two such lifts differ by a map $P_0(M) \to P'$ vanishing on $P_1(M)$ and hence on the quotient M, so they represent the same morphism in the comma category. Thus ε_M is initial. Consequently $\mathcal{R}(M)$ and therefore f/M are contractible. □

The full resolution theorem follows by applying Proposition 8.4 to the filtration of subcategories $\mathcal{P}_n$ of objects admitting a $\mathcal{P}$-resolution of length at most n and then passing inductively from $\mathcal{P}_n$ to $\mathcal{P}_{n+1}$.

8.5. Dévissage with the layer poset written explicitly

The dévissage proof is one of the cleanest places in Quillen's paper where a comma category collapses to a combinatorial ordered set.

Proposition 8.5 (Proof of dévissage)**.** *With the hypotheses of Theorem 7.19, the inclusion $Q\mathcal{B} \to Q\mathcal{A}$ is a homotopy equivalence.*

Proof. Let $f : Q\mathcal{B} \to Q\mathcal{A}$ be the inclusion. By Theorem A it suffices to prove that f/M is contractible for each $M \in \mathcal{A}$.

Step 1: identify f/M with a layer poset. Let $J(M)$ be the ordered set of pairs (M_0, M_1) of subobjects of M with $M_0 \subseteq M_1 \subseteq M$ and $M_1/M_0 \in \mathcal{B}$, ordered by

$$(M_0, M_1) \leq (M_0', M_1') \iff M_0' \subseteq M_0 \subseteq M_1 \subseteq M_1'.$$

Given an object of f/M, represented by a roof

$$N \xleftarrow{p} U \xrightarrow{i} M \qquad (N \in \mathcal{B}),$$

with p an admissible epimorphism and i an admissible monomorphism, define

$$M_0 := i(\ker p), \qquad M_1 := i(U) \subseteq M.$$

Because $U/\ker p \cong N \in \mathcal{B}$, the pair (M_0, M_1) lies in $J(M)$. Conversely, from a layer (M_0, M_1) we obtain a roof

$$M_1/M_0 \xleftarrow{\pi} M_1 \xrightarrow{\iota} M,$$

where π is the quotient map and ι is the inclusion. These two assignments are inverse on objects and on morphisms, so f/M is equivalent to the ordered set $J(M)$.

Step 2: compare $J(M)$ with the layer poset of a subobject. Suppose $M' \subseteq M$ and $M/M' \in \mathcal{B}$. Let $i : J(M') \hookrightarrow J(M)$ be the obvious inclusion. Define functors

$$r : J(M) \to J(M'), \qquad r(M_0, M_1) = (M_0 \cap M',\, M_1 \cap M'),$$

$$s : J(M) \to J(M), \qquad s(M_0, M_1) = (M_0 \cap M',\, M_1).$$

These are well-defined because there are canonical inclusions

$$\frac{M_1 \cap M'}{M_0 \cap M'} \hookrightarrow \frac{M_1}{M_0 \cap M'} \hookrightarrow \frac{M_1}{M_0} \times \frac{M'}{M_0 \cap M'}$$

and the right-hand side lies in $\mathcal{B}$ since $\mathcal{B}$ is closed under subobjects, quotients, and finite products. Moreover one has

$$r \circ i = \mathrm{id}_{J(M')}$$

and natural transformations of functors on the ordered set $J(M)$,

$$i \circ r \Longrightarrow s \Longleftarrow \mathrm{id}_{J(M)},$$

represented objectwise by the inequalities

$$(M_0 \cap M', M_1 \cap M') \leq (M_0 \cap M', M_1) \geq (M_0, M_1).$$

By Proposition 2 of Quillen's Section 1, such natural transformations induce homotopies of classifying spaces; hence Bi is a homotopy equivalence.

Step 3: induct along a filtration. Choose a filtration

$$0 = M^{(0)} \subset M^{(1)} \subset \cdots \subset M^{(n)} = M$$

with successive quotients in $\mathcal{B}$. Applying Step 2 repeatedly to the inclusions $M^{(k-1)} \subset M^{(k)}$ shows that

$$BJ(M^{(0)}) \xrightarrow{\sim} BJ(M^{(1)}) \xrightarrow{\sim} \cdots \xrightarrow{\sim} BJ(M).$$

Since $J(M^{(0)}) = J(0)$ is a point, $BJ(M)$ is contractible. Therefore $B(f/M)$ is contractible for every M, and Theorem A implies that $Q\mathcal{B} \to Q\mathcal{A}$ is a homotopy equivalence. $\square$

8.6. The final step in localisation as a diagram chase in Theorem B

The reductions in Subsection 5.4 already prove the two hard ingredients of localisation: the equivalence $Q\mathcal{B} \simeq 0\backslash Qs$ and the base-change equivalences on all comma categories of Qs. Once those are in place, the last step is completely formal and deserves to be written as one clean diagram.

Proposition 8.6 (Localisation from Theorem B)**.** *Let $e : \mathcal{B} \to \mathcal{A}$ be the inclusion of a Serre subcategory and $s : \mathcal{A} \to \mathcal{A}/\mathcal{B}$ the quotient functor. Assume:*

(i) *the factorisation $Qe : Q\mathcal{B} \to 0\backslash Qs \to Q\mathcal{A}$ identifies $Q\mathcal{B} \to 0\backslash Qs$ as a homotopy equivalence;*

(ii) *for every morphism $u : V \to V'$ in $Q(\mathcal{A}/\mathcal{B})$, the induced base-change functor $u^* : V'\backslash Qs \to V\backslash Qs$ is a homotopy equivalence.*

Then there is a homotopy fibration

$$BQ\mathcal{B} \longrightarrow BQ\mathcal{A} \longrightarrow BQ(\mathcal{A}/\mathcal{B}),$$

and hence Quillen's localization long exact sequence.

Proof. Apply Theorem B to the functor $Qs : Q\mathcal{A} \to Q(\mathcal{A}/\mathcal{B})$ and to the distinguished object $0 \in Q(\mathcal{A}/\mathcal{B})$. Hypothesis (ii) is exactly the base-change hypothesis of Theorem B. Therefore the square

$$\begin{tikzcd}
0\backslash Qs \arrow[r] \arrow[d] & Q\mathcal{A} \arrow[d, "Qs"] \\
0\backslash Q(\mathcal{A}/\mathcal{B}) \arrow[r] & Q(\mathcal{A}/\mathcal{B})
\end{tikzcd}$$

becomes homotopy-cartesian after applying classifying spaces. But $0\backslash Q(\mathcal{A}/\mathcal{B})$ has initial object $(0, \mathrm{id}_0)$, hence contractible. Thus the upper-left corner is a model for the homotopy fibre of $BQ\mathcal{A} \to BQ(\mathcal{A}/\mathcal{B})$ over the vertex 0. By hypothesis (i), the comparison functor $Q\mathcal{B} \to 0\backslash Qs$ is a homotopy equivalence, so the fibre is also $BQ\mathcal{B}$. Consequently we obtain a homotopy fibration sequence

$$BQ\mathcal{B} \longrightarrow BQ\mathcal{A} \longrightarrow BQ(\mathcal{A}/\mathcal{B}).$$

Passing to homotopy groups and using Quillen's convention $K_i(_) = \pi_{i+1}(BQ(_))$ gives the exact sequence

$$\cdots \to K_{i+1}(\mathcal{A}/\mathcal{B}) \to K_i(\mathcal{B}) \to K_i(\mathcal{A}) \to K_i(\mathcal{A}/\mathcal{B}) \to \cdots .$$

This is precisely the localization theorem. □

Remark 8.7. The point of Proposition 8.6 is that once the comma categories have been understood, localisation is not mysterious at all: it is simply Theorem B applied to the quotient functor. What makes the theorem hard is not the final passage to the long exact sequence, but the prior categorical reductions that identify the fibre and verify the base-change hypothesis.

9. Waldhausen's $S_\bullet$-Construction

9.1. Categories with cofibrations and weak equivalences

Definition 9.1 (Waldhausen [Wa85]). A **Waldhausen category** (or *category with cofibrations and weak equivalences*) is a category $\mathcal{C}$ with a zero object $*$ and two distinguished subcategories:

- $\mathrm{co}\mathcal{C} \subseteq \mathcal{C}$: **cofibrations** (written $A \hookrightarrow B$),
- $w\mathcal{C} \subseteq \mathcal{C}$: **weak equivalences** (written $A \xrightarrow{\sim} B$),

satisfying:

(W1) All isomorphisms are cofibrations and weak equivalences.

(W2) The map $* \to A$ is a cofibration for every A.

(W3) Cofibrations are closed under pushout: if $A \hookrightarrow B$ is a cofibration and $A \to C$ is any map, then $C \to B \cup_A C$ is a cofibration.

(W4) The **gluing lemma**: if

$$\begin{tikzcd} A \arrow[d, "\sim"'] & B \arrow[l, "\sim"'] \arrow[r] \arrow[d, "\sim"'] & C \arrow[d, "\sim"] \\ A' & B' \arrow[l] \arrow[r] & C' \end{tikzcd}$$

has all vertical maps weak equivalences and horizontal maps cofibrations, then $B \cup_A C \xrightarrow{\sim} B' \cup_{A'} C'$ is a weak equivalence.

Example 9.2. Any exact category $\mathcal{C}$ becomes a Waldhausen category by taking cofibrations to be admissible monomorphisms and weak equivalences to be isomorphisms. The $S_\bullet$-construction then reproduces Quillen's K-groups.

Example 9.3. Let R be a ring. The category $\mathrm{Ch}^b(\mathbf{Proj}(R))$ of bounded chain complexes of finitely generated projective R-modules is a Waldhausen category with cofibrations = degree-wise split monomorphisms and weak equivalences = quasi-isomorphisms. This is the starting point for the K-theory of derived categories.

9.2. The $S_\bullet$-construction

Definition 9.4. Let $\mathbf{Ar}[n]$ denote the category with objects (i,j), $0 \le i \le j \le n$, and a unique morphism $(i,j) \to (i',j')$ when $i' \le i$ and $j \le j'$.

For a Waldhausen category $\mathcal{C}$, define the simplicial category $S_\bullet\mathcal{C}$ by:

$$S_n\mathcal{C} := \mathrm{Fun}_{\mathrm{cof}}(\mathbf{Ar}[n], \mathcal{C}),$$

the full subcategory of functors $A : \mathbf{Ar}[n] \to \mathcal{C}$ (objects A_{ij} for $0 \le i \le j \le n$) such that:

- $A_{ii} = *$ for all i;
- each $A_{ij} \hookrightarrow A_{ij'}$ (for $j \leq j'$) is a cofibration;
- each square $A_{ij} \hookrightarrow A_{ij'}, A_{i'j} \hookrightarrow A_{i'j'}$ (for $i \leq i'$) is a pushout (so $A_{i'j'} \cong A_{ij'} \cup_{A_{ij}} A_{i'j}$).

Face and degeneracy maps are given by restriction and diagonal-insertion along $\mathbf{Ar}[n-1] \to \mathbf{Ar}[n]$.

Definition 9.5. Define

$$|w\mathcal{C}| := |\mathrm{N}(w\mathcal{C})|, \quad |wS_\bullet\mathcal{C}| := |n \mapsto \mathrm{N}(wS_n\mathcal{C})|,$$

and the **K-theory space** of $\mathcal{C}$ as

$$\mathrm{K}(\mathcal{C}) := \Omega|wS_\bullet\mathcal{C}|.$$

Set $\mathrm{K}_n(\mathcal{C}) := \pi_n(\mathrm{K}(\mathcal{C}))$.

Theorem 9.6 (Waldhausen [Wa85])**.** *For $\mathcal{C}$ a Waldhausen category, $|wS_\bullet\mathcal{C}|$ is a loop space delooping: $|wS_\bullet\mathcal{C}| \simeq \Omega|wS_\bullet S_\bullet\mathcal{C}|$, so that $\mathrm{K}(\mathcal{C}) = \Omega|wS_\bullet\mathcal{C}|$ is an infinite loop space and the K-groups form an Ω-spectrum.*

Proof. The simplicial space $n \mapsto wS_n\mathcal{C}$ is a special Γ-space in the sense of Segal [Se74]: the Segal maps $wS_n\mathcal{C} \to (wS_1\mathcal{C})^n$ given by the inclusions of each $(i-1, i)$-piece are weak equivalences. (This is the content of the *Extension Lemma*, which states that a cofibration sequence $A \hookrightarrow B \twoheadrightarrow B/A$ is equivalent data to an object A, an object $C = B/A$, and the cofibre square.) By Segal's theory, a special Γ-space gives a homotopy-commutative group-like H-space and hence an infinite loop space. The explicit delooping is $|wS_\bullet\mathcal{C}| \simeq \Omega|wS_\bullet S_\bullet\mathcal{C}|$. □

9.3. Additivity theorem

Theorem 9.7 (Additivity, Waldhausen [Wa85])**.** *Let $\mathcal{C}, \mathcal{D}$ be Waldhausen categories and let $0 \to F' \to F \to F'' \to 0$ be an exact sequence of exact functors $\mathcal{C} \to \mathcal{D}$ (i.e. $F(A) \simeq F'(A) \cup_{F'(A)} F''(A)$ naturally). Then*

$$F_* = F'_* + F''_* : \mathrm{K}_*(\mathcal{C}) \to \mathrm{K}_*(\mathcal{D}).$$

Proof. The proof proceeds via the *Additivity Lemma* at the level of the $S_\bullet$-construction. One constructs a Waldhausen category $E(\mathcal{D})$ whose objects are cofibration sequences $A \hookrightarrow B \twoheadrightarrow C$ in $\mathcal{D}$, with natural functors $s, q : E(\mathcal{D}) \to \mathcal{D}$ (source and quotient) and $t : E(\mathcal{D}) \to \mathcal{D}$ (total). The claim reduces to showing that $(s, q) : E(\mathcal{D}) \to \mathcal{D} \times \mathcal{D}$ induces an equivalence on K-theory, which follows from a direct application of the $S_\bullet$-construction to the diagram and the observation that the simplicial space $n \mapsto wS_nE(\mathcal{D})$ splits as a product. □

Remark 9.8 (Why Waldhausen's machine is broader). Quillen's exact categories are already flexible enough for rings and many abelian settings, but derived and homotopical phenomena demand a framework in which weak equivalences

are not just isomorphisms. Waldhausen's formalism accomplishes exactly this: it replaces literal exactness by cofibration sequences and replaces strict equality by weak equivalence. This is why perfect complexes, spaces, and eventually stable ∞-categories all fit naturally into the same K-theoretic picture.

9.4. Approximation theorem and comparison with Quillen

Theorem 9.9 (Approximation, Waldhausen [Wa85])**.** *Let $F : \mathcal{A} \to \mathcal{B}$ be an exact functor of Waldhausen categories. Suppose:*

(i) F induces an equivalence on homotopy categories: $\operatorname{ho} F : \operatorname{ho} \mathcal{A} \xrightarrow{\sim} \operatorname{ho} \mathcal{B}$.

(ii) (Approximation property) *For every $A \in \mathcal{A}$, $B \in \mathcal{B}$ and map $FA \to B$, there exists a morphism $A \to A'$ in $\mathcal{A}$ and a weak equivalence $FA' \xrightarrow{\sim} B$ such that the composite $FA \to FA' \simeq B$ is the given map.*

Then F induces an equivalence $\mathrm{K}(\mathcal{A}) \simeq \mathrm{K}(\mathcal{B})$.

Corollary 9.10. *For a ring R, the exact category* $\mathbf{Proj}(R)$ *and the Waldhausen category* $\mathrm{Ch}^b(\mathbf{Proj}(R))$ *with quasi-isomorphisms have the same K-theory. More generally, this is the content of the Gillet–Waldhausen comparison theorem: passing from an exact category to bounded complexes does not change K-theory.*

Proof. One does *not* argue that every bounded complex of projectives is quasi-isomorphic to a complex concentrated in degree zero; that would be false in general. Instead, one applies the Gillet–Waldhausen theorem, whose proof uses the approximation theorem together with a careful analysis of acyclic complexes and mapping cylinders. The point is that the derived enlargement from objects to bounded complexes introduces no new K-theoretic content: it only makes homological constructions functorial and flexible enough for localisation and descent arguments. □

10. Thomason–Trobaugh: K-Theory of Schemes

The correct place of Thomason–Trobaugh in the story is *after* Quillen's formal and scheme-theoretic arguments and *after* Waldhausen's homotopical viewpoint. Quillen's Q-construction gives localisation for coherent sheaves and regular situations; Waldhausen explains how one should do K-theory in categories with weak equivalences; Thomason and Trobaugh then combine these insights to show that perfect complexes, rather than vector bundles alone, carry the right localisation and descent formalism for general schemes. Read in this order, Thomason's work is not an appendix to Quillen, but the point at which Quillen's geometric calculations acquire the flexibility needed for modern algebraic geometry.

10.1. Perfect complexes and the derived category

Let X be a quasi-compact quasi-separated (qcqs) scheme.

Definition 10.1. A complex $\mathcal{F}^\bullet$ of $\mathcal{O}_X$-modules is **perfect** if locally on X it is quasi-isomorphic to a bounded complex of locally free sheaves of finite rank. Denote by $\mathrm{Perf}(X)$ the full triangulated subcategory of $D(\mathcal{O}_X\text{-}\mathbf{Mod})$ consisting of perfect complexes.

$\mathrm{Perf}(X)$ is a Waldhausen category with:

- cofibrations = degree-wise split monomorphisms (as a dg-category, or equivalently strict monomorphisms of complexes of injective resolutions),
- weak equivalences = quasi-isomorphisms.

Definition 10.2. The **K-theory of perfect complexes** on X is

$$\mathrm{K}(X) := \mathrm{K}(\mathrm{Perf}(X)),$$

defined via Waldhausen's $S_\bullet$-construction.

Theorem 10.3 (Thomason–Trobaugh [TT90])**.** *For a scheme X that is qcqs and has an ample family of line bundles (e.g. quasi-projective over an affine scheme), the natural map*

$$\mathrm{K}(\mathrm{Vect}(X)) \to \mathrm{K}(\mathrm{Perf}(X))$$

is a homotopy equivalence.

Proof. By the approximation theorem (theorem 9.9) it suffices to show: (i) every perfect complex on X is quasi-isomorphic to a bounded complex of locally free sheaves (this uses the ample family hypothesis), and (ii) the approximation property holds (a standard dg-categorical argument using truncations and the fact that locally free sheaves are a generating class).

The conceptual point is that vector bundles are often too rigid to behave well under localisation, whereas perfect complexes are flexible enough to glue and restrict correctly. The theorem says that on a sufficiently nice scheme this extra flexibility costs no K-theoretic information: the passage from bundles to perfect complexes is a change of language, not a change of invariant. □

10.2. The Thomason–Trobaugh localisation theorem

Theorem 10.4 (Localisation, Thomason–Trobaugh [TT90])**.** *Let $i : Z \hookrightarrow X$ be a closed immersion and $j : U \hookrightarrow X$ the open complement. There is a homotopy fibration sequence of spectra*

$$\mathrm{K}(Z) \xrightarrow{i_*} \mathrm{K}(X) \xrightarrow{j^*} \mathrm{K}(U)$$

and hence a long exact sequence

$$\cdots \to \mathrm{K}_n(Z) \xrightarrow{i_*} \mathrm{K}_n(X) \xrightarrow{j^*} \mathrm{K}_n(U) \xrightarrow{\partial} \mathrm{K}_{n-1}(Z) \to \cdots .$$

Proof. The key observation is that $j^* : \mathrm{Perf}(X) \to \mathrm{Perf}(U)$ is exact and surjective on quasi-isomorphism classes (every perfect complex on U extends to X using the ample family). The kernel $\ker(j^*)$ consists of perfect complexes on X supported scheme-theoretically on Z, i.e. those in the full subcategory

$$\mathrm{Perf}_Z(X) := \{P \in \mathrm{Perf}(X) \mid j^*P \simeq 0\}.$$

Step 1: Show $i_* : \mathrm{Perf}(Z) \to \mathrm{Perf}_Z(X)$ is an equivalence of triangulated categories. (For i a regular closed immersion, this is a dévissage using the Koszul complex on the ideal sheaf $\mathcal{I}_Z \subset \mathcal{O}_X$.)

Step 2: The sequence $\mathrm{Perf}_Z(X) \hookrightarrow \mathrm{Perf}(X) \xrightarrow{j^*} \mathrm{Perf}(U)$ is a *Thomason localisation sequence*: there is a Waldhausen exact functor from a model for $\mathrm{Perf}(U)$ back to $\mathrm{Perf}(X)$ (a splitting up to homotopy) using the ample line bundle.

Step 3: The Waldhausen additivity theorem (theorem 9.7) and the fibration theorem (if $F : \mathcal{A} \to \mathcal{B}$ is an exact functor with $\mathrm{K}(\ker F) \simeq \mathrm{K}(\mathcal{A})$ then $\mathrm{K}(\mathcal{B}) \simeq *$) give the homotopy fibration. This is the *Fibration Theorem* of Waldhausen:

$$\mathrm{K}(\mathcal{A}') \to \mathrm{K}(\mathcal{A}) \to \mathrm{K}(\mathcal{A}/\mathcal{A}')$$

is a fibration sequence for $\mathcal{A}' \subseteq \mathcal{A}$ a full Waldhausen subcategory with the right saturation and extension properties. □

Remark 10.5 (Why perfect complexes are the correct geometric objects). A vector bundle is locally free and therefore easy to understand pointwise, but cokernels, extensions, and support conditions naturally produce complexes rather than bundles. Perfect complexes form the smallest stable class containing vector bundles and closed under the operations that geometry forces upon us. Thomason–Trobaugh's contribution is to show that once this passage is made, localisation and descent become genuinely natural statements.

The paper of Thomason and Trobaugh is often remembered for the slogan that one should replace vector bundles by perfect complexes. That slogan is accurate but not yet sufficiently precise. What their localisation theorem really says is that support conditions on perfect complexes behave functorially enough to produce fibre sequences of K-theory spectra, and that this mechanism propagates through open covers, Nisnevich squares, and coefficient systems. In this way the geometric applications cease to be isolated tricks and become formal consequences of a robust exact structure on the derived category.

Proposition 10.6 (A technical core of Thomason–Trobaugh localisation)**.** *Let X be qcqs with an ample family of line bundles, let $j : U \hookrightarrow X$ be an open immersion with closed complement $i : Z \hookrightarrow X$, and let* $\mathrm{Perf}_Z(X) = \{P \in \mathrm{Perf}(X) \mid j^*P \simeq 0\}$*. Then:*

(i) $\mathrm{Perf}_Z(X)$ *is a thick Waldhausen subcategory of* $\mathrm{Perf}(X)$*;*

(ii) *the restriction functor* $j^* : \mathrm{Perf}(X) \to \mathrm{Perf}(U)$ *induces an exact functor of Waldhausen categories whose kernel is* $\mathrm{Perf}_Z(X)$*;*

(iii) *after replacing* $\mathrm{Perf}(U)$ *by a quasi-equivalent strict model, the induced map on* K*-theory fits into a homotopy fibre sequence*

$$\mathrm{K}(\mathrm{Perf}_Z(X)) \longrightarrow \mathrm{K}(X) \xrightarrow{j^*} \mathrm{K}(U).$$

Proof. The first point is formal: if two terms in a distinguished triangle of perfect complexes are supported on Z, then so is the third, and the same is true for direct summands. For the second point, exactness of j^* comes from exactness on complexes, while the description of the kernel is the definition of support in the triangulated category of perfect complexes.

The third point is where Thomason–Trobaugh's innovations enter. One chooses a strict model of perfect complexes for which cofibrations and weak equivalences satisfy Waldhausen's axioms, and one shows that restriction to U is essentially surjective after passing to the derived category. The ample-family hypothesis is used precisely here: it gives enough global line bundles to extend perfect complexes from U to X and to strictify those extensions. Once essential surjectivity and identification of the kernel are in place, Waldhausen's fibration theorem applies and yields the asserted homotopy fibre sequence. The hardest strictification and extension steps are exactly those supplied in [TT90, Sections 1, 7, 8]. □

Corollary 10.7 (Mayer–Vietoris from localisation)**.** *If* $X = U \cup V$ *is a qcqs scheme with an ample family of line bundles, then there is a homotopy cartesian square*

$$\begin{tikzcd} \mathrm{K}(X) \arrow[r] \arrow[d] & \mathrm{K}(U) \arrow[d] \\ \mathrm{K}(V) \arrow[r] & \mathrm{K}(U \cap V). \end{tikzcd}$$

Proof. Apply Proposition 10.6 twice, first to the pair (X, U) and then to the pair $(V, U \cap V)$. The support subcategory for $X \setminus U$ can be identified with the support subcategory for $V \setminus (U \cap V)$ after restriction to V, because both describe perfect complexes supported on the same closed subset. One therefore obtains a morphism of fibre sequences with identical left-hand term, and the resulting 3×3 comparison of fibres gives the stated homotopy-cartesian square. This is the geometric heart of the Mayer–Vietoris statements emphasized by Thomason–Trobaugh. □

Remark 10.8 (Why Neeman enters the story)*.* The localisation theorem of Thomason–Trobaugh has a second life inside triangulated category theory. Neeman showed that the compact objects in a compactly generated localisation sequence are controlled by exactly the same formal pattern that appears in K-theory localisation; this is the bridge from Thomason–Trobaugh to the classification of smashing localisations in derived categories. For these notes the philosophical point is that K-theory localisation is not merely a statement about groups or spectra: it is the shadow of a structural decomposition in the ambient triangulated world.

10.3. G-theory and the comparison for regular schemes

Definition 10.9. For a Noetherian scheme X, define

$$\mathrm{G}(X) := \mathrm{K}(\mathrm{Coh}(X)^{\mathrm{Wald}}),$$

where $\mathrm{Coh}(X)$ is viewed as a Waldhausen category with cofibrations = admissible monomorphisms and weak equivalences = isomorphisms, or equivalently via the abelian structure. Then $\mathrm{G}_n(X) = \pi_n(\mathrm{G}(X))$ is Quillen's G-theory.

Theorem 10.10 (Thomason [TT90])**.** *If X is a regular Noetherian scheme, then the natural map*

$$\mathrm{K}(X) \xrightarrow{\sim} \mathrm{G}(X)$$

is a homotopy equivalence.

Proof. There is a sequence of equivalences:

$$\begin{aligned} \mathrm{K}(X) &= \mathrm{K}(\mathrm{Perf}(X)) \\ &\simeq \mathrm{K}(D^b(\mathrm{Coh}(X))) && \text{(regularity: every coh. sheaf has finite proj. dim.)} \\ &\simeq \mathrm{G}(X) && \text{(devissage in the derived category).} \end{aligned}$$

The first equivalence uses the approximation theorem: since X is regular, every coherent sheaf has a finite resolution by locally free sheaves of bounded length (bounded by $\dim X$), so $D^b(\mathrm{Coh}(X)) \simeq D^b(\mathrm{Vect}(X)) = \mathrm{Perf}(X)$ as triangulated categories. The second uses the derived devissage: the natural inclusion $\mathrm{Coh}(X) \hookrightarrow D^b(\mathrm{Coh}(X))$ (as complexes in degree zero) induces an equivalence on K-theory, which is the statement that $\mathrm{K}(\mathcal{A}) \simeq \mathrm{K}(D^b(\mathcal{A}))$ for any abelian category $\mathcal{A}$ (proved using the approximation theorem and the Gillet–Waldhausen resolution theorem). □

10.4. Nisnevich descent and the Brown–Gersten spectral sequence

Definition 10.11. The **Nisnevich topology** on Sch/S has covers $\{f_i : U_i \to X\}$ consisting of étale morphisms such that for every (scheme-theoretic) point $x \in X$, there exists i and a point $u \in U_i$ with $f_i(u) = x$ and $\kappa(u) \xrightarrow{\sim} \kappa(x)$ (residue field isomorphism).

Theorem 10.12 (Thomason–Trobaugh Descent [TT90])**.** *The presheaf of K-theory spectra $X \mapsto \mathrm{K}(X)$ satisfies descent for the Nisnevich topology. That is, for any Nisnevich hypercovering $U_\bullet \to X$, the natural map*

$$\mathrm{K}(X) \xrightarrow{\sim} \operatorname{holim}_\Delta \mathrm{K}(U_\bullet)$$

is a homotopy equivalence.

Proof. By the localisation theorem (theorem 10.4) applied to elementary Nisnevich squares

$$\begin{tikzcd} V \arrow[r] \arrow[d] & U \arrow[d, "f"] \\ Z \arrow[r, "i"] & X \end{tikzcd}$$

(where $Z \hookrightarrow X$ is a closed immersion, $f : U \to X$ étale, and $f^{-1}(X \setminus Z) \xrightarrow{\sim} X \setminus Z$), the square of K-theory spectra is homotopy Cartesian. Since the Nisnevich topology is generated by such squares (Morel–Voevodsky), descent follows by induction on the complexity of the cover using the homotopy Cartesian squares and a standard descent argument for sheaves of spectra. □

Corollary 10.13 (Brown–Gersten spectral sequence)**.** *For a Noetherian scheme X of finite Krull dimension d, there is a strongly convergent spectral sequence*

$$E_2^{p,q} = H^p_{\mathrm{Nis}}(X, \mathcal{K}_{-q}) \Rightarrow \mathrm{K}_{-p-q}(X),$$

where $\mathcal{K}_n$ is the Nisnevich sheaf associated to $U \mapsto \mathrm{K}_n(U)$.

10.5. K-theory with finite coefficients and the telescope conjecture

Definition 10.14. For a prime ℓ and integer $\nu \geq 1$, set $\mathrm{K}_n(X; \mathbb{Z}/\ell^\nu) := \pi_n(\mathrm{K}(X) \wedge M(\mathbb{Z}/\ell^\nu))$, where $M(\mathbb{Z}/\ell^\nu)$ is the mod-ℓ^ν Moore spectrum.

Theorem 10.15 (Thomason [Th85])**.** *Let X be a regular Noetherian $\mathbb{Z}[1/\ell]$-scheme of finite Krull dimension. There is a descent spectral sequence*

$$E_2^{p,q} = H^p_{\text{ét}}(X, \mathbb{Z}/\ell^\nu(q/2)) \Rightarrow \mathrm{K}_{-p-q}(X; \mathbb{Z}/\ell^\nu)$$

relating étale cohomology and K-theory with finite coefficients. (Here $\mathbb{Z}/\ell^\nu(m)$ is the m-th twist of the constant sheaf.)

This spectral sequence is the algebraic analogue of the Atiyah–Hirzebruch spectral sequence. Its E_2 term is étale cohomology, which is often computable (Galois cohomology for fields, étale cohomology via comparison theorems).

Remark 10.16 (Beilinson–Lichtenbaum and Voevodsky). When $\ell \neq \mathrm{char}(k)$, Voevodsky's proof of the Bloch–Kato conjecture (now Voevodsky–Rost theorem) establishes that the spectral sequence of theorem 10.15 degenerates at E_2 rationally and the natural map from motivic cohomology to étale cohomology is an isomorphism modulo torsion, yielding the algebraic Atiyah–Hirzebruch spectral sequence

$$H^p_M(X, \mathbb{Z}(q)) \Rightarrow \mathrm{K}_{2q-p}(X).$$

11. Applications of Quillen's Theorems and Later Developments

The classical story developed in these notes reaches a natural culmination with Thomason–Trobaugh: by that point K-theory has become local on schemes, it has acquired a spectrum-level formulation, and it is flexible enough to be attached to perfect complexes rather than only to vector bundles. Modern work pushes each of these three themes further. First, one asks what remains true for singular objects and in negative degrees. Second, one asks for the exact scope of localisation and excision beyond the exact settings originally treated by Quillen and Thomason. Third, one asks whether K-theory itself can

be characterised by a universal property inside a larger motivic or categorical world.

A useful way to organise the landscape is to keep in mind the following slogan: *regular objects are controlled by vector bundles, singular objects are controlled by perfect complexes, and genuinely modern questions are controlled by descent, traces, and universal properties.* The purpose of this final section is not to reproduce the deepest recent theorems line by line, but to explain how they grow out of the material already developed above.

11.1. Gersten complexes, Bloch–Ogus, and the coniveau machine

One of the most fertile applications of Quillen's localisation theorem is the coniveau filtration on the K-theory of a regular scheme. The filtration by codimension of support, already visible in Quillen's Section 7, produces an exact couple and hence a spectral sequence whose first page is built from residue fields. Bloch and Ogus recast this mechanism in a far more conceptual language: given a cohomology theory with supports satisfying the expected exactness, homotopy, and excision axioms, one gets a Cousin-type complex of sheaves and a coniveau spectral sequence computing the theory from its local pieces.

Theorem 11.1 (Bloch–Ogus applied to algebraic K-theory)**.** *Let X be a smooth scheme over a field. Quillen's Gersten exactness on the local rings of X yields a flasque resolution of the Zariski sheaf $\mathcal{K}_n$ by the Gersten complex*

$$0 \to \mathcal{K}_n \to \bigoplus_{x \in X^{(0)}} i_{x*} K_n(\kappa(x)) \to \bigoplus_{x \in X^{(1)}} i_{x*} K_{n-1}(\kappa(x)) \to \cdots ,$$

and hence a coniveau spectral sequence whose E_2-page is Zariski cohomology of the sheaves $\mathcal{K}_q$.

Proof. Quillen's codimension filtration identifies the E_1-page with the direct sum over points of fixed codimension of the K-groups of their residue fields. Gersten exactness for regular local rings, proved in the equicharacteristic case by Quillen and extended in geometric settings by later work, shows that the associated Cousin complex is exact off degree 0. Sheafifying this exact complex produces a flasque resolution of $\mathcal{K}_n$, so its hypercohomology computes the Zariski cohomology of $\mathcal{K}_n$. The Bloch–Ogus formalism then identifies the resulting exact couple with the coniveau exact couple and gives the spectral sequence. The verification of the full Bloch–Ogus axioms is technical and is best taken from [BlOg74]; the point relevant here is that Quillen's localisation theorem is the input that makes the Cousin resolution exact. □

Corollary 11.2 (Bloch's formula revisited)**.** *For a smooth variety X over a field and $p \geq 0$, one has*

$$H^p_{\mathrm{Zar}}(X, \mathcal{K}_p) \cong \mathrm{CH}^p(X).$$

The significance of this result is hard to overstate. It says that Chow groups are not external companions to algebraic K-theory; they are recovered from the

coniveau filtration on K-theory itself. In this precise sense Quillen's localisation theorem already contains the seed of the later cycle-theoretic and motivic formalisms.

11.2. Totaro and the cycle-theoretic face of higher K-theory

After Bloch introduced higher Chow groups, a natural question was whether the pieces of K-theory predicted by the coniveau filtration could be described by algebraic cycles in a direct way. For a field, Totaro's theorem gives one of the cleanest answers: Milnor K-theory is the "linear" part of higher Chow groups.

Theorem 11.3 (Totaro)**.** *For every field F and every $n \geq 0$, there is a natural isomorphism*

$$K_n^M(F) \cong \mathrm{CH}^n(F, n).$$

Proof. The map is defined on symbols by sending $\{a_1, \ldots, a_n\}$ to the class of the graph cycle cut out by the coordinates a_i inside Bloch's cubical cycle complex. The Steinberg relation $\{a, 1-a\} = 0$ becomes rational equivalence of a very explicit curve, already visible in low degree, so the symbol map descends to Milnor K-theory. Surjectivity is proved by moving higher Chow cycles into a linear position and showing that every class is represented by a cycle of this restricted type. Injectivity requires the deeper comparison with the field case of higher Chow groups and the norm arguments developed around Suslin's and Nesterenko–Suslin's calculations. The geometric content, however, is already transparent: the simplest nontrivial part of algebraic K-theory is generated by very controlled cycle classes rather than by arbitrary algebraic cycles. The technical comparison is carried out in [To92]. □

This result explains why Milnor K-theory repeatedly appears at the first nontrivial layer of motivic spectral sequences. It is the part of Quillen K-theory that survives after one passes to symbols, residues, and the first coniveau quotients.

11.3. Voevodsky's insight: motivic cohomology as the conceptual shadow of Quillen's filtration

At this point one can see why Voevodsky's work changed the status of Quillen's ideas. Quillen had produced the coniveau filtration, its residue-theoretic E_1-page, and the exactness phenomena now called the Gersten property. Bloch had introduced higher Chow groups as a cycle complex extending classical Chow groups into a homological direction. Voevodsky's decisive insight was that these two lines of thought are not merely parallel. They belong to a single motivic framework in which higher Chow groups are the correct cohomology theory measuring the graded pieces of algebraic K-theory, and in which the formal exactness properties discovered by Quillen are explained by $\mathbb{A}^1$- homotopy invariance, Nisnevich descent, and transfers.

One may summarize the conceptual jump as follows. In Quillen's paper, the coniveau filtration is a filtration on K-theory whose quotients are described pointwise in terms of residue fields. In the motivic world, one packages those residue-theoretic pieces into a cohomology theory $H^i(X, \mathbf{Z}(n))$ represented in a triangulated category of motives. The identification with Bloch's cycle complex then shows that the graded pieces of the coniveau machine are not ad hoc correction terms but bona fide motivic invariants. This is the sense in which Voevodsky turns Quillen's spectral-sequence technology into a conceptual theory of *weights*, *transfers*, and *motivic cohomology*. See [MV99, Vo00].

Theorem 11.4 (Voevodsky's motivic interpretation of higher Chow groups)**.** *Let k be a perfect field and let X be a smooth quasi-projective k-scheme. Then motivic cohomology and higher Chow groups agree:*

$$H^p(X, \mathbf{Z}(q)) \cong \mathrm{CH}^q(X, 2q - p).$$

In particular,

$$H^{2q}(X, \mathbf{Z}(q)) \cong \mathrm{CH}^q(X), \qquad H^q(k, \mathbf{Z}(q)) \cong K_q^M(k).$$

Proof. The construction of the motivic complexes $\mathbf{Z}(q)$ is arranged so that they satisfy homotopy invariance, Nisnevich descent, and functoriality with transfers. On the other hand, Bloch's cubical cycle complexes $z^q(X, *)$ already produce a candidate for the weight-q cohomology of X. The comparison theorem identifies the hypercohomology of the motivic complex with the homology of Bloch's cycle complex. At the technical level one proves that both theories satisfy the same basic formal properties and then uses moving arguments to compare generators and relations. The essential geometric input is that cycles on $X \times \Delta^n$ can be moved into sufficiently transverse position for the cubical boundary operators to compute the correct homotopy invariant information. The precise comparison is proved in the motivic literature; for the present notes the point is that Voevodsky's framework upgrades Bloch's cycle complexes from a powerful construction to the intrinsic cohomology theory predicted by the Quillen–Bloch–Ogus machine. See [Bl86, MV99, Vo00]. □

Remark 11.5 (Why this matters for the story after Quillen)*.* The theorem above reorganizes several earlier facts into one picture.

1. Totaro's identification $K_n^M(F) \cong \mathrm{CH}^n(F, n)$ becomes the field-valued weight-n case of motivic cohomology.

2. Bloch's formula $H^p_{\mathrm{Zar}}(X, \mathcal{K}_p) \cong \mathrm{CH}^p(X)$ becomes the degree-$2p$ edge of a motivic description rather than an isolated coincidence.

3. The Gersten complexes arising from Quillen localisation become Cousin resolutions for motivic sheaves, so the Bloch–Ogus formalism is no longer a secondary afterthought: it is built into the structure of motivic cohomology itself.

From this point onward the phrase *after Quillen* means more than stronger localisation statements. It means that algebraic K-theory is being viewed inside a larger homotopy-theoretic universe where higher Chow groups, Milnor K-theory, and étale comparison theorems occupy the same conceptual diagram.

Remark 11.6 (Beilinson–Lichtenbaum, norm residues, and arithmetic consequences). The motivic interpretation also explains why Voevodsky's work on norm residues feeds back into the classical K-theoretic story. Once motivic cohomology with finite coefficients is identified with higher Chow groups, comparison with étale cohomology in the stable range becomes a statement about weights and residues. Voevodsky's proof of the norm residue isomorphism theorem therefore tells us that the Milnor-symbol layer appearing on Quillen's E_1-page already knows the correct mod-ℓ arithmetic information. This is one of the deepest reasons that coniveau, Gersten, and higher Chow groups remain central long after the original exact-category formalism has been established.

11.4. The motivic spectral sequence and the slice-filtration viewpoint

Voevodsky did not merely reinterpret Quillen's filtration after the fact. He proposed that the spectral sequence from motivic cohomology to algebraic K-theory should arise from a genuinely homotopy-theoretic tower inside the motivic stable homotopy category. In his 2002 note *A possible new approach to the motivic spectral sequence for algebraic K-theory*, the guiding idea is that the algebraic K-theory spectrum KGL should admit a motivic Postnikov-type filtration whose layers are Eilenberg–MacLane objects for motivic cohomology. In that form, Quillen's coniveau machine would cease to be an external filtration built from supports and would become the shadow, on homotopy groups, of an intrinsic tower in motivic homotopy theory. See [Vo02S, FS02, Le08, GP15].

There are really three stages in this picture.

1. Friedlander–Suslin construct a strongly convergent spectral sequence from motivic cohomology to K-theory by analysing the tower given by codimension of supports in the simplicial schemes $\Delta^\bullet \times X$.

2. Voevodsky observes that this tower should admit a conceptual reformulation inside the motivic stable homotopy category via the slice filtration of KGL.

3. Levine and, from a complementary angle, Garkusha–Panin show that the resulting towers are compatible with the homotopy coniveau tower and with Grayson's tower, so that the motivic spectral sequence is not an accident of one construction but the common output of several equivalent filtrations.

The importance of this point for the story after Quillen is hard to overstate. Quillen's original spectral sequence was already geometrically rich, but it was still built at the level of exact categories, supports, and localisation. The slice-filtration viewpoint says that these phenomena are organised by the ambient homotopy category itself. The filtration is no longer merely a tool

for computations: it becomes a structural feature of the motivic K-theory spectrum.

Theorem 11.7 (Motivic spectral sequence for K-theory)**.** *Let X be a smooth quasi-projective scheme over a field. There is a strongly convergent spectral sequence*

$$E_2^{p,q} = H^{p-q}(X, \mathbf{Z}(-q)) \Longrightarrow K_{-p-q}(X),$$

multiplicative and functorial in X. Under the identification

$$H^{p-q}(X, \mathbf{Z}(-q)) \cong \mathrm{CH}^{-q}(X, -p-q),$$

this is the spectral sequence relating higher Chow groups to algebraic K-theory.

Proof. The starting point is the tower of spectra obtained by filtering coherent sheaves on $\Delta^\bullet \times X$ by codimension of support. Friedlander and Suslin show that the exact couple attached to this tower yields a strongly convergent spectral sequence whose E_2-page is motivic cohomology. The identification of the E_2-term with higher Chow groups comes from the comparison between motivic cohomology and Bloch's cycle complexes. Multiplicativity is inherited from the pairings on the tower, and functoriality comes from pullback on the support conditions and on the associated cycle complexes. The deep inputs needed to identify the fibers with motivic cohomology are part of the Friedlander–Suslin package; once those identifications are granted, the spectral sequence is the usual exact-couple spectral sequence of a filtered object. See [FS02, Theorem 13.13 and Proposition 13.17]. □

Proposition 11.8 (Slice-theoretic reading of the motivic spectral sequence)**.** *Suppose one has a tower on the motivic K-theory spectrum KGL whose q-th slice is*

$$s_q(KGL) \simeq \Sigma^{2q,q} H\mathbf{Z}.$$

Then the exact couple of the slice tower yields a spectral sequence with

$$E_2^{p,q} \cong H^{p-q}(X, \mathbf{Z}(-q))$$

abutting to the homotopy groups represented by KGL, hence to algebraic K-theory.

Proof. For any filtered spectrum, the associated exact couple is obtained by applying the cohomological or homotopical functor represented by X to the tower and then passing to the long exact sequences of the successive fibers. In the slice tower of KGL, the successive quotients are by assumption the slices $s_q(KGL)$. Replacing each slice by $\Sigma^{2q,q} H\mathbf{Z}$, one identifies the E_1- or E_2-page (depending on indexing conventions) with motivic cohomology groups of the corresponding weights. The convergence target is the homotopy class represented by the inverse limit of the tower, which is KGL itself. Thus the spectral sequence produced by the slice tower has the same shape as the Friedlander–Suslin spectral sequence. The technical work lies in proving the slice identification and the comparison between the various towers; the formal passage from slices to the spectral sequence is exactly the standard exact-couple construction. □

Remark 11.9 (What later work clarified). Voevodsky's 2002 note states the slice-theoretic picture as a program rather than as a closed proof. The subsequent literature supplies the missing bridges.

1. Levine identifies the homotopy coniveau tower with Voevodsky's slice filtration and obtains a spectral sequence reminiscent of the Atiyah–Hirzebruch spectral sequence.

2. Garkusha and Panin show that the Grayson tower for algebraic K-theory agrees with the slice tower for KGL, thereby confirming that the spectral sequence obtained from Grayson's model is the same motivic spectral sequence coming from the slice tower.

3. In this way the original coniveau filtration of Quillen, the cycle-theoretic filtration of Friedlander–Suslin, and the motivic slice tower become different realizations of the same phenomenon.

This is one of the most persuasive examples of a general pattern in modern algebraic K-theory: a classical filtration first appears through exact-category geometry, is then reinterpreted by cycles and residues, and finally acquires its cleanest conceptual form inside motivic homotopy theory.

11.5. The homotopy coniveau tower as the bridge from supports to slices

The most delicate step in the passage from Quillen's filtration to Voevodsky's picture is not the identification of the E_2-page alone, but the recognition that the *entire filtered object* can be reconstructed inside motivic homotopy theory. Quillen's original tower is built from supports; Bloch and Friedlander–Suslin reinterpret the successive quotients through cycle complexes; Levine shows that, after the appropriate homotopy-invariant replacement, one really has a canonical tower attached to a cohomology theory on smooth schemes, and that for algebraic K-theory this tower agrees with Voevodsky's slice filtration. This is the bridge on which the modern theory walks from exact categories to motivic spectra.

For a presheaf of spectra E on smooth k-schemes, the homotopy coniveau tower is obtained by filtering $E(X)$ according to codimension of supports. Roughly speaking, the q-th stage $E^{(q)}(X)$ remembers only those classes that can be represented with support in codimension at least q, and the successive quotient $E^{(q/q+1)}(X)$ measures what is lost when one tightens the support condition from q to $q+1$. In Quillen's paper these quotients appear through residue fields on the E_1-page; in Levine's reformulation they are promoted to honest layers of a functorial tower.

Theorem 11.10 (Levine's bridge theorem, schematic form)**.** *Let k be a field and let E be a cohomology theory on smooth k-schemes satisfying the standard homotopy-invariance and excision hypotheses needed to construct the homotopy coniveau tower. Then:*

1. *the assignment $X \mapsto E^{(q)}(X)$ is functorial and yields a tower*

$$\cdots \to E^{(q+1)}(X) \to E^{(q)}(X) \to \cdots \to E^{(0)}(X) \simeq E(X);$$

2. *for algebraic K-theory, the layer $K^{(q/q+1)}(X)$ is represented by motivic cohomology in weight q;*

3. *after passage to the $\mathbb{P}^1$-stable motivic homotopy category, the homotopy coniveau tower identifies with Voevodsky's slice filtration.*

Proof. The first point comes from a systematic version of the support filtration already implicit in Quillen. One constructs a simplicial diagram of closed subsets of $X \times \Delta^n$ of codimension at least q, evaluates E with supports on that diagram, and then passes to the homotopy colimit. The nontrivial issue is functoriality: an arbitrary morphism of smooth schemes does not preserve support conditions on the nose. Levine resolves this by a refined moving argument, extending Chow's moving lemma so that the resulting spectra become strictly functorial over fields and functorial in the homotopy category more generally.

For $E = K$, the description of the layer $K^{(q/q+1)}$ is the motivic refinement of Quillen's old observation that codimension-q strata contribute residues of weight q. One compares the homotopy cofiber of successive support conditions with Bloch-type cycle objects and proves that the latter are Eilenberg–MacLane in the required range. This yields motivic cohomology as the representing spectrum for the layer. Finally, stabilization in the $\mathbb{P}^1$-direction shows that the coniveau tower is characterized by the same universal properties as Voevodsky's slice tower; the comparison theorem in Levine identifies them. The hard work is geometric—moving, supports, and localization—while the formal outcome is exactly the existence of a functorial filtered object whose layers are motivic. See [Le08]. □

Corollary 11.11 (What this means for algebraic K-theory)**.** *For algebraic K-theory on smooth schemes over a field, the classical support filtration of Quillen, the Friedlander–Suslin tower, the homotopy coniveau tower, and the slice filtration of KGL all give rise to the same motivic spectral sequence after the corresponding comparison maps are identified.*

Proof. Quillen provides the support filtration and the first residue-theoretic exact couple. Friedlander and Suslin identify a functorial spectral sequence whose E_2-page is motivic cohomology. Levine shows that the homotopy coniveau tower recovers the same layers for K-theory and that, after stabilization, it agrees with the slice tower. Garkusha–Panin then prove that the Grayson model produces the same motivic spectral sequence as the slice tower of KGL. Each step compares filtered objects rather than merely their abutments, so the resulting spectral sequences agree from the stage at which the exact couples become identified. See [FS02, Le08, GP15]. □

The comparison can be visualized as follows. Once Levine's bridge has been established, the geometric support tower and the motivic slice tower become

two presentations of the same filtered object:

$$\begin{tikzcd}
\cdots \arrow[r] & f_{q+1}KGL \arrow[r] \arrow[d, "\simeq"'] & f_qKGL \arrow[r] \arrow[d, "\simeq"'] & s_q(KGL) \arrow[d, dashed] \\
\cdots \arrow[r] & K^{(q+1)} \arrow[r] & K^{(q)} \arrow[r] & K^{(q/q+1)}
\end{tikzcd}$$

Applying $[\Sigma^\infty_{\mathbb{P}^1} X_+, -]$ to the top row, or evaluating the bottom row on X, produces the same exact couple. In that precise sense the motivic spectral sequence is not merely analogous to Quillen's coniveau machine; it is Quillen's machine lifted into the stable motivic world.

Theorem 11.12 (Zero slice, motivic cohomology, and module structure)**.** *In the $\mathbb{P}^1$-stable motivic homotopy category over a field, the zero slice of the motivic sphere spectrum is the motivic cohomology spectrum $H\mathbf{Z}$. Consequently, the slices of any $\mathbb{P}^1$-spectrum carry canonical $H\mathbf{Z}$-module structures. In particular, once the slices of KGL are identified as suspensions of $H\mathbf{Z}$, the motivic spectral sequence becomes an Atiyah–Hirzebruch-type spectral sequence with coefficients in motivic cohomology.*

Proof. Levine proves that the $\mathbb{P}^1$-stable homotopy coniveau tower agrees with Voevodsky's slice filtration and that the zeroth layer of the motivic sphere is motivic cohomology. Since the slice tower is functorial and the sphere is the tensor unit, the zeroth slice acts on every layer of every spectrum. Thus each slice acquires a canonical $H\mathbf{Z}$-module structure. For KGL this means that the successive quotients of the motivic filtration are not just abstract spectra but motivic Eilenberg– MacLane objects, so the associated spectral sequence has coefficients in motivic cohomology exactly as in a generalized Atiyah–Hirzebruch spectral sequence. The extra input needed to identify the q-th slice of KGL with $\Sigma^{2q,q}H\mathbf{Z}$ comes from the later slice-comparison literature, but the formal module-structure statement already follows from Levine's theorem on the zero slice. □

Remark 11.13 (Why the zero slice matters)*.* The slogan $s_0(\mathbf{1}) = H\mathbf{Z}$ is more than a convenient mnemonic. It says that motivic cohomology is not attached to algebraic K-theory from the outside; it is the coefficient theory that the slice filtration sees at the very first level of the motivic stable homotopy category. This is the structural reason one repeatedly encounters motivic cohomology as the first approximation to K-theory, why higher Chow groups control the first nontrivial layers of filtrations, and why residue morphisms in coniveau constructions look formally like boundary maps in Eilenberg–MacLane objects.

Remark 11.14 (Convergence and modern refinements)*.* One conceptual difficulty in Voevodsky's original proposal was convergence: a filtration by slices is useful only if the associated tower reconstructs the spectrum one started from. For the classical motivic spectral sequence this issue is handled in the geometric framework of Friedlander–Suslin for smooth schemes. Later work of Bachmann, Elmanto, and Østvær proves a derived variant of Voevodsky's convergence

conjecture for the slice spectral sequence in an étale setting and uses it to establish a Thomason-style étale descent theorem for the Bott-inverted motivic sphere. This does not replace the classical convergence results; rather, it shows that the slice-filtration philosophy continues to govern new descent phenomena far beyond the original range of Quillen's exact categories and Bloch's cycles.

11.6. From Quillen to Kerz–Strunk–Tamme and Weibel

Quillen's proof of Gersten exactness for regular local rings containing a field solved the geometric prototype of a much larger problem: how far can localisation and coniveau exactness survive beyond smooth or regular situations? The modern answer goes in two directions. First, one extends Gersten- type exactness and Cousin formalisms to broader cohomology theories. Second, one replaces exactness on the nose by pro-descent statements for blow-up squares, which are strong enough to control negative K-groups of singular schemes.

Theorem 11.15 (Kerz–Strunk–Tamme)**.** *Algebraic K-theory satisfies pro-cdh descent for abstract blow-up squares of noetherian schemes. As a consequence, if X is a noetherian scheme of Krull dimension d, then*

$$K_i(X) = 0 \qquad \text{for } i < -d.$$

Proof. The proof begins by replacing a singular scheme by a blow-up square in which one term has better singularities and the remaining terms are infinitesimal thickenings of the exceptional locus. Pro-cdh descent says that nonconnective K-theory sends the resulting pro-system of blow-up squares to a homotopy cartesian square. One then argues by induction on the dimension, using the fact that the exceptional pieces have strictly smaller dimension and that regular schemes have vanishing negative K-groups. The formal shape of the argument is therefore still Quillenian: one deduces global vanishing from localisation, comparison, and induction on support. The difficult new input is that birational and infinitesimal corrections must be inserted, and this is exactly the content of [KST18]. □

Remark 11.16. Weibel's book remains the standard synthesis of these developments from the classical side: it makes clear how Bass's fundamental theorem, Quillen's localisation, Thomason's nonconnective refinements, and Schlichting's exact-category methods fit into a single long-range picture. In that picture, negative K-theory is not an afterthought but the inevitable extension of Quillen's exact functorial machine to singular geometry.

11.7. Kelly, descent without resolution, and motivic comparison

A different line of development concerns the comparison between K-theory, higher Chow groups, and motivic or étale cohomology in positive characteristic. Voevodsky's original proofs often relied on resolution of singularities through the cdh-topology. Kelly showed that Gabber's alterations can be used instead, via the ldh-topology, to recover large parts of the motivic comparison formalism over a perfect field after inverting the exponential characteristic.

Proposition 11.17 (Kelly, in broad form)**.** *Working with* $\mathbf{Z}[1/p]$*-coefficients over a perfect field of exponential characteristic* p*, one can replace cdh-descent arguments based on resolution of singularities by ldh-descent arguments based on alterations. In particular, comparisons between Bloch's higher Chow groups and étale cohomology that formerly required resolution hypotheses become unconditional in this setting.*

Proof. The central step is to compare cdh and ldh sheafifications for presheaves endowed with transfers or, in Kelly's language, with traces. Gabber's theorem on alterations supplies the finite covers needed for ldh descent, while the trace formalism controls passage back to cdh information. Once the relevant motivic spectra are shown to satisfy ldh descent, Suslin-type comparison theorems and parts of the motivic spectral sequence survive without assuming resolution of singularities. The proof is highly technical, so we cite Kelly for the descent machinery and emphasize only the conceptual outcome: the cycle-theoretic side of the Quillen story extends much farther in positive characteristic than one could previously justify. □

11.8. Negative K-theory and the boundary case

Bass's fundamental theorem already suggests that K-theory should continue below degree 0. Indeed, once one has the exact sequence relating $\mathrm{K}_n(R)$, $\mathrm{K}_n(R[t])$, $\mathrm{K}_n(R[t,t^{-1}])$, and $\mathrm{K}_{n-1}(R)$, it is difficult to avoid the conclusion that negative groups ought to exist if localisation is to remain formally stable. From this perspective, negative K-theory is not an exotic appendix: it is the correction term required for localisation to remain true in all degrees.

For regular noetherian rings these negative groups vanish, so in the classical world they are largely invisible. For singular schemes, however, they measure how far perfect complexes fail to behave as though they were built from vector bundles. One may think of K_{-1}, K_{-2}, and so on as recording the obstructions to continuing Quillen's delooping indefinitely while preserving the same formal exactness properties.

Schlichting's work made this precise by constructing nonconnective K-theory for exact and triangulated settings and by proving localisation, additivity, resolution, and devissage statements in negative degrees as well [Sch06]. This is conceptually important for the present notes: the Thomason–Trobaugh localisation sequence is not merely a final theorem about positive homotopy groups, but the connective shadow of a larger localising invariant.

The sharp geometric statement is Weibel's vanishing conjecture, proved by Kerz–Strunk–Tamme [KST18]: if X is a noetherian scheme of Krull dimension d, then

$$\mathrm{K}_n(X) = 0 \qquad \text{for } n < -d.$$

The theorem is best understood as a boundary result. It says that singularities can force negative K-groups to survive, but only up to the geometric dimension. Thus $\mathrm{K}_{-d}(X)$ is the borderline group; it may be nonzero and often contains meaningful singular information, whereas all lower groups must vanish.

Remark 11.18. A good mental model is the following. For a regular d-dimensional scheme, all negative groups vanish, so the Bass–Thomason exact machinery is invisible below degree 0. For a singular d-dimensional scheme, nontrivial groups may appear exactly in the range $-d \leq n \leq -1$. The size of this range is thus a quantitative measure of how much singular geometry remains after one passes from vector bundles to perfect complexes.

11.9. Excision, blow-ups, and descent beyond the classical formalism

The original localisation theorems of Quillen and Thomason are already descent statements in disguise: they say that K-theory can be reconstructed from exact subquotient data. Modern developments sharpen this by asking for descent with respect to genuinely geometric squares, especially blow-up squares and pullback squares of rings or ring spectra.

A first decisive extension is Tamme's excision theorem [Ta18]. In a pullback square of ring spectra satisfying a Tor-unitality hypothesis, any localizing invariant — hence in particular nonconnective K-theory — carries that square to a homotopy cartesian square. This theorem explains why excision should be regarded not as an accidental property of K-theory but as a structural consequence of its universal localising nature.

From the algebro-geometric side, Kerz–Strunk–Tamme prove pro-descent for abstract blow-up squares [KST18]. This result may be viewed as a deep refinement of the localisation philosophy of Thomason–Trobaugh. Whereas the classical localisation sequence compares a scheme with an open complement and a closed support condition, pro-cdh descent compares a singular scheme with all of its infinitesimal thickenings along a blow-up square. In other words, one no longer asks merely for exactness under removing a closed subset; one asks for a controlled passage through birational modifications and infinitesimal neighbourhoods.

Remark 11.19. This is one of the places where the passage from vector bundles to perfect complexes is decisively vindicated. Vector bundles alone do not see singular schemes stably enough for such descent results. Perfect complexes, by contrast, are robust under derived and infinitesimal constructions, which is precisely why Thomason–Trobaugh became the correct framework for later progress.

11.10. Trace methods and the cyclotomic approximation to K-theory

A second modern leitmotif is that K-theory is often inaccessible directly but is remarkably close to topological cyclic homology TC. The bridge is the cyclotomic trace

$$\mathrm{K} \longrightarrow \mathrm{TC},$$

whose computational force comes from the fact that TC is built from structured fixed-point data and is often approachable by homotopy-theoretic, p-adic, and filtration methods.

The guiding philosophy is subtle. K-theory is usually the genuinely universal invariant, so it is the one with the strongest formal properties. But TC is often the calculable approximation. One proves deep structural results by showing that the fibre of the cyclotomic trace is small, nilpotent, or even eventually trivial in suitable ranges.

A landmark theorem of Clausen–Mathew–Morrow shows that for a henselian pair (R, I) the relative K-theory and relative topological cyclic homology with finite coefficients agree via the cyclotomic trace [CMM21]. This recovers and extends earlier rigidity theorems and implies that for large classes of p-adic rings, K-theory is extremely well approximated by TC in high degrees. Conceptually, the theorem says that in henselian situations the gap between algebraic and topological trace invariants becomes nearly invisible.

This viewpoint has reshaped the subject. Instead of trying to compute $\mathrm{K}(R)$ directly, one often computes $\mathrm{TC}(R)$ using filtrations coming from Hochschild-type theories, Nygaard or prismatic structures, and then transfers the result back to K-theory through comparison theorems. Even when no complete equivalence is available, the trace map isolates the genuinely K-theoretic error term and turns it into something geometrically interpretable.

11.11. Universal properties and noncommutative motives

The classical definitions of Quillen and Waldhausen are constructional: one builds K-theory by means of the plus construction, the Q-construction, or the $S_\bullet$-construction. Modern higher-categorical work reveals that these are not merely clever recipes. They realise K-theory as the universal additive or localising invariant on appropriate stable categories.

Blumberg–Gepner–Tabuada show that connective algebraic K-theory is the universal additive invariant of small idempotent-complete stable ∞-categories, while nonconnective K-theory is the universal localising invariant [BGT13]. This result recasts additivity and localisation as part of the defining essence of the theory. It also explains conceptually why so many comparison maps out of K-theory — to Hochschild homology, topological Hochschild homology, and topological cyclic homology — are canonical: they are induced by universal properties rather than by ad hoc constructions.

From this perspective, Waldhausen's additivity theorem and Thomason–Trobaugh localisation are not isolated miracles. They are the first visible instances of a general principle: K-theory is initial among invariants that convert exact or localising decompositions into fibre sequences.

11.12. Motivic refinements and algebraic cobordism

There is an equally powerful geometric reformulation of the same idea. In motivic homotopy theory, algebraic K-theory is represented by the motivic spectrum KGL, and the motivic spectral sequence identifies the associated graded pieces with motivic cohomology. Thus the filtration behavior that looked mysterious from the point of view of rings becomes geometrically natural when viewed inside the motivic stable homotopy category.

Recent work of Annala and Iwasa gives a universal description of K-theory in a broad motivic setting without assuming $\mathbb{A}^1$-invariance from the outset [AI22]. Combined with subsequent work of Annala–Hoyois–Iwasa on algebraic cobordism and the Conner–Floyd isomorphism for algebraic K-theory [AHI25], this shows that K-theory continues to sit at the meeting point of orientation theory, projective bundle formulae, and periodicity phenomena, closely paralleling the role of topological complex K-theory in stable homotopy theory.

For the reader of the present notes, the main conceptual lesson is the following. Grothendieck's K_0 began as the universal additive receptacle for vector bundles. Modern motivic theory shows that this universality persists at the spectral level and interacts with algebraic cobordism much as Bott periodic K-theory interacts with complex cobordism in topology. The old and new pictures are not separate subjects; they are different levels of the same architecture.

11.13. Outlook

Several active borderlines remain.

(i) **Borderline negative groups.** Vanishing below $-d$ is now known, but the borderline group $\mathrm{K}_{-d}(X)$ and the groups just above it still encode delicate singular information and remain difficult to describe in a concrete geometric way.

(ii) **Trace methods in mixed characteristic.** The comparison between K-theory and TC is now powerful enough to import methods from p-adic Hodge theory, and this interaction continues to produce new structural results and new computational tools.

(iii) **Derived, logarithmic, and spectral geometry.** The correct home for modern K-theory is often a derived or spectral enhancement of classical algebraic geometry, where perfect complexes are more natural than ordinary sheaves and where localisation becomes compatible with deeper homotopical structures.

(iv) **A-theory and manifold topology.** Waldhausen's vision that K-theory of spaces should control manifold invariants remains one of the deepest bridges in the subject, and the relation between algebraic, geometric, and topological K-theories continues to be a major source of difficult problems.

We may therefore summarise the subject in one final sentence. Grothendieck identified the additive shadow of geometry; Quillen discovered its higher homotopy content; Waldhausen extended it to homotopical categories; Thomason and Trobaugh made it local on schemes; and modern work shows that this same theory sits simultaneously at the boundary of singularity theory, trace methods, motivic homotopy theory, and noncommutative geometry. The enduring force of algebraic K-theory is precisely that it remains recognisably the same invariant through all of these enlargements while continually revealing new layers of structure.

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